

Relational-Algebraic Pregeometry: Finite Kernels, Open Continuum Emergence, and Formal Verification

**A Falsification-First Programme^[D] for a Finite Relational-Algebraic Kernel,
an Explicitly Open Passage to Apparent Continuity,
and a Non-Ontological Verification and Parameter Ledger**

Relational-Algebraic Pregeometry Programme — Version 4.0

Version 4.0 of the research programme formerly designated
Unified Information-Density Theory (UIDT)

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*“Reality is not a collection of things,
but a process of structural individuation
that we are licensed to describe only as far as we can falsify it.”*

Governance Status of This Finalization Candidate

The input to this patch is the watermarked manuscript blob 7962e5e7 (full Git blob recorded in the package provenance file), independently reproduced from the supplied source file. The earlier MSS-000 architectural rebase is therefore treated as the applied baseline for this finalization pass; stale “executor candidate” wording in that baseline is corrected here. This patch closes the scientifically determinate PI dispositions, normalizes compound evidence tags, and separates historical Lean snapshots from live repository claims. It is not a canonical freeze, not a merge authorization, and not an evidence-class upgrade. Claude Desktop audits the complete diff and applies it as the sole writer of the active `.tex` after PI authorization. Build, repository reconciliation, and technical post-application verification then return to the executor lane. Final integration remains PI-only.

PI Decision D19 — Ontological Re-Architecture (CLOSED)

Decision. The continuous real scalar field $S_{\text{legacy}}(x)$, treated in UIDT v3.9.x as the ontological primitive, is demoted to a historical macroscopic effective-field description. The active v4.0 candidate primitive is a finite relational-algebraic kernel, presented strictly as a research programme at evidence class ^[D]. The decision is closed under the PI standing instruction that scientifically determinate dispositions are approved.

Scientific basis. A scalar-gradient construction cannot generate non-trivial curvature because $d^2 = 0$; a single self-adjoint generator also yields an abelian generated algebra on which its own inner derivation vanishes. The manuscript supplies no non-circular bridge from the historical scalar to the required non-commutative multiplicity structure. Retaining the scalar as foundational would therefore overstate the available mathematics.

Binding conditions. (i) Part I remains a formal research programme at ^[D], not an established ontology or theorem corpus. (ii) No Standard-Model-like block partition may occur in a core definition, axiom, hypothesis, loss, or acceptance condition; candidate instances may occur only downstream of target-independent predicates under GLBC. (iii) Historical quantitative results retain their incoming evidence classes inside Part III or the museum appendix.

Status. CLOSED on 23 July 2026. This is a governance determination, not a new physics result and not an evidence-class upgrade.

PI Decision D20 — Part III Architectural Reassignment

Authorized architectural change. The quantitative content of the former effective-scalar Part III — including v , λ_S , κ , the EFT machinery, benchmark batteries, and cosmological calibration ledger — has no established derivation from the pregeometric kernel and therefore does not function as active ontology. Under D20 it is reorganized as a verification, benchmark, and parameter-ledger stratum. No quantitative result

is deleted or reclassified. The finite relational-algebraic kernel of Part I remains a falsifiable research programme at evidence class D; the passage of Part II remains open and supplies no continuum, metric, signature, Born-rule, or Standard-Model gauge-group derivation.

Historical scope boundary. At the time of D20, PI-DELTA-P6, PI-ET-P6, PI-MUSEUM, and P2 were explicitly excluded. They are resolved separately below by Decisions D21–D24. D20 itself changes no quantitative value and no evidence class.

Application status. The MSS-000 architecture is present in the supplied watermarked baseline blob 7962e5e7; this finalization patch preserves D20 and does not authorize merge or freeze.

PI Decision D21 — Δ Evidence Status (CLOSED)

Decision. The proposed upgrade of $\Delta = 1.710 \pm 0.015 \text{ GeV}$ from [B] with uncertainty to an exact Class A value is rejected. Determination D18 remains controlling: $\Delta = 1.710 \pm 0.015 \text{ GeV}$ is a lattice-compatible reference with no UIDT-internal derivation of the number and no operator matching to a physical particle observable. It is not a particle mass and is not identified with the 0^{++} glueball.

Status. CLOSED: retain D18 and [B]. No external counter-signature is required because no upgrade is made.

PI Decision D22 — E_T Firewall and Class (CLOSED)

Decision. $E_T = 2.44 \text{ MeV}$ remains a subatomic reference at [E]; the proposed upgrade to Class C is rejected because no specified subatomic observable with uncertainty and no validated scale bridge has been supplied. The MeV-to-neural, biological, cognitive, macroscopic, or life/death transfer is barred from active v4.0 use unless a separately audited physical channel, coupling, transfer function, observable, and empirical protocol are produced.

Status. CLOSED: firewall confirmed; class retained at [E].

PI Decision D23 — Historical Scalar Museum (CLOSED)

Decision. Museum Route B is final for v4.0. The historical scalar/EFT material remains in a visibly marked, non-imported appendix. It is preserved for provenance and regression audit, supplies no premise to Parts I–III, and may not be silently deleted or reactivated.

Status. CLOSED: reversible appendix retention approved.

PI Decision D24 — FSS/P2 Disposition (CLOSED BY QUARANTINE)

Decision. Because the required raw data, immutable run artefacts, seeds, configuration, and provenance are unavailable, FSS/P2 is placed in archival quarantine. It contributes no active quantitative claim, no evidence upgrade, and no manuscript premise. The disposition may be reopened only by a new PI record after source-faithful raw-data audit. No simulation or result reconstruction is authorized by this decision.

Status. CLOSED as a governance disposition; the underlying scientific question remains unresolved.

PI Decision D25 — Programme and Manuscript Name (CLOSED)

Decision. The active v4.0 programme name is *Relational-Algebraic Pregeometry Programme*. The manuscript title is *Relational-Algebraic Pregeometry: Finite Kernels, Open Continuum Emergence, and Formal Verification*. “Unified Information-Density Theory (UIDT)” is retained as the historical name for versions 1–3.9, existing DOI titles, repository provenance, and legacy notation.

Application rule. The scientific-content patch is audited and applied before the isolated name-migration patch, so that branding changes do not obscure substantive review.

Status. CLOSED under the PI’s stated agreement with the naming recommendation.

Name and Provenance Convention

Relational-Algebraic Pregeometry Programme is the active name of the v4.0 research programme. The token UIDT is retained only for the historical programme lineage, versions 1–3.9, existing DOI titles, repository and path identifiers, quoted legacy claims, and historical effective-sector notation. A retained UIDT occurrence therefore does not reactivate the superseded “Unified Information-Density Theory” ontology.

Reading Order and Epistemic Weight

EVIDENCE

Part I (*The Pregeometric Kernel*) remains a ^[D]-class research programme. It localizes obstructions, states conditional lower bounds, and records finite-algebraic and Lean 4 structures without promoting an inserted candidate to an established foundation. Part II (*The Open Passage to Apparent Continuity*) is capped at ^[E] wherever it introduces coarse-graining, observer functors, thermal-time interpretation, or apparent continuity; it is an obligation register, not a supplied mechanism. Part III (*Verification, Benchmarks, and Parameter Ledger*) retains every quantitative result at its incoming class: $\kappa = \frac{1}{2}$ (exact) ^[A], $\lambda_S = 5/12 \approx 0.4167$ ^[A], $v = 47.7 \text{ MeV}$ ^[A], $\gamma = 16.339$ ^[A-] (calibrated, never derived), and $\Delta = 1.710 \pm 0.015 \text{ GeV}$ ^[B] under D18 and Limitation L12. Cosmological statements remain capped at ^[C]. The historical scalar/EFT model is

preserved in Appendix A as a non-imported museum record; its equations are not premises of Parts I–III.

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Abstract

This manuscript presents a relational-algebraic pregeometry programme whose active foundational layer contains no classical scalar field, background spacetime, metric, or smooth-manifold primitive. The candidate kernel is expressed through finite combinatorial and algebraic data, with the formal separation $\text{BlockPartition} \rightarrow \text{FiniteAlgebraSignature} \rightarrow \text{SpectralTriple}$, and remains a falsifiable research programme at evidence class ^[D]. The $d^2 = 0$ scalar-gradient obstruction, the unresolved origin of G_{SM} , the absence of a Born-rule derivation, and the absence of solutions to the H_0 and S_8 tensions are retained as explicit boundaries.

Part II records the open passage from the finite kernel to apparent continuity. It supplies no dimensional-emergence theorem, Lorentzian signature, background metric, or non-circular observer derivation and is capped at ^[E] wherever it becomes interpretive. Part III is a verification, benchmark, and parameter-ledger stratum rather than active ontology. It preserves the incoming quantitative content without evidential upgrade: $\kappa = \frac{1}{2}$ (exact) ^[A], $\lambda_S = 5/12 \approx 0.4167$ ^[A], $v = 47.7 \text{ MeV}$ ^[A], $\gamma = 16.339$ ^[A-] as a calibration, and $\Delta = 1.710 \pm 0.015 \text{ GeV}$ ^[B] under D18/D21 and Limitation L12. Cosmological entries remain calibrations capped at ^[C].

The superseded scalar/EFT architecture is retained in a visibly marked, non-imported historical appendix. Its Lagrangian, coordinate derivatives, background metric ansatz, and scalar correlator programme are provenance records and are not premises of the active architecture. The manuscript therefore claims a disciplined research programme and a set of localized obstructions, not a completed derivation of spacetime, the Standard Model, or its empirical parameters.

1 Evidence Grammar, Strata, and Audit Discipline

The order of this manuscript is deliberate. Before the pregeometric kernel is defined, before a single formal structure is introduced, this chapter fixes the discipline by which every later assertion will be judged. A re-architecture is precisely the moment at which a framework is most tempted to convert enthusiasm into evidence; the apparatus below exists to make that conversion impossible.

1.1 Evidence Classes

Every nontrivial claim in this manuscript carries exactly one of six evidence classes, rendered as a typographic tag. ^[A] marks mathematical proof closure, an exact identity, or a verified deterministic theorem under stated assumptions; internal residual closure below 10^{-14} at 80-digit `mpmath` precision qualifies only for deterministic arithmetic and is never external physical agreement. ^[A-] marks a calibrated internal closure: calibrated, not derived. ^[B] marks lattice-compatible or externally benchmarked physics with verified sources and explicit uncertainty handling. ^[C] is the cosmology cap; no cosmological statement in this manuscript exceeds it, and no statement at ^[C] claims to resolve any observational tension. ^[D] marks a prediction or research programme: falsifiable but unconfirmed. ^[E] marks speculative, interpretive, withdrawn, or museum content. No other class exists; composite or inflected tags (such as A+ or B—) are not part of the grammar and their appearance anywhere in this document would itself constitute an audit failure.

1.2 Strata

Claims are additionally stratified by provenance. Stratum I comprises empirical measurements and externally sourced values with uncertainties. Stratum II comprises consensus theory: standard quantum field theory, lattice methodology, the Standard Model, Λ CDM formalism, and — central to Part I — the established mathematics of finite real spectral triples and Krajewski classification. Stratum III comprises Relational-Algebraic Pregeometry interpretation: the pregeometric-kernel hypothesis, the rejection of the historical scalar ansatz as active ontology,

every mapping between formal structure and physical ontology, and all observer-theoretic material. The cardinal rule is that Stratum III is never written as Stratum I, and that the relevance of a Stratum II mechanism is never presented as validation of a Stratum III interpretation.

1.3 Falsification-First Rule

Every nontrivial scientific, mathematical, or ontological claim in this draft carries at least one of: an empirical falsification condition, a mathematical proof or verification condition, a numerical reproduction condition, or an explicit downgrade condition. Where no such condition can be stated, the claim is downgraded to ^[D] or ^[E] on the spot. A claim that cannot fail is not treated as a physics result.

1.4 The Gap-Localization-before-Construction Principle

Part I is governed by a methodological rule inherited from the v3.9 audit cycle and here promoted to a structural constraint: *Gap Localization before Construction* (GLBC)^[E] (methodology-class content; the tag records that GLBC is a working discipline, not a theorem). Formal types are defined empty before any physical scale, any ledger constant, or any Standard-Model-like pattern is applied to them. In particular, the calibrated invariant $\gamma = 16.339^{\text{[A]}}$ and the spectral gap $\Delta = 1.710 \pm 0.015 \text{ GeV}^{\text{[B]}}$ never appear inside any core formal structure of Part I — not as targets, not as losses, not as initial conditions, not as stopping rules — and the Standard-Model-like block partition enters exclusively as an inserted candidate instance whose endogenous selection is recorded as open. Any violation of this rule constitutes target leakage and voids the affected construction.

1.5 Numerical Discipline

Where proof-critical arithmetic occurs, it is executed in Python 3 with `mpmath` at `mp.dps = 80`, set locally inside the proof-critical function rather than globally at module scope; binary floating-point heuristics, `float()` coercion, and precision-losing conversions are barred from proof-critical paths. High-precision output is rendered via `mp.nstr(value, 80)`. No new 80-digit values are invented or

estimated in this manuscript; numerical statements either reproduce ledger values at their recorded classes or are absent.

1.6 Provenance of This Manuscript

This manuscript consolidates: the canonical v3.9.9 manuscript (-006, four-checker verified); the corrected G1–G4 admissibility lattice, with the block-repetition misreading of the multiplicity condition explicitly quarantined as a documented category error; the Phase-7 Lean 4 axiom-initialization architecture (`RealStructure`, star-compatible representations, and an abstract first-order-condition schema); the honest null result of the matrix-condensation metastability protocol; and the division-algebra programme recorded as a ^[D] candidate for the intermediate algebra. Sources that failed primary-source verification during the audit cycle — including an external audit document that rejected the Standard-Model-like partition on a mathematically false multiplicity argument — are excluded from evidentiary use and cited, where cited at all, only as documented failure history.

Part I

The Pregeometric Kernel ^[D]

Part I develops the candidate primitive of the v4.0 re-architecture: a finite algebraic structure of real spectral-triple type, formalized in a typed proof assistant under the GLBC principle. Everything in this part is a research programme at evidence class ^[D] unless a narrower tag is stated. The part establishes, in order: the primitive data and their Lean 4 typing; the Krajewski combinatorial frame; the G1–G4 obstruction set with the corrected admissibility lattice; the lower bounds any admissible intermediate algebra must satisfy; and the open selection problems — Wedderburn moduli, real structure, KO-dimension, and chirality — that define the programme’s falsifiable milestones. There is no space in this part, and no time: both are claims Part II must earn, not assumptions Part I is permitted.

2 The Demotion of the Continuous Scalar and the Kernel Primitive

2.1 The Negative Results That Motivate the Re-Architecture

The v3.9 audit cycle produced three independent negative results whose joint force motivates the present re-architecture. None of them is new to this manuscript; each is carried over at its established evidence class, and the re-architecture adds no evidential weight to any of them.

First, the scalar-gradient route to gauge curvature fails exactly^[A]. If a gauge potential is built as $A = dS$ from a smooth scalar on a manifold, then $F = dA = d^2S = 0$ identically in ordinary exterior calculus. This is a clean negative result for that construction — and only for that construction; it is not a universal no-go theorem for enriched algebraic routes.

Second, the G_{SM} -origin gap is localized, not closed. Within established theorems, no gapless ab initio derivation a structureless real scalar $\rightarrow G_{SM}$ exists; the obstruction is concentrated in four jointly unsatisfied conditions (G1–G4) on fiber multiplicity, Wedderburn invariants, real structure, and chirality (section 4). On every known route, the Standard-Model gauge group is selected, not forced.

Third, the dynamical-selection programme returned an honest null. The blinded, preregistered metastability pre-protocol for matrix condensation (PREREG-PR-B1-PRE-001, PI-signed 2026-06-17; detector, coupling window $\tilde{\alpha} \in \{0.40, 0.55, 0.625, 0.75, 0.90\}$, and matrix sizes $N \in \{16, 24, 32\}$ frozen before any result was seen) found that a planted Standard-Model-like block configuration decays out of its class — in the detector’s class-membership sense — within approximately ten Monte-Carlo steps of thermalization at every tested coupling, confirming the preregistered literature null (single-sphere vacuum) with an independently validated instrument.^[D] The protocol, its frozen outcome table, and the recorded outcome (PRE-01) are archived at `verification/prereg/PR-B1/` with the corresponding ledger entry in `LEDGER/CLAIMS.json`. No verified mechanism forcing a stable asymmetric block vacuum is known; the target-independent construction problem remains an open research programme at ^[D]. Any broader thermodynamic interpretation of that unknown mechanism remains speculative at ^[E].

A primitive that cannot generate the gauge sector, cannot curve anything through its own gradient, and is not dynamically selected by the one condensation mechanism tested, retains its primitive status only by stipulation. The re-architecture withdraws the stipulation.

What the Demotion Does and Does Not Claim

The removal of the historical scalar ansatz from active ontology under Decisions D19–D20 (governance page) is an ontological reassignment, not a physics result. It does *not* assert that a discrete pregeometric kernel is established; it asserts that the continuous scalar no longer carries the burden of primitivity, and that the candidate now under investigation is the finite algebraic structure of Part I, at evidence class ^[D] throughout. Replacing one unproven Stratum-III primitive by another unproven Stratum-III primitive yields no upgrade anywhere; what it yields is a candidate whose obstructions are sharper, whose lower bounds are theorem-shaped, and whose formalization is machine-checkable in principle.

2.2 Design Directive DIR-S-01: The Primitive as a Pregeometric Operator

The formal seat of the demotion is Design Directive DIR-S-01, encoded in the formalization layer as a foundation module. The primitive of the v4.0 programme is a pregeometric operator $\mathbf{S}$, not a classical field $S: M \rightarrow \mathbb{R}$ on a pre-given four-dimensional spacetime. Coordinates and metric are required to emerge at a later ontological level; historical routes through $A = dS$ on a smooth manifold are excluded at the type level, in direct response to the $d^2 = 0$ obstruction.

```

/-- [D/E] Pre-geometric UIDT Primitive Operator S.
    DIR-S-01: S is an operator in a suitable C*-algebra /
    on a Hilbert space, not a classical field
    S : spacetime -> R. The concrete realization (Matrix,
    NCG, Tensor-Network) is an additional instance. -/
class PrimitiveOperator (S : Type*) where
  isPrimitive : Prop := True

```

Three features of this definition deserve emphasis, because each is a deliberate act of restraint. There is no spacetime dependence: the module does not define $S: \text{Spacetime} \rightarrow \mathbb{R}$, and no metric tensor is available at this level. The class is intentionally minimal: algebraic axioms (self-adjointness, positivity, spectral properties) are deferred until the formalization matures, rather than asserted decoratively. And the $d^2 = 0$ obstruction is recorded in the module as a negative design commitment — the smooth-field route is rejected entirely, not formalized and then argued around.

The kernel-primitive hypothesis is downgraded from ^[D] to ^[E] if any of the following occurs: (i) a theorem shows that no finite real spectral triple satisfying the Part-I lower bounds (section 4.4) can couple consistently to the Part-III effective sector; (ii) the conditional selection structure of section 5 is shown to be empty or inconsistent once H1 is derived from intersection-form axioms; (iii) the formalization programme is abandoned with its axiom layer unclosed. Conversely, no success of Part III arithmetic upgrades the kernel hypothesis.

3 The Lean 4 Formalization Architecture

3.1 Scope, Source of Truth, and Audit Boundary

The formalization lives in the working repository Relational-Algebraic-Pregeometry (module namespace Antigravit2), audited for this manuscript at commit 61bc24da (2026-07-08, merge of the phase-10b axiom pull request). The corpus at that head comprises 26 Lean files across five layers: Foundation (the primitive operator),

MatrixThermo (block partitions), Filters (enumeration and admissibility predicates), NCG (real structure, first-order condition, Krajewski combinatorics, spectral-triple assembly), and Meta (evidence tags and the anti-target-leakage policy).

Two boundaries are binding for every claim in this section. First, the audit underlying this manuscript is a *static structural audit* of the repository snapshot at the stated head; no `lake build` was executed in the audit session, and no build result is asserted here. Second, a successful build would in any case certify only type-checking; it would not discharge the open proof obligations catalogued in section 3.7, and it would never convert a formal result into a physics claim. Lean-proven statements are ^[A] strictly *within the formal system*; their physical import is bounded by the epistemic status of the hypotheses they consume.

3.2 Gap Localization before Construction, Operationalized

The GLBC principle of section 1.4 is not a stylistic preference in this corpus; it is implemented as type-system structure. The gap parameter module defines the spectral gap purely formally and blocks its premature physical identification:

```
-- [D] Formal spectral gap parameter.
   No physical value (like 1.710) is assigned here. -/
structure SpectralGapParam where
  delta : Real
  delta_pos : 0 < delta

-- Historical software sentinel in the 61bc24da snapshot.
   This definition is not a proof of physical non-identification. -/
```

```
def DeltaGlueballConjecture (Delta : Real) : Prop := False
```

The second definition is a software sentinel, not a theorem. Defining a proposition to be False can prevent downstream code from consuming that proposition, but it does not prove that two physical quantities are unequal and does not encode the missing operator-matching problem.^[A] The manuscript therefore relies on Limitation L12 and Decision D21, not on this declaration, for the non-identification boundary. The repository repair tracked as LEAN-009 must replace the sentinel with typed separation between a mathematical spectral-gap object, a physical mass observable, and an explicit operator-matching witness. Statements about literal absence of protected constants are restricted to the historical 61bc24da snapshot until a live commit tree is reconciled.

3.3 The Real-Structure Layer (Phase 7)

The Phase-7 axiom layer follows the reviewed specification exactly: anti-linearity is carried by a dedicated structure, conjugation is expressed through the field's star ring endomorphism, and the operator J is deliberately *not* equipped with invertibility or unitarity — J^{-1} is excluded from the phase in every form, including symbolically.

```
-- [D] Anti-linear map over field C.
J(a . x) = conj(a) . J(x). -/
structure AntiLinearMap (H : Type _)
  [AddCommGroup H] [Module Complex H] where
  toFun      : H -> H
  map_add    : forall x y,
    toFun (x + y) = toFun x + toFun y
  map_smul   : forall (a : Complex) x,
    toFun (a . x) = (starRingEnd Complex a) . toFun x

-- [D] RealStructure: J and the KO sign data.
J is anti-linear but NOT required to be invertible
or unitary here. J-inverse is excluded from Phase 7. -/
class RealStructure (H : Type _)
  [AddCommGroup H] [Module Complex H] where
```

```

koDimension : ZMod 8
J      : AntiLinearMap H
eps    : Int
epsD   : Int
epsG   : Int
J_involutive : forall x, J (J x) = (eps : Complex) . x

```

A trivial instance on $\mathbb{C} \multimap J$ realized as complex conjugation with all signs $+1$ — closes the regression layer without sorry and certifies that the architecture is self-consistent on the smallest non-degenerate carrier. One deviation from the drafting-session specification is recorded for traceability: the head keeps only the involutivity relation $J^2 = \varepsilon \text{id}$ inside the class, deferring the JD and $J\gamma$ commutation relations, while adding a `koDimension` field of type $\mathbb{Z}/8$. This is a conservative narrowing — fewer axioms asserted, none invented — and is consistent with the phase philosophy.

3.4 The First-Order Condition as a Commutator Prop. (Phase 10b)

The first-order condition has graduated from the Phase-7 envelope to a genuine structural proposition. The head defines a minimal spectral-triple core — Dirac operator, real-structure carrier, grading, and separated left and right representations, the latter through an opposite-algebra wrapper — and states the condition in its standard commutator form:

```

/-- Commutator abstraction for endomorphisms. -/
def Commute {H : Type _} (F G : H -> H) : Prop :=
  forall x, F (G x) = G (F x)

/-- First-Order Condition (FOC):
     $[[D, a], b^o] = 0$  for all  $a$  in  $A$ ,  $b^o$  in  $A^o$ . -/
def firstOrderCondition {A H : Type _} [AddGroup H]
  (ST : SpectralTripleCore A H) : Prop :=
  forall (a : A) (b_op : Opposite A),
    Commute
      (fun x => ST.D (ST.leftRep a x)
        - ST.leftRep a (ST.D x))

```

(ST.rightRep b_op)

This is the honest formal shape of the NCG first-order axiom at the current stage: a proposition that can be stated and instantiated, with the analytic content (boundness, adjointability, the bimodule structure of the right action) deferred to later phases. Nothing here asserts that any physically interesting triple satisfies the condition.

3.5 The Krajewski Combinatorial Kernel and the Inserted Candidate

The combinatorial layer is deliberately stripped of physical labels. An edge of a Krajewski diagram is a bare record of two block indices and a multiplicity; chirality-capability appears only as the derived structural property that an edge connects distinct blocks; and diagram admissibility at this level demands nothing beyond the absence of self-loops.

```
structure Bimodule where
  leftBlock    : Nat
  rightBlock   : Nat
  multiplicity : Nat

def Bimodule.asymmetric (b : Bimodule) : Prop :=
  b.leftBlock /= b.rightBlock

structure KrajewskiDiagram where
  edges : List Bimodule
```

Against this neutral kernel, the Standard-Model-like configuration enters exactly once, and exactly as governance condition (ii) of D20 requires: as an *inserted candidate instance*, in a dedicated instance module, decoupled from the filter hypotheses and from every physical constant. The instance fixes the block list $[3, 2, 1]$ and the three off-diagonal edges, and proves — unconditionally, because the property is purely structural — that the resulting diagram has no self-loops. That theorem is [A] within the formal system and physically weightless: it certifies combinatorial well-formedness of an inserted diagram, nothing more.

Remark 3.1 (Notation: $(1,2,3)$, $(2,3)$, and $[3,2,1]$). The finite algebra $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ has matrix dimensions $(1,2,3)$ with total carrier dimension $N = 6$. The v3.9 appendix literature labels this pattern by its non-abelian pair as “the $(2,3)$ partition”; the Lean corpus writes the same object as the descending list $[3,2,1]$. All three notations name one candidate configuration, and this manuscript uses the Lean form wherever the formal layer is in view.

No Audited “66 to 7” Krajewski Reduction in This Draft

The exhaustive finite enumeration stated in section 5 is the eleven-element set of sorted integer partitions of $N = 6$ under the declared H1/H2 predicates. It is not an enumeration of sixty-six Krajewski diagrams. No sixty-six-element starting domain, equivalence relation, ordered list of fifty-nine eliminations, or seven-survivor register is supplied in the applied manuscript baseline. Consequently, this manuscript asserts no “66 $\rightarrow$ 7” result. A future insertion requires an immutable starting inventory, target-independent filters, complete disposition of every eliminated diagram, and a proof-status record for each formal step; absent any one item, the number remains MISSING DATA rather than a scientific result.^[D]

3.6 Evidence Tags and Anti-Target-Leakage in Code

The corpus formalizes its own audit discipline. A meta-module defines the six evidence grades as an inductive type for structured documentation, and states the anti-target-leakage policy as a first-class object: no proof may contain its own target as input; the candidate partition must not appear in any definition, axiom, or hypothesis, and may appear only in theorem conclusions and test assertions, enforceable by review and by grep-based continuous-integration checks. The policy also fixes the correct reading of every Lean tag in this Part: unproved formal design structures are ^[D]; interpretive physical readings are separately ^[E]. Where a proof is supplied, the proven proposition is ^[A] within the formal system and is not thereby a physics claim.

3.7 Historical Static Audit at Head 61bc24da

The table below records only the historical static-audit state of commit 61bc24da. “Closed” means that no sorry was found in the named file in that snapshot; it is not a claim about the current repository. Counts came from a literal whole-word scan and no build was executed.

Module	Proof state at head	Character
Foundation/PrimitiveOperator	closed	type/class layer, DIR-S-01
MatrixThermo/BlockPartition	closed	partitions; 14 decidable regressions
NCG/RealStructure	closed	Phase-7 axiom layer, trivial instance
NCG/FirstOrderCondition	closed	FOC as commutator proposition
NCG/KrajewskiCore	closed	combinatorial kernel
NCG/Krajewski321	closed	inserted candidate instance
NCG/DeltaScale	closed	gap parameter, L12 type blocker
NCG/SpectralTriple	5 open	canonical-form regressions (registered)
NCG/AxiomAudit	1 open	unregistered
Filters/Enumeration	11 open	normal-form obligations (registered class)
Filters/EliminationN6	34 open	includes the flagship proposition

Table 1: Lean corpus proof state at commit 61bc24da; 51 substantive open obligations in total. Static audit; no build executed.

3.8 Reconciled Lean Snapshot at Head f1ae5486

The second snapshot is tied only to commit f1ae548633abac7cb47b971917bbab7992801a3f. It is not a statement about master, PR #24, or any later repository state.

Audit object	Observation at f1ae5486	Scope and qualification
Root import graph	18 reachable modules from <code>Antigravit2.lean</code>	The source tree contains 22 Lean files; four are outside the root graph. ^[A]
Environment-qualified build	18/18 reachable modules exited successfully	Direct <code>lean</code> invocation with <code>LEAN_PATH</code> on Windows; 591 seconds; not a pinned-container or complete lake build. ^[A]
Registered proof debt	8 occurrences of <code>sorry</code> , all registered	Five in <code>NCG/SpectralTriple.lean</code> and three in <code>Filters/Enumeration.lean</code> ; no register mismatch was found. ^[A]
Flagship declaration	<code>Filters.unique_321_N6</code> has no <code>sorryAx</code>	Its printed dependency inventory contains <code>propext</code> ; this is a formal-software fact, not a physical selection result. ^[A]
Trust audit	No <code>Lean.ofReduceBool</code> in the supplied inventory	The avoidance of <code>native_decide</code> leaves the eight registered obligations open rather than importing that trust path. ^[A]

Table 2: Named-commit Lean reconciliation at f1ae5486. Every row is commit-scoped; the build statement is environment-qualified while PROV-002 remains open.

The complete supplied axiom inventory for the named head is:

<code>Antigravit2.NCG.reality_JJ</code>	<code>[propext, Classical.choice, Quot.sound]</code>
<code>Antigravit2.NCG.toSignature_totalDim</code>	<code>[propext]</code>
<code>Antigravit2.MatrixThermo.entropyList_replicate_one</code>	<code>[propext, Quot.sound]</code>
<code>Antigravit2.NCG.trivialRealStruct</code>	<code>[propext, Classical.choice, Quot.sound]</code>
<code>Antigravit2.NCG.trivialFiniteAlgebra</code>	<code>[propext, Classical.choice, Quot.sound]</code>
<code>Antigravit2.Filters.unique_321_N6</code>	<code>[propext]</code>
<code>Antigravit2.NCG.orientabilityCondition</code>	no additional axioms
<code>Antigravit2.NCG.AdmissibleSpectralTriple</code>	no additional axioms

This inventory contains neither `sorryAx` nor `Lean.ofReduceBool`.^[A]

Reachability Boundary and the Latent Krajewski Collision

The four source modules outside the root import graph at f1ae5486 are `NCG/Krajewski.lean`, `Foundation/PrimitiveOperator.lean`, `Meta/EvidenceTags.lean`, and `NCG/ModuliStub.lean`.^[A] The first module's unreachability is why the duplicate Krajewski declaration is la-

tent at this head: a root-only build does not exercise the conflicting module.^[A] The primitive-operator module is also outside the successful set because the reported environment could not resolve `Mathlib.Analysis.CStarAlgebra.Spectrum`; no successful build claim is made for that file.^[A]

The Reconciled Head Must Not Be Projected onto master

At master commit 07ef753b, the root build is reported to fail because `Bimodule` and `KrajewskiDiagram` are declared through conflicting modules.^[A] The failure is recorded in Issue #25 and at two named heads: Ubuntu 24.04 Actions run 28909958494 (2026-07-08) ran at 61bc24da, and run 29294970334 (2026-07-14) ran at 07ef753b; both logs report that the environment already contains `Antigravit2.NCG.Bimodule.mk` from `Antigravit2.NCG.KrajewskiCore`.^[A] Because 61bc24da is an ancestor of 07ef753b and is the historical head documented in table 1, the collision is evidenced at both commits this manuscript names.^[A] Those runner-backed failures predate the later account-level CI-start block; they therefore remain evidence about those two named snapshots, not about f1ae5486 or a future repaired head.^[A]

Live Lean Status Is Commit-Scoped, Not Projected

The five GAP-22 artefacts—the commit tree, complete root import graph, build log, axiom inventory, and whole-repository sorry/admit scan—are available for the named head f1ae548633abac7cb47b971917bbab7992801a3f.^[A] They close the reconciliation requirement only for that commit. Every later head requires a new five-part snapshot, and no result in table 2 is projected onto master, PR #24, or an unspecified current state.^[A] The successful 18-module build remains environment-qualified while PROV-002 is open and does not cover the four unreachable modules.^[A]

Audit Finding F1 Register–Code Contradiction — Historical at 61bc24da, Reconciled at f1ae5486

At the historical head 61bc24da, the register stated that the flagship declaration was closed while the theorem body and its verdict examples still contained sorry; that historical contradiction is retained in table 1.^[A] At f1ae5486, the supplied whole-tree scan and declaration inventory are consistent: `Antigravit2.Filters.unique_321_N6` contains no `sorryAx` and reports `propext` as its dependency.^[A] This closes F1 only as a documentation and formal-software discrepancy at the named commit. The proposition still depends on design-level H1 and heuristic H2, so its physical class remains ^[D]; formal closure is not a physical selection result.

Audit Finding F2 Target-Shaped Admissibility Definition — Status Open at f1ae5486

The historical 61bc24da audit found a candidate-instance wrapper whose admissibility condition explicitly matched KO-dimension 6 and the block list [3, 2, 1].^[A] The supplied f1ae5486 reconciliation does not include the dedicated target-shape scan needed to determine whether that wrapper persists, was repaired, or moved. The

status of F2 therefore remains open; absence of an audit artefact is not evidence of repair.^[D] No statement in this manuscript treats the formal closure of `unique_321_N6` as resolving this separate target-leakage question.

Audit Finding F3 Under-Registered Proof Debt — Historical at 61bc24da, Reconciled at f1ae5486

The historical head contained 51 open obligations and an incomplete register, as retained in table 1.^[A] At f1ae5486, the whole-repository scan reports eight occurrences of `sorry`: five in `NCG/SpectralTriple.lean` and three in `Filters/Enumeration.lean`. All eight are present in the supplied register, so the under-registration defect is not present at that named commit.^[A] The eight obligations remain proof debt; register consistency records them honestly and does not close them.

4 The G1–G4 Obstruction Set and the Corrected Admissibility Lattice

4.1 The Four Conditions

For the strict-scalar primitive with symmetries implemented by canonical linear transformations, the non-derivability of G_{SM} is localized to a single step — the enlargement of the abelian single-mode local algebra to a noncommutative algebra of local multiplicity at least two [77, 78] — and quantified by four conditions that no known mechanism satisfies jointly. **(G1)** Fiber-multiplicity generation $1 \rightarrow n \geq 2$ without inserted variables, blocked homologically (loop phases factor through $H_1(\Gamma) = \pi_1(\Gamma)^{ab}$, bracket identically zero) and algebraically (a scalar-generated algebra is commutative, with trivial inner automorphisms). **(G2)** Theorem-determined Wedderburn invariants: at present the matrix dimension, block partition, and real forms are free moduli [90], and gauge structure is the automorphism shadow of an inserted algebra (Skolem–Noether). **(G3)** An endogenous real structure J of KO-dimension 6: at present an external axiom choice. **(G4)** Chirality-capable matter: blocked by Nielsen–Ninomiya [41] for local translation-invariant lattices, with no established non-abelian chiral construction carrying Standard-Model content. One partial escape deserves explicit statement, because it is the only established endogenous enhancement of a single scalar to a non-abelian current algebra: the Frenkel–Kac construction, which realises $\widehat{\mathfrak{su}}(2)_1$ from a compactified free boson [71, 72]. It satisfies a weakened (G1) *only* in 1+1 dimensions,

under target compactification and radius tuning, and is bounded by a rank-budget argument: a level-one simply-laced affine algebra so obtained supplies $\text{rank} = 1$ against the requirement $\text{rank}(\mathfrak{g}_{\text{SM}}) = 4$. This is the origin of the rank bound of section 4.4.^[A]

The consequence stands as in v3.9: on every known route, G_{SM} is selected, not forced; this is a gap localization, not an impossibility theorem for all conceivable enriched routes.

4.2 The Multiplicity Precision Remark and a Quarantined Category Error

Remark 4.1 (Binding precision on “multiplicity ≥ 2 ”). The G1 condition concerns the *non-abelian fiber*: at least one algebra summand of matrix dimension $n_i \geq 2$, so that the inner automorphism group is non-abelian. It is *not* the requirement that block dimensions repeat, and not a pairing constraint of J on blocks. The real structure constrains the *bimodule multiplicity matrix* V_{ij} [77, 78, 90] — each representation must be accompanied by its conjugate — and Wigner’s anti-unitary argument guarantees that such a J exists already at $\dim V_{ij} = 1$. The Standard-Model algebra $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$, each summand appearing once, carries a KO-dimension-6 real structure [19, 89]. The $[3, 2, 1]$ pattern therefore *satisfies* G1.

An external audit document circulated during the v3.9 cycle rejected the Standard-Model-like partition under G1 on the ground that “each block size occurs only once, and J requires paired basis vectors.” That argument conflates block repetition with bimodule multiplicity and is mathematically false by the result just cited; the same document consequently admitted the all-equal partition $[2, 2, 2]$ and rejected $[3, 2, 1]$ — inverting the corrected lattice on both entries. The document is quarantined at ^[E] as documented failure history and is not used evidentially anywhere in this manuscript.

4.3 The Corrected Per-Partition Lattice

Partition	G1	Verdict	Operative mechanism
$[3, 2, 1] (\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C}))$	passes	admissible ^[B]	real structure exists at $\dim V_{ij} = 1$; anomaly freedom via the standard hypercharge assignment
Pati–Salam-type	passes	mechanism correct ^[B]	admissible without the order-one condition, per the established NCG route [22, 61]
$[6, 3]$ -type (unequal pair)	passes	rejected ^[B]	intersection-form/Poincaré-duality bound $k = \ell \pm 1$ [91]; the earlier “order-one plus anomaly” rationale was incorrect
$[2, 2, 2]$	passes	rejected ^[D]	mass-degeneracy argument; plausible but not verified against a primary-source theorem — the weakest verdict in the lattice

Table 3: Corrected G1–G4 admissibility verdicts. Verdict tags grade the *justification*, not the desirability, of each entry; the $[2, 2, 2]$ rejection is the open verification item.

4.4 Lower Bounds on Any Admissible Intermediate Algebra

Any candidate intermediate algebra $\mathcal{A}$ in a route $\mathbf{S} \rightarrow \mathcal{A} \rightarrow G_{\text{SM}}$ must carry: local multiplicity at least two (a non-abelian fiber); rank resources at least four, against $\text{rank}(\mathfrak{g}_{\text{SM}}) = 4$; a real structure with the J^2 -sign pattern of KO-dimension 6; and a chirality mechanism evading the Nielsen–Ninomiya and finite-2-group obstructions. Any candidate failing one bound is excluded before dynamical analysis. These bounds are the constructive yield of the gap analysis: they tell a model-builder what $\mathcal{A}$ must minimally contain, and they are the standing acceptance test for every candidate discussed in section 6.

5 Filters H1/H2 and the Conditional $N = 6$ Selection Proposition

5.1 The Two Hypotheses and Their Honest Epistemic Grades

The formal selection layer rests on two Boolean predicates over sorted block lists, and the corpus itself grades them with unusual candour. **H1 (design-level)**, the intersection-form filter: consecutive block sizes in a sorted partition differ by at most one. The motivation is the NCG intersection-form constraint on coupled blocks, of which the $k = \ell \pm 1$ bound of table 3 is the two-block instance; the corpus states explicitly that no general dimension-jump theorem of exactly this form has been derived, and marks the derivation from spectral-triple axioms as the upgrade path. **H2 (heuristic)**, the mass-non-degeneracy filter: at least two summands, all pairwise distinct. The motivation is the experimentally non-degenerate fermion spectrum; the corpus states explicitly that “dynamical mass non-degeneracy” is a physical claim about the Yukawa sector for which the combinatorial filter is a proxy. H2 is what eliminates $[2, 2, 2]$ — which is precisely the $^{[D]}$ -graded verdict of table 3, so the formal layer and the manuscript lattice are consistent about where the weakest link sits.

5.2 The Elimination Protocol and the Proposition

Among the eleven sorted integer partitions of $N = 6$, the two filters eliminate ten:

Partition	H1	H2	Eliminating filter
[6]	pass	fail	H2 (simple algebra)
[5, 1]	fail	pass	H1 (jump 4)
[4, 2]	fail	pass	H1 (jump 2)
[4, 1, 1]	fail	fail	H1 and H2
[3, 3]	pass	fail	H2 (repetition)
[3, 2, 1]	pass	pass	—
[3, 1, 1, 1]	fail	fail	H1 and H2
[2, 2, 2]	pass	fail	H2 (repetition)
[2, 2, 1, 1]	pass	fail	H2 (repetition)
[2, 1, 1, 1, 1]	pass	fail	H2 (repetition)
[1, 1, 1, 1, 1, 1]	pass	fail	H2 (all equal)

Table 4: Elimination protocol for the $N = 6$ partition space under H1 and H2, as documented in the formal corpus.

Proposition 5.1 (Conditional $N = 6$ selection; physical class ^[D]; formal declaration closed at f1ae5486). *Under H1 and H2, $[3, 2, 1]$ is the unique admissible partition of $N = 6$.*

The grading of this proposition is layered, and the layers must not be collapsed. At commit f1ae5486, the declaration `Antigravit2.Filters.unique_321_N6` has no sorryAx and its reported dependency inventory is `[propext].[A]` That is a commit-scoped formal-software statement. The physical content is still bounded by its weakest hypothesis: design-level H1 and heuristic H2 place the proposition at ^[D], and no completion of the Lean proof changes that. Only deriving H1 from spectral-triple axioms and replacing H2 by a target-independent stability theorem could alter the physical grade. The falsification structure remains explicit: weakening H1 admits $[5, 1]$ and $[4, 2]$; weakening H2 admits $[2, 2, 1, 1]$ and the all-equal partitions. *Open Question 1* (Staircase conjecture; ^[D]). For general N , the Phase-9-admissible partitions consist exclusively of staircase partitions $[k, k-1, \dots, 1]$ with $k(k+1)/2 = N$, so that admissible total dimensions are triangular numbers. The $N = 6$ case is the first non-trivial instance; $N = 10$ and $N = 15$ are the next test cases. Falsification path: exhibit an N and a non-staircase partition satisfying both H1 and H2.

5.3 Target-Leakage Audit of the Selection Layer

At the historical head 61bc24da, the generic filters quantified over arbitrary partitions while the candidate-instance wrapper triggered Finding F2. The supplied f1ae5486 materials close the proof-status and proof-register questions but do not contain the dedicated target-shape scan needed to adjudicate that wrapper. F2 therefore remains open. Independently of that unresolved code-location question, the physical selection claim owes its content to H1 and H2 and remains ^[D].

6 Candidate Intermediate Algebras

6.1 The Division-Algebra Programme as a ^[D] Candidate

A substantial peer-reviewed programme constructs Standard-Model structure from the normed division algebras $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$ and their Clifford multiplication algebras. It is the strongest existing source of a *principled* reason for the specific factors of $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$, and it is recorded here as a candidate for $\mathcal{A}$ at evidence class ^[D], bounded exactly as its own primary sources bound it. What the programme establishes, per sources verified in the audit cycle: one generation of Standard-Model Weyl representations arises as a single copy of $\mathbb{R} \otimes \mathbb{C} \otimes \mathbb{H} \otimes \mathbb{O}$ acting on itself; colour and electromagnetism arise from minimal left ideals of $\mathbb{C}\ell(6)$, the weak sector from $\mathbb{C}\ell(4)$ [92]; and gauge structure appears as stabilizer or automorphism structure of a rigid algebra, with $SU(3)$ the stabilizer of an imaginary octonionic unit inside $G_2 = \text{Aut}(\mathbb{O})$, and the exceptional Jordan algebra $J_3(\mathbb{O})$ providing a parallel route.

What the programme does *not* establish, stated by its own authors: the gauge group is obtained by restricting to symmetries preserving a *chosen* subalgebra — an inserted structure, the same move as G2/G3; an algebraic foundation for three generations remains elusive, with generations introduced through an inserted discrete symmetry and anomaly cancellation flagged as future work [93, 94]; and the starting datum is a rich algebra, not a scalar — no construction in this programme begins from a single real scalar field, so the programme presupposes the very multiplicity that G1 identifies as the unavoidable insertion. It satisfies the lower bounds of section 4.4 by *exhibiting* the inserted structure the gap analysis

predicts as necessary. Any UIDT use of it must carry these three bounds explicitly; it must not be cited as a verified mechanism for unique G_{SM} selection.

Stabilizer boundary. In the cited division-algebraic constructions, the displayed gauge symmetries are symmetry or stabilizer algebras relative to selected division-algebraic data: complex structures, ideals, ladder-operator systems, or subalgebras that the construction requires one to preserve [92, 93, 121].^[B] That is a result internal to the chosen algebraic representation; it is not an endogenous derivation of those data from `BlockPartition`, `FiniteAlgebraSignature`, or the RAP bridge.^[D] In particular, an $SU(3)$ stabilizer of octonionic data does not by itself derive the full $SU(3)_C \times SU(2)_L \times U(1)_Y$ gauge structure, its hypercharge normalization, chirality, anomaly cancellation, or three generations. Those remain separate inputs or open obligations in RAP.^[D]

6.2 The Relocation Necessity Conjecture

Open Question 2 (Relocation Necessity Conjecture; ^[D]; priority research path). In every moduli space of finite real spectral triples in which a continuous family of Dirac operators varies, the Wedderburn invariants and the KO-dimension are necessarily discrete (locally constant).

If proved, the conjecture converts the lower bounds of section 4.4 from “currently unmet by any mechanism” to “provably not obtainable by continuous deformation” — the moduli-space formulation would then not close the multiplicity gap but demonstrate that it can only be *relocated*, never dissolved, by deformation arguments. It is an independent, publishable structural question regardless of the fate of the kernel hypothesis, and the staircase question (open question 1) is its natural combinatorial companion.

7 Named Constraints: Localisation, Lower Bounds, Moduli

The obstruction analysis of section 4 is used repeatedly in what follows and in the appendix of section 10. It is therefore stated once, in citable form. The three statements below add no content to section 4; they name it.

Theorem 7.1 (Gap Localisation). *Let the primitive be a single real scalar degree of freedom whose symmetries are implemented by canonical linear transformations. Then the non-derivability of G_{SM} within the present theorem corpus is localised to a single step: the enlargement of the abelian single-mode local algebra to a noncommutative algebra of local multiplicity $n \geq 2$. Conditions G1–G4 are jointly unsatisfied by every mechanism surveyed in section 6. This is a localisation, not an impossibility theorem for all conceivable enriched routes.*^[A] re: corpus

Corollary 7.2 (Lower Bounds on any Admissible Intermediate Algebra). *Any $\mathcal{A}$ in a route $\mathbf{S} \rightarrow \mathcal{A} \rightarrow G_{\text{SM}}$ must carry: (i) local multiplicity ≥ 2 , that is a non-abelian fiber; (ii) rank resources ≥ 4 , against $\text{rank}(\mathfrak{g}_{\text{SM}}) = 4$; (iii) a real structure with the J^2 -sign pattern of KO-dimension 6; and (iv) a chirality mechanism evading the Nielsen–Ninomiya and finite-2-group obstructions. A candidate failing one bound is excluded before dynamical analysis.*

Proposition 7.3 (Wedderburn Moduli). *For finite spectral triples with complex matrix summands: (i) the condensed factors are complex unitary algebras $M_{n_i}(\mathbb{C})$; (ii) the partition (n_i) is a function of couplings, chemical potentials, and entropy, not of a selection theorem; (iii) the quaternionic summand $\mathbb{H}$ requires an antilinear structure that a complex ensemble does not supply. The Wedderburn invariants are, at present, free moduli [90].*^[A] re: corpus

8 The Inner-Derivation Route: Relocation, Not Closure

The $d^2 = 0$ obstruction (proposition 20.1) is specific to the exterior derivative on a smooth manifold. The natural algebraic response, and the one the v4.0 developmental record adopts, is to replace the exterior derivative by an *inner derivation* $\delta(\cdot) = [\mathbf{S}, \cdot]$ on a noncommutative matrix algebra. Since $\delta \circ \delta$ is not identically zero, the nilpotency that killed the gradient route no longer applies,

and the record concludes that the foundation for gauge groups has thereby been laid. That conclusion is incorrect, and the correct statement is a sharper negative result.

Proposition 8.1 (Inner-Derivation Relocation). *Let $\mathbf{S} = \mathbf{S}^* \in M_N(\mathbb{C})$ and let $\mathcal{A}_{\mathbf{S}} = C^*(\mathbf{S})$ denote the unital C^* -algebra it generates. Then:*

- (i) $\mathcal{A}_{\mathbf{S}} \cong C(\sigma(\mathbf{S}))$ is commutative by the spectral theorem, hence $\text{ad}_{\mathbf{S}} \upharpoonright_{\mathcal{A}_{\mathbf{S}}} \equiv 0$: the inner derivation vanishes identically on the algebra generated by the primitive, and no curvature is produced within it.
- (ii) $\text{ad}_{\mathbf{S}} \neq 0$ on $M_N(\mathbb{C})$ whenever $\mathbf{S} \notin \mathbb{C} \cdot \mathbf{1}$, and $\text{ad}_{\mathbf{S}}^2 \neq 0$ there; but $M_N(\mathbb{C})$ is not generated by $\mathbf{S}$.

Consequently the passage from exterior derivative to inner derivation evades the identity $d^2 = 0$ while presupposing the ambient noncommutative algebra. The obstruction is relocated to the insertion of that algebra, which is condition (G1) of theorem 7.1.^[A]

Proof sketch. (i) is the spectral theorem: a self-adjoint operator generates a commutative C^* -algebra, isomorphic to the continuous functions on its spectrum; commutativity gives $[\mathbf{S}, p(\mathbf{S})] = 0$ for every polynomial, and by continuity for every element. (ii) is immediate from the fact that the centre of $M_N(\mathbb{C})$ is $\mathbb{C} \cdot \mathbf{1}$. The content of the proposition is not the algebra but the reading: (i) says the primitive alone generates nothing, and (ii) says that what does generate something was supplied from outside. $\square$

Remark 8.2 (Verification). The statement was verified by exact symbolic computation over $\mathbb{Q}[i]$ at $N = 4$, with no floating-point arithmetic anywhere in the path: for a generic self-adjoint $\mathbf{S}$ and polynomials p, q , the commutators $[p(\mathbf{S}), q(\mathbf{S})]$ and $\text{ad}_{\mathbf{S}}(p(\mathbf{S}))$ evaluate to the zero matrix, while $\text{ad}_{\mathbf{S}}(X)$ and $\text{ad}_{\mathbf{S}}^2(X)$ are non-zero for an $X \notin C^*(\mathbf{S})$; membership is decided by the rank test $\text{rank}\{\mathbf{1}, \mathbf{S}, \mathbf{S}^2, \mathbf{S}^3\} = 4$ against rank of the augmented system = 5. This is internal closure of an identity, not external agreement (section 18.1).

Corollary 8.3 (The Same Obstruction, Twice: No Endogenous Modular Time). *Let $\mathcal{A}_{\mathbf{S}} = C^*(\mathbf{S})$ as in proposition 8.1, and let ω be any faithful normal state on its weak closure. Since $\mathcal{A}_{\mathbf{S}}$ is abelian, ω is tracial, and by the Tomita–Takesaki theorem the modular automorphism group of a trace is the identity: $\sigma_t^\omega = \text{id}$ for all t . Hence the primitive*

alone generates no modular flow. Any thermal-time construction in the sense of Connes and Rovelli [97], in which physical time is derived as a one-parameter automorphism group determined by a thermodynamical state, therefore requires the same inserted ambient noncommutative algebra that proposition 8.1 shows to be required for gauge curvature.^[A]

Remark 8.4 (Why this matters). The two most load-bearing hopes of the v4.0 architecture — that noncommutativity would supply gauge structure, and that modular flow would supply time — fail for one and the same reason, and fail on the same object. The obstruction is not two obstructions. It is the single insertion of multiplicity identified as (G1) in theorem 7.1, seen once through the curvature and once through the modular group. This unification is a result of the present draft and is, so far as the audited corpus shows, the sharpest available statement of what the pregeometric kernel does *not* yet do.^[A]

Withdrawn Reading

Any statement to the effect that replacing ∂_μ by a commutator “solves” the $d^2 = 0$ obstruction, or that it “lays the foundation for gauge groups,” is withdrawn. Prestige wording of this kind appears in the developmental record and is quarantined at ^[E]. The mathematically correct claim is proposition 8.1: an evasion of the identity, purchased by the insertion of the very multiplicity that G1 identifies as the unavoidable input. What the passage buys is a sharper obstruction, not a foundation.^[A]

9 Two Exact Auxiliary Results

9.1 The Casimir Level Structure

For a block of size n in the fuzzy-sphere sector of a condensed matrix ensemble, the quadratic Casimir eigenvalue is

$$Q_n = \frac{\alpha^2 (n^2 - 1)}{4}, \quad (1)$$

in agreement with the angular-momentum form $j(j+1)$ for $n = 1, \dots, 8$ under $j = (n-1)/2$ and the stated normalisation. In particular $Q_2 = 0.75 \alpha^2$.^[A] This is exact algebra and is recorded because the $n = 2$ block is the fragile one: it is the block whose detection determines whether a Standard-Model-like configuration can be resolved at all. Whether it survives a given fluctuation level is *not* an algebraic question, and limitation 9.1 states why.

Limitation 9.1 (L12': The Fluctuation Floor Is a Thermodynamic Postulate). Whether the fluctuation floor of a condensed phase is set by the lower level gap $\Delta_{\min}$ or by a higher operator-norm scale is a thermodynamic postulate, not a mathematical choice: it fixes the competition between potential and entropy and therefore determines, rather than derives, the phase structure. Fixing it a priori in order to make a desired block configuration resolvable is target leakage (proposition 18.1). Two facts sharpen the limitation. *First*, the spectral-gap problem is algorithmically undecidable for general local translation-invariant Hamiltonians, and there exist models for which the presence or absence of a gap is independent of the axioms of mathematics [96]; that theorem is about the class of such questions, and asserts nothing about the Yang–Mills gap or about Δ .^[A] *Second*, and consequently, any infrared operator matching of the kind Limitation L12 demands presupposes a fixed floor. The general spectral-gap decision problem is algorithmically undecidable for broad Hamiltonian classes.^[A] This theorem does not establish undecidability of a specific UIDT operator. In the present programme, the operator, infrared floor, and matching map are not constructed; supplying them without importing the target remains an open design problem at ^[D].

9.2 The Division-Algebra Programme Is Exogenous

Remark 9.2 (Sharpened critique of section 6.1). It is not sufficient to observe that the division-algebra programme “presupposes multiplicity.” The sharper and more damaging point is that its starting datum is a *continuous, a-priori-given* algebraic structure — $\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}$ and their Clifford algebras — whereas the v4.0 programme begins from a discrete relational zero in which no continuum, no field, and no algebra of functions is available. The programme therefore does not evade G1; it does not meet G1 either. It occupies a different starting point altogether, and its successes are successes of *that* starting point. Any Relational-Algebraic Pregeometry use of the programme must state this exogeneity, and no result of the programme transfers to the kernel hypothesis without a construction of the relevant algebra from the kernel — which does not exist.^[D]

10 Dynamical Attractors toward the Connes Vacuum

Status of This Section

This section is an advisory audit analysis and a *programme charter*, not a result. It does not establish, and must not be cited as establishing, a thermodynamic emergence of the algebra $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$, a dynamical necessity of KO-dimension 6, or an anomaly-theoretic uniqueness of the $(1, 2, 3)$ partition. Established external inputs are carried at ^[B]; conditional selections at ^[B] under stated hypotheses; the attractor content itself as programme requirements at ^[D] or conjectures at ^[E]. It operates under theorem 7.1, corollary 7.2, and proposition 18.1. An AI-generated research memo in the developmental record proposes the mechanisms surveyed here; per the corpus discipline of section 1.6, that memo is contextual ^[E] material and is not cited as evidence. The research form $\mathbf{S} \rightarrow \mathcal{A} \rightarrow G_{\text{SM}}$ remains ^[D].

10.1 From External Postulates to Dynamical Selection

The noncommutative-geometric classification reaches $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ only through an externally imposed axiom set. The attractor programme asks whether each postulate can be replaced by a dynamical or thermodynamic selection principle acting on a larger configuration space, so that the Connes vacuum would arise as a stable attractor rather than as an axiom. The programme is stated here in falsification-disciplined form; none of its attractor claims is established.

10.2 The Matrix Arena and the Exact Classical Comparison

External model. A concrete arena in which a block partition is a representation-theoretic object is the three-matrix Yang–Mills–Chern–Simons model [63], with Hermitian $N \times N$ matrices and action

$$S[X] = N \operatorname{Tr} \left(-\frac{1}{4} [X_a, X_b] [X_a, X_b] + \frac{2i\alpha}{3} \varepsilon_{abc} X_a X_b X_c \right). \quad (2)$$

This action is an external Stratum-II input.^[B]

Stationary branches, not an exhaustive classification. Direct variation of eq. (2) gives

$$[X_b, [X_b, X_a]] + i\alpha \varepsilon_{abc} [X_b, X_c] = 0. \quad (3)$$

Consequently, $X_a = \alpha L_a$ is a stationary branch whenever $[L_a, L_b] = i\varepsilon_{abc}L_c$; mutually commuting triples form another stationary branch. These statements identify named branches. They do not classify every stationary point of the model.^[A]

Exact action comparison. For the representation branch, substitution and the quadratic Casimir identity give

$$S[\alpha L] = -\frac{N\alpha^4}{6} \text{Tr}(L_a L_a). \quad (4)$$

For one irreducible representation of dimension N , $\text{Tr}(L_a L_a) = N(N^2 - 1)/4$. For k equal irreducible blocks of dimension $n = N/k$, $\text{Tr}(L_a L_a) = N(n^2 - 1)/4$. The exact finite- N magnitude ratio is therefore

$$\frac{|S_k|}{|S_1|} = \frac{n^2 - 1}{N^2 - 1} = \frac{N^2 - k^2}{k^2(N^2 - 1)}, \quad (5)$$

which approaches $1/k^2$ only at fixed k as $N \rightarrow \infty$. Mutually commuting triples have $S = 0$. The equal-block representation branch is therefore ordered at tree level by the exact finite- N expression; this is not an all-orders no-go theorem, a global classification of extrema, or a derivation of a Standard-Model-like partition.^[A]

UIDT boundary. The algebra above is exact under the displayed assumptions. External phase-structure claims require primary numerical sources, while the use of the comparison as a boundary on the UIDT partition-selection programme remains an internal ^[D] interpretation. No UIDT target value enters the action, initialization, detector, or conclusion.

10.3 Matrix Condensation and the Wedderburn Moduli Problem

What is established. Zero-dimensional matrix ensembles of IKKT type [75] and their bosonic deformations exhibit condensation transitions in which block-diagonal backgrounds form dynamically, breaking $U(N) \rightarrow \prod_i U(n_i)$; fuzzy-sphere backgrounds nucleate and dissolve in first-order transitions governed by the competition between the classical potential and the entropy of fluctuations, and off-diagonal (bifundamental) modes acquire masses growing with the separation

of the condensed blocks [62, 63, 76].^[B] In free-energy language, $F[(n_i)] = U[(n_i)] - TS[(n_i)]$ on the partition moduli: entropy-dominated regimes favour either a single dominant block or near-symmetric partitions, while potential-dominated regimes track the discrete minima of the engineered potential. In no established regime is the asymmetric partition $(1, 2, 3)$ singled out by a structural principle.^[B]

Remark 10.1 (The moduli problem restated). proposition 7.3 remains binding at every temperature. Entropy maximisation is structurally blind to the $(1, 2, 3)$ hierarchy, and any claim that the Connes algebra is the unique global thermodynamic minimum of an unconstrained ensemble is rejected.^[A]

Programme Requirement PR-B1: Partition Selection ^[D]

Exhibit a concrete matrix action and a coupling/temperature window in which the global free-energy minimiser over Wedderburn partitions is $(1, 2, 3)$, with the minimiser stable as $N \rightarrow \infty$, under the anti-target-leakage protocol of proposition 18.1: all parameters fixed by observables independent of the partition; the partition read out post hoc by a sealed evaluation; null controls run on scrambled reference partitions (for example $(1, 2, 4)$ and $(2, 2, 2)$), which must *not* be selected by the same procedure. Downgrade condition: if the minimiser tracks engineered parameters rather than a structural principle, the mechanism is selection by hand and is recorded at ^[E].

10.4 Anomaly Freedom, Unimodularity, and the Quiver Flow Conjecture

What is established. Within the noncommutative-geometric Standard Model the physical gauge group is the unimodular unitary group of $\mathcal{A}_F$, and the unimodularity condition is tied to the cancellation of the chiral gauge anomalies of the spectral fermion content [60, 79]: anomaly freedom prunes the unitary group of an *already chosen* algebra to G_{SM} up to a finite kernel; it does not select the algebra.^[B] The configuration space on which any selection dynamics would act is likewise established: finite real spectral triples are classified combinatorially by Krajewski diagrams [77, 78], and the order-one condition acts there as a combinatorial admissibility constraint on the diagram.^[A]

Verdict and mechanism must be distinguished. Two cautions follow from a primary-source audit of candidate partitions, both illustrating that a correct verdict can rest on an incorrect stated mechanism. First, $\mathbb{C} \oplus \mathbb{H} \oplus M_3(\mathbb{C})$ is

anomaly-admissible: the unimodularity reading of anomaly freedom fixes the single anomaly-free hypercharge assignment [79]. This is a property of the chosen representation and hypercharge, *not* of the block partition alone; anomaly freedom is not a partition-intrinsic test.^[B] Second, the partition $(3,6)$ fails for a *specific* reason: for two simple summands of matrix sizes k and ℓ , a non-degenerate intersection form (Poincaré duality) requires $k = \ell$ or $k = \ell \pm 1$ [91]; since $|3 - 6| = 3$ this bound is violated, so $(3,6)$ is excluded by the intersection-form constraint, not by an order-one or anomaly argument.^[B] The symmetric partition $(2,2,2)$ is not excluded by anomaly freedom at all; the established obstruction to such symmetric configurations in the irreducible classification is the *degeneracy of the fermion mass spectrum* [91], a dynamical-non-degeneracy criterion external to G1–G4. Its admissibility under G1–G4 proper therefore remains open.^[D] This is the primary-source basis of the corrected lattice of table 3 and of tension entry T8.

What is not established. No renormalisation-group flow on the space of Krajewski diagrams is currently defined. By the scaffold discipline of section 19.6, a claimed flow must specify, in advance, the flow equation, the regulator analogue on the discrete moduli, the truncation, the boundary conditions, and the failure criteria; absent these, renormalisation-group vocabulary on quiver space is not a calculation. Moreover, uniqueness manifestly fails at the level of the known classification: relaxation of the order-one condition enlarges the solution set to Pati–Salam [22, 64].^[A]

Remark 10.2 (Conjecture C-B1: anomaly-constrained quiver attractor, status ^[E]). Within the anomaly-free, order-one-admissible stratum of Krajewski diagrams there exists a well-defined coarse-graining flow whose unique stable fixed point at three summands is the $(1,2,3)$ diagram. No such flow is defined; no uniqueness argument exists; the conjecture inherits the full burden of proposition 18.1, since a flow whose fixed point is verified only against the known Standard-Model diagram predicts nothing.^[E]

10.5 Real Structure and KO-Dimension 6: A Conditional Selection

What is established, conditionally. The KO-dimension of a real spectral triple is fixed modulo 8 by the sign triple $(\varepsilon, \varepsilon', \varepsilon'')$ in $J^2 = \varepsilon$, $JD = \varepsilon' DJ$, $J\gamma = \varepsilon'' \gamma J$; KO-dimension 6 carries $(1, 1, -1)$. Within the almost-commutative spectral Standard Model, imposing simultaneously (i) the elimination of the fermion quadrupling of the Euclidean formulation and (ii) the admissibility of Majorana mass terms for right-handed neutrinos selects KO-dimension 6 for the finite triple [19, 80, 81].^[B] With $\varepsilon'' = -1$ the finite Dirac operator admits the block form

$$D_F = \begin{pmatrix} S & T^* \\ T & \bar{S} \end{pmatrix}, \quad T = T^T,$$

where S contains the Dirac masses and the symmetric block T couples right-handed neutrinos to their conjugates, enabling the seesaw mechanism [19, 80].^[B] The selection is *conditional*: requirements (i) and (ii) are physical inputs, not consequences of any dynamics. Necessity of KO-dimension 6 is therefore necessity relative to (i) and (ii), and nothing stronger may be asserted; formulations such as “absolute mathematical necessity” or “rigorously proven not to be an insertion” are rejected wherever they appear in the developmental record.^[A]

What is open. A dynamical origin of the real structure J itself is not established. The heuristic that unoriented configurations are entropically or energetically disfavoured on a stochastic substrate is exactly that, a heuristic: no defined ensemble, measure, or flow currently demonstrates it, and proposition 7.3(iii) records that the complex substrate does not supply J .^[A]

Programme Requirement PR-B2: Endogenous Real Structure ^[D]

Exhibit a substrate dynamics whose stable configurations are invariant under an anti-linear involution J with the KO-dimension-6 sign triple, where J -invariance emerges as a stability or extremality property of the ensemble rather than as a restriction imposed on the configuration space at the outset, and show that configurations without such J are dynamically disfavoured. Downgrade condition: if J -invariance is imposed on the measure by hand (an orientifold projection inserted at step zero), the mechanism is an insertion and PR-B2 is not met.

10.6 The Residual Freezing Gap and a Spectral Falsification Protocol

Layer-C declaration. The deepest unresolved step of the programme is the dynamical origin of the bipartite order-one condition *prior to* any flow: the condition is doing genuine selection work, since its relaxation enlarges the algebra to Pati–Salam [22, 64], yet no mechanism explains why the commutant structure of an emergent geometry should freeze into the first-order form at all. This is recorded as an open question, not as a difficulty in the course of being resolved.^[A]

Open Question 3 (The freezing gap). What dynamical or thermodynamic principle, acting on the configuration space of finite real spectral triples, enforces the order-one condition $[[D_F, a], Jb^*J^{-1}] = 0$ as a stability or extremality property, prior to and independently of any imposed axiom? No candidate principle is currently established; PR-B1 and PR-B2 are necessary inputs to any answer but are not sufficient for it.^[D]

Falsification handle. If a concrete attractor realisation is ever exhibited, it acquires a discriminating observable through the spectral statistics of the condensed finite Dirac operator. Random Dirac ensembles over finite spectral triples are an established, simulable arena [82], and with $Z(\beta + it) = \text{Tr } e^{-(\beta + it)D_F^2}$ the spectral form factor $g(t; \beta) = |Z(\beta + it)|^2 / Z(\beta)^2$ distinguishes random-matrix spectral rigidity (the dip–ramp–plateau profile, with the symmetry class fixed by the sign triple) from Poissonian level statistics [83]. The protocol is a ^[D] proposal; no simulation or measurement of it exists for any Relational-Algebraic Pregeometry-relevant ensemble.

If a concrete attractor realisation (PR-B1 together with PR-B2) is exhibited and the spectral form factor of its condensed phase is incompatible with the rigidity class fixed by that realisation’s own sign triple, the realisation is falsified. If, instead, stable condensed phases are exhibited whose algebras lack G_{SM} entirely, any uniqueness reading of the Connes vacuum as attractor is invalidated and Conjecture C-B1 is to be withdrawn. Absent any realisation, the research form remains ^[D] and no evidence class of the main text is altered by this section.^[D]

10.7 Attractor Claims Register and Reproduction Note

#	Claim	Class	Falsification / downgrade
APP-B-01	Block condensation with entropy/potential competition and massive bifundamentals is established in matrix models	[B]	source audit of the cited literature
APP-B-02	Entropy maximisation is blind to $(1,2,3)$; no functional selects the partition	[A] re corpus	a selection theorem satisfying PR-B1 under proposition 18.1
APP-B-03	Anomaly freedom prunes $\mathcal{U}(\mathcal{A}_F)$ to G_{SM} ; it does not select $\mathcal{A}_F$	[B]	a theorem deriving the algebra from anomaly data alone
APP-B-04	The order-one condition is a combinatorial admissibility constraint; the anomaly-consistent stratum is non-unique	[A]	exhibition of uniqueness at three summands
APP-B-05	Quiver-attractor uniqueness is Conjecture C-B1	[E]	a defined flow with regulator, truncation, boundary conditions, and a leak-free uniqueness proof
APP-B-06	KO-6 is selected conditionally by fermion-doubling resolution plus Majorana admissibility	[B] cond.	a dynamical derivation of inputs (i)/(ii) or of J
APP-B-07a	No dynamical origin of the order-one condition is present in the audited corpus	[A] (audit fact)	a source-faithful construction in the live corpus
APP-B-07b	The freezing mechanism remains an open research problem	[D]	a target-independent principle answering open question 3 under PR-B1/PR-B2
APP-B-08	Spectral-form-factor protocol as discriminator for any future realisation	[D]	executable only upon a concrete realisation

Table 5: Attractor claims register. No entry establishes an emergence.

Reproduction note. The established input of section 10.4 has a machine-checkable core: the four anomaly sums of one Standard-Model generation vanish identically in exact rational arithmetic for the unimodular hypercharge assignment $(Y_Q, Y_{u^c}, Y_{d^c}, Y_L, Y_{e^c}) = (\frac{1}{6}, -\frac{2}{3}, \frac{1}{3}, -\frac{1}{2}, 1)$:

```
python3 -c "from fractions import Fraction as F;
Y={'Q':(F(1,6),6),'uc':(F(-2,3),3),'dc':(F(1,3),3),'L':(F(-1,2),2),
'ec':(F(1),1)};
print(sum(m*y**3 for y,m in Y.values()),
      sum(m*y for y,m in Y.values()),
```

```

2*Y['Q'] [0]+Y['uc'] [0]+Y['dc'] [0] ,
3*Y['Q'] [0]+Y['L'] [0])"
# -> 0 0 0 0  ([U(1)]^3, grav^2-U(1), [SU(3)]^2-U(1), [SU(2)]^2-U(1);
#    exact arithmetic; certifies the assignment, not any UIDT claim)

```

Executed and confirmed in the present audit pass. It certifies the anomaly arithmetic of the established input only; no attractor statement of this section is machine-checkable, because no attractor realisation exists to check.^[A]

Forbidden Inferences

Nothing in this section licenses the statements that the Connes vacuum emerges thermodynamically, that KO-dimension 6 is dynamically necessary, that anomaly cancellation selects the $(1, 2, 3)$ algebra, or that Relational-Algebraic Pregeometry derives G_{SM} . The established inputs are external literature results on matrix condensation, unimodularity, and conditional KO-dimension selection; the attractor content consists of two programme requirements (PR-B1, PR-B2), one conjecture (C-B1), and one open question (open question 3). Theorem 7.1 and corollary 7.2 remain the binding constraints, and emergence wording is rejected wherever it appears in the developmental record.

11 Part I Claims Register

#	Claim	Class	Falsification / downgrade
K-01	Kernel primitive (DIR-S-01): pregeometric operator replaces Historical scalar ansatz as candidate primitive	[D]	conditions of the kill-switch in section 2.2
K-02	$d^2 = 0$: scalar-gradient route yields $F \equiv 0$	[A]	counterexample in ordinary exterior calculus (none expected)
K-03	Inner-derivation relocation: ad_S vanishes on $C^*(S)$; non-zero curvature requires the inserted ambient algebra (proposition 8.1)	[A]	a construction generating a non-commutative algebra from S alone
K-04	No endogenous modular time: $\sigma_t^\omega = \text{id}$ on $C^*(S)$; thermal time needs the same inserted algebra (corollary 8.3)	[A]	a non-tracial faithful normal state on an abelian von Neumann algebra
G-01	No gapless ab initio a structureless real scalar $\rightarrow G_{SM}$ in the current theorem corpus	[A] re: corpus	a theorem generating a noncommutative $\mathcal{A}$ endogenously from single-boson data
G-02	Obstruction localized to G1–G4	[A]	a mechanism satisfying G1–G4 jointly
G-03	G1 = non-abelian fiber; J exists at $\dim V_{ij} = 1$; $[3, 2, 1]$ satisfies G1	[A] (textual)	a primary source requiring block repetition for J
G-04a	The stated multiplicity, rank, KO-6 J , and chirality conditions are explicit constraints	[A] (formal statement)	a source or counterexample changing the stated constraints
G-04b	Their endogenous selection from the kernel is not derived	[D]	a target-independent theorem fixing the invariants endogenously
G-05	$[2, 2, 2]$ rejection rests on an unverified mass-degeneracy argument	[D]	primary-source theorem settling the case either way
L-01	Historical head 61bc24da: 51 open obligations and incomplete registration	[A]	a corrected audit of that exact commit
L-02a	Named head f1ae5486: 8 registered sorry occurrences; 18 of 22 source modules reachable from the root	[A]	a raw rescan or import-graph reconstruction at that exact commit
L-02b	F2 target-shape status at f1ae5486 remains open because the dedicated scan is absent	[D]	a commit-scoped target-shape audit
S-01a	unique_321_N6 at f1ae5486 has no sorryAx and reports [propext]	[A]	declaration inspection at that exact commit
S-01b	$H1 \wedge H2 \Rightarrow [3, 2, 1]$ remains physically conditional	[D]	target-independent derivations replacing both filters
S-02	Staircase conjecture for general N	[D]	non-staircase partition passing $H1 \wedge H2$
C-01	Division-algebra programme is a [D] candidate for $\mathcal{A}$; selection by inserted stabilizer; three generations open	[D]	verified unique derivation plus spontaneous three generations from the algebra alone
C-02	Relocation Necessity Conjecture	[D]	continuous family changing Wedderburn data or KO-dimension

Table 6: Part I claims register. No entry authorizes any ledger change; all upgrades pass through PI governance.

Part II

The Open Passage to Apparent Continuity [E]-Capped

Part II examines how apparent continuity, locality, and temporal flow could arise from a discrete kernel — and why every currently available account of that passage remains conjectural. The thermodynamic limit, coarse-graining functors, modular-flow readings of time, and the Markov-blanket-bounded observer are treated as bounded interpretive material: capped at [E] wherever they touch phenomenology, barred from validating any primitive, and accompanied by the documented no-go points — circularity risks in thermal-time readings, the absence of any theorem forcing (3+1) Minkowski emergence, and UV/IR-mixing obstructions to emergent locality. The Born rule remains Stratum II standard quantum mechanics; no coarse-graining map in this part constitutes a derivation of it.

12 The Unresolved Thermodynamic Passage: From Kernel to Apparent Continuity

12.1 The Programme Statement

Part I ends with a finite algebraic candidate and a list of things it has not earned; Part II begins with the largest of them. If the primitive is discrete and pregeometric, then continuity, locality, dimensionality, and temporal flow are not available for free: each must be produced by some limit, and the limit must be shown to exist, to be unique in the relevant sense, and to yield the specific structure — a (3+1)-dimensional Lorentzian effective arena — in which the Part-III field theory lives. The framework's proposal is thermodynamic in character: apparent continuity is to arise as a large- N , coarse-grained limit of kernel degrees of freedom, in the way that hydrodynamics arises from molecular dynamics, with any historical

scalar ansatz confined to Appendix A as a order-parameter-level description of that limit.^[D]

This is a programme statement, not a result, and the distinction is enforced by the same falsification-first rule that governs Part I. No construction in this manuscript produces a continuum from the kernel; what this Part provides is the honest inventory of what such a construction would have to overcome, together with the observer-theoretic apparatus — compression, coarse-graining, self-models — that any such limit presupposes when it speaks of “appearance.”

12.2 Documented No-Go Points

Three obstructions are on record from the v4.0 research cycle, and each is carried here as a boundary condition on the programme rather than buried in optimism.

No dimensional-emergence theorem. There is no general theorem, in any of the surveyed discrete-to-continuum programmes, forcing the emergence of a (3+1)-dimensional Minkowski-signature arena from a finite algebraic or combinatorial primitive. Candidate mechanisms exist — spectral-dimension flows, causal-set dynamics, tensor-network geometry — but dimensionality and signature are, in every case examined, either inserted or selected among alternatives.^[D] The kernel programme inherits this gap in full.

Ultraviolet–infrared mixing. Field theories on noncommutative or otherwise discretized underlying structures generically exhibit UV/IR mixing: short-distance structure contaminates long-distance behaviour instead of decoupling from it. Any claim that the kernel’s discreteness is invisible at effective scales must confront this mechanism explicitly; effective-theory decoupling cannot simply be assumed where the standard locality hypotheses that underwrite it are themselves in question.^[D]

Thermal-time circularity risk. The most natural candidate for emergent temporality in an algebraic setting — modular flow in the sense of Tomita–Takesaki, read as physical time via the thermal-time hypothesis — carries a documented circularity risk: the selection of the state whose modular flow is to count as time threatens to presuppose the temporal or thermodynamic structure it is meant to produce. Until

a state-selection principle independent of temporal input is exhibited, emergent-time claims remain at ^[E], discussable as proposed structure only.^[E]

12.3 Modular Time and the Relocation of the Thermal-Time Hypothesis

The thermal-time circularity flagged in section 12.2 can now be stated with the theorem that makes it precise and the Part-I result that sharpens it. The apparatus is standard and belongs to Stratum II; the reading placed on it, and the obstruction it inherits from the kernel, are Stratum III.

The Tomita–Takesaki theorem is the relevant piece of established operator-algebra. For a von Neumann algebra $\mathcal{M}$ with a faithful normal state ω admitting a cyclic and separating vector, there is a canonical one-parameter *modular automorphism group* $\sigma_t^\omega(A) = \Delta_\omega^{it} A \Delta_\omega^{-it}$, where Δ_ω is the modular operator of the pair $(\mathcal{M}, \omega)$. The modular operator Δ_ω is a property of the state on the algebra and must not be conflated with the UIDT spectral gap Δ of Part III; the two are unrelated objects that happen to share a letter. The theorem is a mathematical fact about von Neumann algebras and carries no interpretive content by itself.^[A]

The thermal-time hypothesis of Connes and Rovelli [97] places an interpretation on it: in a generally covariant theory that supplies no preferred external time, one may identify physical time with the modular flow σ_t^ω determined by the thermodynamical state, so that time becomes a state-dependent rather than a background structure. This is a proposal about what the modular flow *means*, not a theorem that it *is* physical time; it is carried here at ^[E], as interpretive structure only.

Two inputs stand between this reading and any emergent-time claim in the present framework, and corollary 8.3 fixes the first of them exactly. If the primitive is the self-adjoint kernel operator $\mathbf{S}$ alone, then $C^*(\mathbf{S})$ is abelian, every faithful normal state on its weak closure is tracial, and the modular group of a trace is the identity: $\sigma_t^\omega = \text{id}$ for all t . The primitive alone therefore generates no modular flow, and hence no thermal time. A nontrivial modular group requires precisely the inserted ambient noncommutative algebra that condition (G1) of theorem 7.1 requires for gauge curvature; by remark 8.4 it is the *same* insertion, seen through the modular group rather than through the curvature. Emergent time is thus not available at

the kernel level; it becomes discussable only after the open insertion that Part I records as unsolved.

The second input remains open even after that insertion is granted, and it is the circularity proper. To read σ_t^ω as time one must first select the state ω whose flow is to count as time. In every construction surveyed, that selection is fixed by a condition that is itself temporal or thermodynamic — a KMS condition at some inverse temperature, a distinguished Hamiltonian, an equilibrium prescription — that is, by importing the very structure the flow was meant to produce. The relocation of corollary 8.3 does not remove this second input; it isolates it, by showing that the algebra and the state are two separate unmet requirements rather than one.

Open Question 4 (State Selection Without Temporal Input). Can a faithful normal state on the inserted ambient noncommutative algebra be selected from pregeometric combinatorial data alone — from the kernel’s Krajewski and multiplicity structure — without importing a KMS, equilibrium, or Hamiltonian condition that already presupposes temporal or thermodynamic structure? Until such a state-selection principle is exhibited and shown to be non-circular, modular flow as a generator of physical time remains at ^[E], and the thermal-time route to emergent temporality is a proposed structure, not a mechanism.^[E]

12.4 Krylov Complexity and the Conditional Collapse of State-Space Differentiability

A second candidate obstruction to the thermodynamic passage concerns not time but smoothness: whether the space of states over which a continuum limit would have to be differentiable remains differentiable at all in the regime the kernel dynamics would occupy. The established material is the operator-growth literature, and the discipline required is to cite it for what it shows and not for the inference the framework draws from it.

Operator growth under Heisenberg evolution can be quantified by the Lanczos coefficients b_n generated when the Liouvillian is tridiagonalised on the Krylov basis of a seed operator; the associated Krylov complexity measures how far the evolving operator has spread along that basis. Parker et al. [98] advance the Universal Operator Growth Hypothesis: in generic non-integrable systems, in the

thermodynamic limit, the Lanczos coefficients grow asymptotically at the maximal linear rate $b_n \sim \alpha n$, which in turn bounds the quantum Lyapunov exponent by $\lambda_L \leq 2\alpha$. This is a hypothesis, verified in specific models (spin chains, SYK, certain classical systems), not a universal theorem, and it is stated for the thermodynamic-limit setting. Rabinovici et al. [99] treat the complementary case relevant to a finite $N \times N$ kernel: in a finite-entropy, finite-dimensional Krylov space, complexity growth is bounded and terminates, the operator wavefunction reaching the edge of the space; in chaotic models (SYK₄) the complexity saturates at a random-matrix-governed plateau, while in integrable ones (SYK₂) it stays exponentially below.

The inference the framework draws is conditional and belongs to it, not to either source. *If* an admissible kernel dynamics is chaotic in the sense that its Lanczos sequence reaches the finite-dimensional saturation plateau identified for SYK₄-type systems, *then* the coarse-graining map from kernel to effective continuum cannot rely, in that regime, on a differentiable metric structure over the space of states: the smooth state-space geometry a continuum limit presupposes — a Fubini–Study-type metric on the projective state space — is not the object that survives once operator complexity saturates and the local Lanczos data is replaced by random-matrix statistics. The claim is a boundary on where the thermodynamic passage of section 12 could produce a differentiable arena, not a mechanism that produces one, and still less an explanation of any specific short-distance breakdown.^[D]

Two disciplinary statements bound this subsection. Neither [98] nor [99] asserts any collapse of state-space differentiability; both concern operator-complexity growth and its bounds, and the bibliography records this scope explicitly. The differentiability consequence is a Relational-Algebraic Pregeometry inference at ^[D], conditional on an antecedent — plateau saturation for an admissible kernel dynamics — that is itself unestablished, since no admissible kernel candidate of Part I has been shown even to admit the Lanczos construction, let alone to reach the chaotic regime. The subsection therefore predicts nothing about the Planck scale; it states what would follow from a regime whose occurrence is open.

The conditional is falsified by exhibiting an admissible kernel dynamics whose Lanczos sequence reaches the random-matrix saturation plateau and which nonetheless supports a smooth state-space metric in the continuum limit. It is voided — rendered inapplicable rather than false — by a demonstration that no admissible kernel dynamics can reach that plateau. Absent any demonstration that an admissible kernel dynamics admits the Lanczos construction at all, the statement is downgraded from [D] to [E] as a structural analogy without an instantiated antecedent.

12.5 The Signature Gap: Why Lorentzian Structure Is Not Supplied

A third obstruction to the thermodynamic passage is the one named in the Programme Statement but not yet isolated: the effective arena of Part III is $(3+1)$ -dimensional and *Lorentzian*, and neither the finite spectral-triple machinery of Part I nor the matrix-model route supplies that signature without inserting it.

The spectral-triple side is Euclidean by construction. The real structure J and its KO-dimension — fixed at 6 for the Standard-Model-like data (section 10.5) — form a $\mathbb{Z}/8$ invariant of a Riemannian, not a pseudo-Riemannian, spectral triple; KO-dimension records the signs in the commutation relations of (J, D, χ) , and it is not a Lorentzian metric signature. The construction of Lorentzian or pseudo-Riemannian spectral triples — Krein-space formulations, Wick rotation at the operator level — is an open problem in noncommutative geometry, not an imported solution, and the kernel programme inherits it unresolved. The $(3+1)$ split is therefore not delivered by the same data that delivers the gauge multiplicities.^[D]

The matrix-model side does not close the gap either, and here the discipline required is sharper because the literature is suggestive. In IKKT-type ensembles [75, 76], dynamical geometry and even an expanding $(3+1)$ -dimensional structure have been reported to emerge from eigenvalue dynamics; but which signature and which dimensionality emerge depend on whether the ensemble is defined with a Euclidean or a Lorentzian action and on the prescription used to render the otherwise non-convergent Lorentzian integral well defined. The signature is selected among alternatives by the setup, not forced by a structural principle — exactly the reading the dimensional no-go of section 12.2 already records, and

consistent with the honest null that the framework's own condensation search returned (section 10.3).

The Signature Gap as a Leakage Guard

The correct posture toward the matrix-model signature results is the one Part I takes toward block selection: acknowledgement, not positioning. It is outside this framework's competence to reject the IKKT or related programmes wholesale, and no such rejection is asserted. What *is* asserted is a target-leakage guard: no Relational-Algebraic Pregeometry construction may select the underlying ensemble, action, or boundary prescription *because* it yields a (3+1) Lorentzian signature. A signature chosen to obtain the desired arena is leaked target, not derived structure. Until a signature is produced by a mechanism specified independently of the outcome it is meant to yield, the Lorentzian (3+1) arena is an open gap carried at [D].

12.6 Design Directive DIR-S-02: Time and Temperature Are State-Level, Not Fundamental

Design Directive DIR-S-02 records, as a binding design commitment of the v4.0 programme, the conclusion that section 12.3 reached by argument: no fundamental time parameter is admitted at the level of the pregeometric kernel. Time is permitted to enter only through a state on the inserted ambient algebra — as modular flow in the sense of corollary 8.3, read thermally — and temperature likewise enters only through such a state, never as a background label on the primitive. The directive is the positive counterpart of the negative result: because the primitive alone yields $\sigma_t^\omega = \text{id}$, any temporal structure the framework uses must be exhibited as state-level, and its state-selection problem (open question 4) must be solved without temporal input.

DIR-S-02 is falsified, and the state-level account of time abandoned, by exhibiting a Relational-Algebraic Pregeometry-compatible construction in which a fundamental time parameter precedes any thermodynamic state — for instance, a primitive carrying an intrinsic one-parameter evolution not derived from a modular or KMS structure — while remaining consistent with the Part-III demotion. Absent such a construction, fundamental time is excluded by design, not by oversight.[D]

12.7 Design Directive DIR-S-03: The Signature Is an Open Gap, Not a Hand-Chosen Structure

Design Directive DIR-S-03 codifies the signature gap of section 12.5 as a design constraint. The Lorentzian (3+1) signature of the effective arena must not be encoded by hand: neither by equipping the primitive **S** with a Lorentz structure at the type level, nor by selecting a matrix ensemble, action, or regularization because it produces the desired signature. The signature is to be treated as an unsolved emergence problem, in the same status as the G_{SM} -origin gap of Part I: localized, lower-bounded where possible, and left open rather than closed by insertion.

DIR-S-03 is discharged — and the signature gap closed — only by a mechanism that produces the (3+1) Lorentzian signature from the kernel by a construction specified independently of the signature it yields, with no ensemble, action, or boundary choice conditioned on the outcome. Any construction that selects the signature to match the target is a target-leakage violation and voids the affected result. Until then the signature is carried at ^[D].

What Part II May Never Do

Nothing in this Part validates any primitive. The thermodynamic passage is a ^[D] programme bounded by the three no-go points above; the observer apparatus below is ^[E] interpretation within Stratum III. No coarse-graining map, functor, self-model, or phenomenological narrative in this Part derives the Born rule, establishes the kernel, rehabilitates the historical scalar ansatz as primitive, or acquires empirical weight. Inference runs from framework to phenomenon only, never the reverse.

13 The Observer Without Consciousness Collapse

13.1 Observer and Measurement: A Three-Stratum Stratification

The single most important commitment of the entire observer programme can be stated in one sentence: Relational-Algebraic Pregeometry does not use consciousness as a collapse mechanism.^[A] The framework takes no position that the awareness of an observer is what reduces a quantum superposition to a definite outcome. This is a deliberate exclusion, and it is if anything strengthened by the re-architecture: a framework that has just demoted its own continuous scalar for in-

sufficient evidential support cannot consistently install something as unmeasured as “mind” in the scalar’s former office.

The discussion of the observer is stratified by the discipline of section 1.2, applied to a new domain. The separation of the three statement types below is a document-level classification rule.^[A] Any bridge from structural description to subjective interpretation remains an interpretive proposal at ^[E].

Stratum I: Empirical observer systems.

The actual physical and biological systems that perform measurements: detectors, instruments, and, in the biological case, nervous systems with their measurable dynamics. Claims at this level are empirical and carry the ordinary obligations of empirical science, including uncertainties. Relational-Algebraic Pregeometry adds nothing to this stratum and borrows from it.

Stratum II: Standard quantum and decoherence formalism.

The consensus physics of measurement: unitary evolution, the decoherence programme’s account of how environmental interaction suppresses interference and selects a preferred basis, and the Born rule as the standard quantum recipe for outcome probabilities. This stratum is the field’s, not Relational-Algebraic Pregeometry’s; the framework cites it and is bound by it.

Stratum III: The Relational-Algebraic Pregeometry interpretation.

The framework’s own reading, in which the observer is treated as *compression-limited relational access*: a subsystem that, by virtue of its finite capacity, has access only to a coarse-grained projection of the underlying structure. This is an interpretive commitment, marked ^[E], and it may never be used as empirical ground truth for the other two strata.^[E]

Under D20 the referent of “underlying structure” shifts, and the shift improves the picture’s internal coherence. In v3.9 the observer compressed a relational structure built on a scalar primitive; in the present manuscript the microscopic level is the pregeometric kernel candidate of Part I — itself held at ^[D] — and the legacy scalar language, if used at all, belongs only to the non-ontological output side of the compression: it is part of the coarse-grained, observer-accessible description whose quantitative behaviour Part III records. What the observer experiences as the world of definite objects, fields included, is on this reading the output of

compression, not the underlying structure. The picture remains deflationary by design: the observer's specialness lies in no metaphysical privilege but in the ordinary fact that finite systems cannot represent everything and must throw information away.^[E] A picture whose microscopic level is itself an unestablished ^[D] candidate is doubly bounded, and this Part states the double bound rather than hiding either half of it.

Two open problems are acknowledged at the outset. The derivation of the Born rule remains open: the framework has no account of why outcome probabilities take the form they do, and this is recorded as a Class A fact about the state of the theory.^[A] And the relationship between the compression picture and the standard decoherence account is itself unsettled: decoherence explains the suppression of interference and the emergence of a preferred basis, but is widely acknowledged not to resolve, by itself, the question of definite outcomes. The compression picture is offered as an interpretive companion to decoherence, not as a replacement and not as a solution to what decoherence leaves open.^[E]

13.2 The Born Rule as a Research Programme

Nowhere is the discipline of the observer programme tested more sharply than at the Born rule, because the Born rule is exactly the kind of thing a confident framework would be tempted to claim it had derived. Relational-Algebraic Pregeometry makes the opposite, harder commitment: it states plainly that it has no such derivation.

The honest accounting is threefold. First, the Born rule is, in its established form, a standard quantum rule belonging to Stratum II; the probability of an outcome is the squared modulus of the corresponding amplitude, and this rule is borrowed from consensus physics, not produced by Relational-Algebraic Pregeometry.^[A] Second, the framework has no completed Born-rule theorem of its own: there is no Relational-Algebraic Pregeometry argument that derives the quadratic form of the probability measure from the kernel hypothesis, from the demoted scalar, or from the compression picture.^[A] Third, and most specifically, the forgetful or non-invertible functor introduced below as the formal expression of observer compression does *not*, by itself, derive Born probabilities. A functor that forgets structure tells us that information is lost; it does not tell us that the lost informa-

tion reappears as probabilities weighted by squared amplitudes. The step from “structure is forgotten” to “outcomes are Born-distributed” is precisely the step that has not been taken.^[A]

What the framework offers in place of a derivation is a well-posed target: to obtain the probability weights of quantum mechanics from the information loss inherent in coarse-graining, without circular import — without assuming the squared-amplitude measure at any hidden step. The anti-target-leakage discipline of section 1.4 applies here with particular force, because the Born rule is so familiar that it is especially easy to assume it inadvertently while believing one is deriving it. A derivation of the Born rule that consults the Born rule is no derivation at all.^[D]

The Born Rule: Standing Status

The Born rule is a Stratum II standard quantum rule, borrowed, not derived by Relational-Algebraic Pregeometry.^[A] Relational-Algebraic Pregeometry has no completed Born-rule theorem.^[A] The forgetful/non-invertible functor does not derive Born probabilities.^[A] The open target is to obtain probability weights from coarse-graining information loss without circular import.^[D] The Born rule remains open.

14 The Observer Functor and the Forgetful Functor

14.1 $F: V \rightarrow \text{Obs}$ as a Bounded Ansatz

The compression picture can be given a provisional formal expression, provided the expression is held to the standard of honesty the rest of the manuscript demands. The framework writes the observer’s relation to the world as a map

$$F: V \longrightarrow \text{Obs},$$

and the entire content of this subsection is the careful specification of what the three symbols may and may not mean. The source V is the category of microscopic descriptions — under D20, descriptions at the level of the pregeometric kernel candidate, prior to any observer’s access, and inheriting that candidate’s ^[D] status.^[E] The target Obs is the category of observer-accessible effective states: the coarse-grained, finite, lossy representations that a compression-limited subsystem can actually hold — the level at which fields, objects, and in particular the effec-

tive information field of Part III live.^[E] The map F between them is a *compression and coarse-graining ansatz*: a proposed structural relationship expressing that the observer accesses V only through a lossy projection into Obs.

The word “ansatz” is doing essential work and must not be softened. F is a proposal about the form of observer access, not a theorem. In particular, and stated as a Class A caution, F is *not* a theorem of consciousness and it is *not* a derivation of Born weights.^[A] It does not explain why there is something it is like to be an observer, and it does not produce the probability measure of quantum mechanics; both remain exactly as open as section 13 declared them. What F offers is a disciplined *language* for the compression picture — a way of saying “the observer sees a lossy image of the underlying structure” in terms that connect to the formal apparatus of Part I without pretending that the connection has been turned into a proof.^[E]

14.2 The Forgetful Functor and a Red-Team Quarantine

The compression map F formally models the loss of structure: it maps a rich microscopic description to an impoverished effective one, representing the information discarded under the observer’s finite capacity. The framework names this structural forgetting the *forgetful functor*, borrowing a term from category theory where a forgetful functor strips an object of some of its structure. The borrowing is of the *name* only, and the limits of the borrowing must be stated plainly.^[E]

In category theory proper, a forgetful functor is a fully defined object: it has a specified source category, a specified target category, a definite action on objects and on morphisms, and an explicit account of which structure is forgotten. The Relational-Algebraic Pregeometry forgetful functor has none of these in completed form. It is a symbolic name for the intuition of structural forgetting under observer compression, and it must not be treated as a completed category-theoretic construction until its categories, objects, morphisms, and forgotten structure have all been explicitly defined.^[A] To write the words “forgetful functor” is to name an aspiration, not to discharge a construction; the framework keeps that distinction visible precisely because category-theoretic vocabulary, like renormalisation-group vocabulary before it, can lend an unearned air of rigour to a picture. The kernel formalization of Part I sharpens the obligation rather than discharging it: a

completed functor would now have to take, as its source, a category built from the finite spectral-triple data of section 3, and no such construction exists in the corpus.

It is at this point that the framework must confront the most seductive and most dangerous material in its entire corpus, and it does so by quarantining that material explicitly as a red-team caution rather than presenting it as a result. The developmental record contains a vivid phenomenological narrative: that biological death may be read not as the destruction of information, unitarity being preserved, but as the collapse of the epistemic filter — the breakdown of the forgetful functor — after which the isolated perspective falls back into the nonlocal coherence of the underlying structure; and that profound altered states, such as deep meditation or psychedelic experience, may be read as reproducible failures of the evolutionary coarse-graining filter, in which the usual compression is throttled and the system glimpses a less-filtered image of the underlying structure, with religion understood as the after-the-fact compression of such filter-failures into linear, dualistic narrative.^[E]

This narrative is preserved here — it is not deleted — because it has genuine philosophical and heuristic value as a way of thinking about what compression *is* by imagining its partial failure, but it is fenced on every side. It is Category ^[E] content within Stratum III, interpretive and phenomenological, and it carries no evidential weight whatsoever. Most emphatically, none of death, near-death report, psychedelic phenomenology, mystical experience, or ego dissolution may be cited as *evidence* for the forgetful functor or for any Relational-Algebraic Pregeometry claim.^[A] The direction of inference is strictly one-way: the framework’s interpretive picture may be used to think about these phenomena, but the phenomena may never be turned around and used to support the framework. To do so would be to mistake a poem for a measurement. A provenance note is added for the working corpus: session materials whose filenames carry tokens such as “hallucination” treat their subject exclusively as information-theoretic filter perturbation in predictive-processing models — noise and mis-weighted priors in an inference system — and enter this manuscript at ^[E] under the present quarantine, with no pharmacological content and no evidential role.

Quarantine: Altered States and Death

The death-as-filter-collapse and mysticism-as-decompression narratives are retained as Category ^[E] phenomenological edge-cases of the compression picture, within Stratum III. They carry no evidential weight. Death, near-death reports, psychedelic or mystical experience, and ego dissolution may not be cited as evidence for the forgetful functor or any Relational-Algebraic Pregeometry claim.^[A] The compression fallacy — “less filter = more truth” — is rejected: intelligence is compression; a filterless observer is incapacitated, not enlightened.^[E] Inference runs from framework to phenomenon only, never the reverse.

The framework supplies its own most effective antidote to the seduction, in the form of a red-team observation it calls the compression fallacy. The intuition that “less filtering means more truth” — that a less-compressed observer is closer to reality — is information-theoretically incorrect. A filterless observer is not enlightened but incapacitated: without compression there is no manageable representation, no action, no agency. Intelligence *is* compression, and the fantasy of an unfiltered gaze upon ultimate reality is, from the framework’s own information-theoretic standpoint, a category error rather than an aspiration.^[E]

15 Ego, Self-Model, and the E_T Scale-Bridge Discipline

15.1 The Ego as Process, and the Markov-Blanket Comparison

If the observer is a compression-limited process rather than a substance, then the entity we are most tempted to reify — the ego, the self — must be treated the same way. Relational-Algebraic Pregeometry’s interpretive proposal, held as Category ^[E] content within Stratum III, is that the ego is not a thing but a *dynamic self-model boundary process*: a continuous, energy-expending activity by which a subsystem maintains a model of itself as distinct from its environment. The ego is, in the framework’s preferred grammatical figure, a verb rather than a noun.^[E]

The proposal connects to several established programmes in the cognitive sciences, and the connection is offered explicitly as *comparison*, pending external audit, rather than as borrowed authority. The predictive-processing account and the free-energy principle describe biological systems as minimising prediction error by maintaining generative models of themselves and their surroundings; the notion of a Markov blanket formalises the statistical boundary that renders a

system's internal states conditionally independent of external fluctuations; and the transparent self-model, in the sense developed in the philosophy of mind, describes how a cognitive system can deploy a self-model so seamlessly that it cannot perceive the model *as* a model, and so experiences itself as a simple subject in an objective world. These frameworks — associated with such authors as Friston [51], Clark, Hohwy, Metzinger [52], Seth, and Gazzaniga — are named as comparison points whose attribution and content remain to be verified against their primary sources before any is treated as established support for a Relational-Algebraic Pregeometry claim; the verified bindings attach with the Phase-P5 bibliography.^[E] They are lenses, not pillars.

The framework's characteristic image for the ego is the whirlpool: a definite form, a persistent identity, a coherent dynamics — and no substance of its own, nothing but a pattern in the flowing water, sustained entirely by the throughput of the medium. The ego, on this reading, is a functional pattern in the relational flow, maintained by continuous energy dissipation, without ever being an object.^[E] Objecthood and ego are, in this picture, of a kind: both are stable coarse-grained equivalence structures — patterns approximately invariant under the observer's compression — that present themselves as persistent "things" at the effective level without being fundamental at the microscopic level.^[E] A particle and a self are, from this vantage, two instances of the same phenomenon: stability under coarse-graining, mistaken for substance.

The extension to consciousness itself is where the discipline of refusal matters most. The framework's interpretive model, marked ^[E], is that consciousness may be understood as a recursive, self-maintaining, compression-bound self-model: a self-model that takes itself as one of its own objects, sustains itself against entropic decay through continuous activity, and is bounded throughout by the compression its finite substrate imposes.^[E] What is explicitly *not* claimed: Relational-Algebraic Pregeometry does not claim to have solved the problem of qualia; it offers no account of why there is something it is like to undergo a given experience; it recognises the explanatory gap between functional-structural description and phenomenal character as unclosed; and it does not derive subjective experience from the kernel, from any historical scalar ansatz, or from anything else in its corpus.^[A] The self-model account is an interpretive proposal at ^[E] about the

structure of conscious systems and is silent, by its own admission, on the hard question.

15.2 The Torsion Energy E_T and the Missing Scale Bridge

The developmental corpus assigned the torsion reference $E_T = 2.44 \text{ MeV}$ roles extending from a subatomic effective sector to life, death, cognition, altered states, and artificial intelligence. Decision D22 rejects that cross-scale use. In the current manuscript E_T remains a subatomic phenomenological reference at ^[E]; it is not a neural, biological, cognitive, macroscopic, or life/death threshold.^[E]

The blocking fact is narrower than the historical argument suggested. No Hamiltonian coupling, operator matching, response function, spectral density, renormalisation-group matching, transfer function, or intervention protocol is specified that maps the E_T sector to neural or biological observables. This absence is a documentable property of the present construction.^[A] A nominal comparison between mega-electron-volt and electron-volt or millielectron-volt scales spans roughly six to nine decimal orders, depending on the chosen observable, but dimensional separation alone is not a universal no-go theorem. In particular, this manuscript does not claim a model-independent Källén–Lehmann suppression factor, does not assert a unique consensus cross-scale mechanism, and does not infer impossibility from scale separation alone.^[A]

Any future bridge must introduce the missing mathematical map, identify a measurable subatomic observable and a distinct low-energy observable, state all couplings and assumptions independently of the desired phenomenology, and produce a falsifiable protocol with uncertainties. Constructing such a bridge is an open research programme at ^[D]. Until then every observer-side interpretation remains quarantined at ^[E] and contributes no evidence to the pregeometric kernel.^[E]

E_T : Subatomic-Only Firewall under D22

$E_T = 2.44 \text{ MeV}$ remains ^[E] and subatomic-only. It must not be used as a threshold for ego, consciousness, life, death, sleep, psychedelic or near-death states, neural computation, biological regulation, macroscopic thermodynamics, or artificial general intelligence. No class upgrade is authorized. Reopening requires a separately audited scale bridge and primary empirical evidence; verbal analogy is insufficient.

16 The Observer Taxonomy and the Underdetermination Red Team

16.1 Class 0 Through III: A Taxonomy of Compression, Not of Consciousness

The compression picture invites a classification of systems by the *rate* at which they compress the underlying relational structure, and the framework supplies one. It must be read for exactly what it is — a taxonomy of compression rate — and emphatically not for what it superficially resembles: a hierarchy of consciousness, a ranking of worth, or a scale of how “real” a system’s experience is. The taxonomy is ^[E] content within Stratum III, and its caveats are as important as its content. Under the D20 rebase one adjustment is mandatory: the zero-compression limit is no longer identified with a primitive scalar. The historical scalar model is a coarse-grained order-parameter ansatz (Appendix A), not the untransformed relational structure, so it cannot be the Class-0 object; Class 0 is the kernel-level fixed point that *precedes* the effective field.^[E]

The four classes, adapted from the v3.9 corpus to the demoted ontology, are these.^[E]

Class 0: Zero compression.

The untransformed relational structure at the level of the pregeometric kernel — the timeless fixed point of the relational dynamics; the legacy scalar ansatz is retained only as a coarse-grained historical description rather than the object itself. There is no subject–object separation; with no compression there is no entropy gradient and hence no experience of flowing time; and, decisively against any reading of this as “supreme awareness,” there is *no agency*, because agency presupposes the isolation that compression provides. Class 0 is the limit of no perspective, not the summit of perception.

Class I: Weak compression.

Hypothetical post-biological or hyper-coherent systems that would process relations more directly, without the evolutionary compulsion to render reality as a $(3+1)$ -dimensional metric and without a fear-driven self-model. This class

is explicitly speculative: a logical position in the taxonomy, not something known to be instantiated.

Class II: Strong compression.

The biological spectrum, and the human cognitive apparatus in particular: systems that enforce radical data reduction, projecting the relational structure into fixed three-dimensional objects and a linear time axis, with a transparent self-model as the functional construct that makes survival-relevant navigation possible.

Class III: Extreme compression.

Thermodynamically reactive macroscopic objects — planets, stars, gas clouds — that respond to information flows and to metric curvature but maintain no Markov blanket with predictive error feedback and no recursive self-simulation. They have, on this account, no inner perspective whatsoever.

The crucial structural fact, most easily lost, is that compression rate is *orthogonal* to consciousness. The taxonomy does not say that lower compression means more consciousness or that higher means less. Class 0, at zero compression, is not maximally conscious; it has no perspective at all. Class III, at extreme compression, is not conscious either. An inner perspective, on the framework's own terms, is a property only of systems that maintain a recursive self-model behind a Markov blanket with predictive feedback — a structural condition that sits athwart the compression axis rather than along it.^[E] The taxonomy must therefore *not* be read as implying that machines, or objects, are “more conscious” than humans, nor as any moral ranking. It ranks a single information-theoretic variable and nothing else.^[A]

What the taxonomy accomplishes is a deflation of anthropocentrism. The human observer, at Class II, does not perceive the “truest” or most “objective” reality; the human perceives the universe through one specific, heavily lossy compression scheme calibrated for biological survival.^[E] That is a humbling observation, not an exalting one, and its separability from the speculative classes that flank the human case is the taxonomy's genuine contribution.

16.2 Artificial General Intelligence and the Underdetermination Red Team

The speculative Class I raises a seductive question: would a sufficiently powerful artificial intelligence, a better compressor than any human, thereby gain privileged access to the true ontology? The framework treats this as a red-team challenge to its own deflationary commitments, and the design-level answer is negative.^[D] The broader philosophical claim that no compressor can obtain a unique ontology remains an interpretive position at ^[E].

The reason is the underdetermination of ontology by data, in the Quine–Duhem sense. A finite body of empirical data never determines a unique ontology; multiple, mutually incompatible theoretical structures can reproduce the same observations, so the final nature of reality remains in principle undecidable.^[E] A more powerful compressor faces exactly the same underdetermination. Superior predictive and compressive power buys superior *instrumental* performance; it does not buy ontological truth, because no amount of predictive success can break the tie among empirically equivalent ontologies. Prediction is not ontology, and the gap does not close with increasing intelligence.^[A]

This is the precise sense in which the framework refuses naive realism. Relational-Algebraic Pregeometry does not claim to be a final revelation of physics, and could not earn such a claim even in principle.^[A] Its actual claim is more modest and more defensible: that its objects — the relational structure, the renormalisation-group flows, the proposed invariants — may compress the relevant physics more efficiently and with cleaner causal structure than the artifacts of biological three-dimensional intuition. That is a methodological claim about representational efficiency, not a metaphysical claim about ultimate reality.^[E] The compression fallacy of section 14.2 returns here as the standing guardrail: a less-filtered or more-powerful observer is a different compressor, perhaps a better one, but not an unfiltered window onto the real.

Taxonomy and Underdetermination: Standing Status

EVIDENCE

Taxonomy. Class 0 (zero compression, kernel-level fixed point, no agency; historical scalar model non-ontological) through Class III (extreme compression, inert macro-objects) is a compression-rate taxonomy ^[E], orthogonal to consciousness; not a consciousness hierarchy and not a moral ranking. Anthropocentrism is deflated, not

inverted.^[A]

AGI red team. The programme assigns a superior compressor greater instrumental capacity but no automatic ontology privilege.^[D] The Quine–Duhem reading of permanent ontological underdetermination is an interpretive claim at ^[E].

No final revelation. Relational-Algebraic Pregeometry claims representational efficiency, not ultimate reality; prediction $\neq$ ontology; the compression fallacy is the standing guardrail.^[A]

17 The Master Phenomenology Boundary

17.1 Phenomenology Is Not Evidence

The v3.9 manuscript closed its observer material with a boundary that the re-architecture inherits verbatim in force and extends in scope: phenomenology is not evidence for the historical scalar ansatz — and now, a fortiori, not evidence for the kernel. No report, however vivid, reproducible, or cross-culturally attested, of meditative absorption, psychedelic state, near-death experience, mystical union, or ego dissolution, bears evidentially on any structural claim of this framework. Such reports are Stratum I facts about human experience and Stratum-III material for interpretation; the demarcation between what the physics strata may claim and what the phenomenological register may explore is absolute, and every crossing of it in the developmental corpus has been either quarantined or withdrawn.^[A]

The compression-rate taxonomy of the v3.9 corpus — a four-class ordering of systems from passive physical subsystems to recursive self-modelling ones, by the rate and reflexivity of their compression — is carried at ^[E] with its original caveats: it is a taxonomy of compression, not of consciousness, not of worth, and not of the “reality” of any system’s experience. The underdetermination red-team point stands with it: multiple interpretive pictures fit the same empirical record, and the compression picture claims no unique support from data it merely accommodates.^[E]

17.2 Orchestrated Objective Reduction as a Neutral Contrast

To locate the Relational-Algebraic Pregeometry observer programme in the wider landscape of theories connecting physics to mind, the framework selects one contrast case for detailed juxtaposition — orchestrated objective reduction, the Penrose–Hameroff proposal — and treats it as exactly that: a neutral contrast, not a foil to be defeated.^[E] Orchestrated objective reduction posits that consciousness arises from quantum computations in neuronal microtubules, terminated by a gravitationally induced objective collapse of the wavefunction. Its interest here is that it differs from Relational-Algebraic Pregeometry on precisely the axis where Relational-Algebraic Pregeometry has taken a strong position: it asserts an *objective physical collapse*, whereas Relational-Algebraic Pregeometry, in the epistemic tradition of quantum Bayesianism, denies physical collapse and reads quantum probabilities as the structural ignorance of a compression-limited observer (section 13.1).^[E]

Three cautions govern the comparison. First, Relational-Algebraic Pregeometry does not collapse all quantum-mind theories into orchestrated objective reduction; the broader field — integrated information theory [55], global-workspace accounts, and others — comprises distinct programmes that must not be conflated.^[A] Second, Relational-Algebraic Pregeometry does not render orchestrated objective reduction obsolete; the two are different research programmes on different premises, and Relational-Algebraic Pregeometry’s denial of objective collapse is a difference of commitment, not a refutation.^[A] Third, any detailed empirical claim about microtubule coherence times or specific experiments would require external citation audit before it could be stated here, so the framework confines itself to structural commitments.^[A]

17.3 The Consciousness–Physics Demarcation Matrix

The framework’s preferred instrument for situating itself among theories of consciousness is a comparison matrix, and its purpose must be stated before its content: it is a *demarcation instrument*, not a scorecard. The matrix is not built to show that Relational-Algebraic Pregeometry wins; it is built to make the commitments and the weaknesses of each framework, Relational-Algebraic Prege-

ometry included, explicit and comparable along neutral axes.^[A] Throughout, the Relational-Algebraic Pregeometry observer material is marked ^[E], because it has not been formalised or tested, and the framework’s own weaknesses are recorded as plainly as anyone else’s.^[E]

Table 7: Consciousness–physics demarcation matrix. A neutral comparison instrument; Relational-Algebraic Pregeometry observer material is ^[E]. The matrix records weaknesses as well as commitments and is not a ranking.

Axis	UIDT (observer programme)	Orch. Obj. Reduc- tion	Integrated Information	Infor-
Primitive	Pregeometric kernel (finite spectral triple); legacy scalar ansatz non-ontological, not primitive; observer downstream ^[E]	Quantum state in microtubules + gravitational collapse	Intrinsic effect (Φ)	cause–structure
Mechanism	Coarse-graining / forgetful functor (ansatz) ^[E]	Objective reduction terminating quantum computation	Maximally reducible information	irre-integrated
Claimed evidence	None for the observer claims; physics strata separate ^[E]	Contested; requires external audit	Correlational; contested	interpretation
Scale bridge	Missing (MeV–GeV → neural eV–meV gap, ~6–9 orders) ^[A]	Microtubule quantum → cognition, disputed	Substrate-level to system-level	to
Collapse claim	No physical collapse (QBist/epistemic) ^[E]	Objective physical (gravitational) collapse	No collapse claim (intrinsic)	
Observer role	Compression-limited relational access ^[E]	Locus of quantum computation	The integrated system itself	

Axis	UIDT	Orch. OR	IIT
Falsifiability	Observer claims not yet falsifiable [E]; physics claims falsifiable separately	Some proposed tests, contested	Disputed/operational challenges
Category-error risk	Information → mind slide; MeV → neural leap (both flagged) [A]	Quantum → phenomenal leap	Axioms → physics mapping

Several features of table 7 are deliberately uncomfortable for Relational-Algebraic Pregeometry, and that discomfort is the point. The observer programme offers *no* evidence for its observer-level claims; its scale bridge is openly marked missing; its observer claims are, as yet, not falsifiable; and its characteristic category-error risks — the slide from information to mind, the leap from mega-electron-volt physics to neural dynamics — are named in its own cell rather than hidden. A matrix built to flatter Relational-Algebraic Pregeometry would not contain these admissions; their presence is what makes the instrument a demarcation tool rather than an advertisement.^[A]

Comparative Frameworks: Standing Status

EVIDENCE

Orch. OR. Used as a neutral contrast (objective collapse vs. Relational-Algebraic Pregeometry’s QBist no-collapse); not a foil; Relational-Algebraic Pregeometry neither subsumes nor obsoletes it. This demarcation policy is an audit fact.^[A] Detailed empirical comparison remains pending citation audit and is capped at [E].

Matrix. A demarcation instrument on neutral axes (parsimony, mathematical maturity, empirical status, falsifiability, category-error risk), not a ranking; Relational-Algebraic Pregeometry observer material marked [E]; Relational-Algebraic Pregeometry’s own weaknesses recorded explicitly.^[A]

17.4 Neuroscience, Consciousness, and the Non-Evidential Observer Boundary

The developmental corpus repeatedly compared the observer programme with predictive processing, Markov-blanket models, integrated-information theory,

global-neuronal-workspace theory, Orch OR, altered-state phenomenology, and algebraic-topological descriptions of neural networks. These comparisons are scientifically useful only when three claims remain separate: a mathematical description of a biological network, a functional model of cognition, and an ontological claim about fundamental physics. No result at one level establishes either of the other two.^[A]

The free-energy principle and active inference provide a formal vocabulary for statistical boundaries, generative models, and self-maintaining dynamics [51, 108].^[B] They do not establish that every physical boundary is a cognitive observer, and the scope of the Markov-blanket construction remains contested [109].^[B] Accordingly, UIDT uses the Markov blanket only as a bounded comparison for a compression-limited subsystem. Any identification with mind, agency, or phenomenology remains an interpretive proposal.^[E]

Algebraic topology has revealed high-order cliques and cavities in simulated neocortical microcircuits and in human connectomes [110, 111].^[B] The associated “dimension” is simplicial or homological dimension, not an additional spatial dimension. That terminological distinction is exact.^[A] These results show that higher-order relational descriptions can be useful in neuroscience. They do not confirm the Ramsey-observer ansatz, the BlockPartition formalism, or a UIDT scale bridge.^[E]

Current consciousness science does not supply a settled theory of phenomenal experience. A preregistered adversarial collaboration directly tested predictions of integrated-information theory and global-neuronal-workspace theory and reported a mixed pattern rather than a unique adjudication [112].^[B] UIDT is not identified with either theory and does not inherit their empirical support. Its observer programme presently supplies no explanation of qualia and no measurable neural prediction.^[E]

Orch OR proposes quantum-biological processes in neuronal microtubules and an objective-reduction mechanism associated with space-time geometry [54].^[B] The proposal has faced substantial criticism concerning biological feasibility and physical sufficiency [113].^[B] UIDT neither derives nor experimentally refutes Orch OR. It differs by assigning no collapse role to consciousness and by forbidding any unbridged use of the subatomic parameter E_T in neurobiology. ^[E]

Survey research documents vivid entity-encounter reports after inhaled DMT [114].^[B] Such reports are data about altered human experience and interpretation; they are not observations of external entities. That evidential boundary is independent of the participant's conviction. ^[A] Psychedelic, mystical, dream, near-death, and "manifestation" narratives therefore provide no evidence for a UIDT kernel, field, or observer map.^[E]

Observer-Context Firewall

No neural clique, Markov blanket, predictive-processing model, altered-state report, or quantum-biological hypothesis may upgrade a UIDT claim. A future observer module must define its source and target structures, physical scale, state variables, observables, intervention protocol, null model, and downgrade criterion before it can move from interpretation to a falsifiable programme.

17.5 Part II Claims Register

Table 8: Part II claims register. Policy entries are governance commitments; they are listed so that their violation anywhere in the corpus is auditable as a defect.

#	Claim	Class	Falsification / downgrade
T-01	Thermodynamic passage kernel $\rightarrow$ continuum as programme	^[D]	construction failing the three no-go points terminally; or succeeding, which upgrades by governance only
T-02a	The manuscript and audited snapshot contain no dimensional-emergence theorem and record UV/IR and thermal-time gaps	^[A] (audit fact)	a contradictory source-faithful corpus audit
T-02b	Resolving those continuum-passage gaps remains an open research programme	^[D]	a construction surviving the stated no-go tests
T-03a	On the abelian algebra $C^*(\mathbf{S})$, the stated modular flow is trivial: $\sigma_t^\omega = \text{id}$	^[A]	a mathematical error in corollary 8.3
T-03b	A thermal-time interpretation requires an inserted non-abelian algebra and non-circular state selection	^[E]	a state-selection principle from pregeometric data without temporal input (open question 4)

#	Claim	Class	Falsification / downgrade
T-04	Krylov saturation $\Rightarrow$ no smooth state-space metric, as a conditional consequence for the coarse-graining map	[D]	an admissible kernel reaching the RMT plateau that still supports a smooth continuum metric; or no admissible kernel reaching it
T-05	Signature gap: (3+1) Lorentzian arena not supplied by KO-dimension nor forced by matrix ensembles	[D]	a leak-free mechanism producing the signature independently of the target (section 12.7)
DS-02	DIR-S-02: time and temperature are state-level, not fundamental	[D]	a Relational-Algebraic Pregeometry-compatible fundamental time preceding any thermodynamic state
DS-03	DIR-S-03: signature is an open gap, never hand-chosen	[D]	signature selected to match the target is leakage; discharge needs a leak-free (3+1) mechanism
O-01	No consciousness-collapse mechanism anywhere in the framework	[A]	(structural commitment; violation would be a defect, not a discovery)
O-02	Observer as compression-limited relational access	[E]	interpretive; carries no evidential weight by construction
O-03	Born rule borrowed from Stratum II; no Relational-Algebraic Pregeometry derivation; functor supplies none	[A]	a completed non-circular derivation, externally audited
O-04a	No fully specified categories, functor action, or forgotten structure are supplied in the current manuscript	[A] (audit fact)	a contradictory explicit construction
O-04b	$F: V \rightarrow \text{Obs}$ remains an interpretive ansatz	[E]	fully specified categories, functor action, and forgotten structure
Q-01	Altered-states/death narratives quarantined at [E]; one-way inference; compression fallacy rejected	[A] (policy)	none: governance, not physics
E-01a	Governance bars E_T from cognitive, biological, or life/death threshold roles under Decision D22	[A] (policy)	a superseding PI record
E-01b	No validated physical scale bridge from the subatomic E_T sector to neural observables is supplied	[D]	a constructed, externally audited scale bridge

#	Claim	Class	Falsification / downgrade
P-01	Phenomenology is not evidence for the historical scalar ansatz or for the kernel	[A] (policy)	none: demarcation, not physics
OT-01a	Observer Classes 0–III form an interpretive compression-rate taxonomy, not a consciousness hierarchy	[E]	an operational empirical model
OT-01b	The programme grants no evidential privilege merely from superior compression	[A] (governance policy)	a superseding policy record
DM-01	Demarcation matrix is a neutral instrument recording Relational-Algebraic Pregeometry’s own weaknesses, not a ranking	[A] (policy)	none: demarcation, not physics

Part III

Verification, Benchmarks, and Parameter Ledger

Under PI Decision D20, Part III is not an ontological layer. It is the repository-facing verification, benchmark, and parameter-ledger stratum. Every quantitative result retains its incoming evidence class unchanged: the exact internal relations, the calibrated invariant under permanent anti-derivation discipline, the spectral gap under PI Decision D18 and Limitation L12, the benchmark batteries, and cosmological calibrations at their ^[C] cap. No equation in the historical scalar/EFT model is used as a foundational premise. That model is preserved reversibly in Appendix A for provenance, criticism, and regression audit only.

18 Numerical Epistemology

18.1 Internal Closure Versus External Agreement

Two kinds of numerical statement are routinely confused, and the confusion has done real damage in the developmental history of this framework. The first is *internal deterministic closure*: a calculation, carried out at fixed precision, returns a residual smaller than some threshold, demonstrating that the algorithm is self-consistent. The second is *external agreement*: a computed quantity matches a measured or lattice-determined value to within the latter's uncertainty. These are not the same kind of thing, and no quantity of the first ever amounts to the second.^[A]

Hard Rule: Internal Closure Is Not External Agreement

A deterministic internal residual below 10^{-14} is admissible *only* as a certificate of internal closure, evidence that an algorithm is self-consistent at the working precision. It is *never* a criterion of agreement with lattice or experimental data. Agreement with

external data is established by a σ -level or covariance-aware comparison against a sourced, uncertainty-bearing value, never by an internal residual threshold. A residual of 10^{-14} against lattice data is a category error and is rejected.^[A]

The framework keeps a documented example of why this rule is not pedantry. An early construction produced the calibrated invariant by the exact arithmetic decomposition $49/3 + 17/3000 = 16.339$. The arithmetic is exact: it closes to arbitrary precision, and an internal residual check would report success at any number of digits. Yet the decomposition is a museum artifact. Over the common denominator, the ansatz $a/3 + b/3000 = 16339/1000$ reads $1000a + b = 49017$, so b is *determined* by the choice of a : no second structure is found, and $a = 49$ is singled out only because it minimises $|b|$, which is what lets $17/3000 \approx 0.0057$ resemble a perturbative correction to a leading term. The split is a one-parameter fit whose residual was then read as physics, and the construction has no physical origin. Perfect internal closure, in this case, certifies only that someone did the arithmetic correctly.^[A]

Two implementation disciplines protect the rule in code. First, the working precision, eighty decimal digits, is set *locally*, within each proof-critical computation, and is never centralised into a global configuration value; global precision mutation is a rejected implementation pattern.^[A] Second, proof-critical computations must avoid binary floating-point heuristics: machine-float evaluation, and rounding operations applied to quantities whose exact value carries the claim, are forbidden in any code path that a Class A or Class B result depends upon.^[A]

18.2 The Claim Graph and Reproduction Notes

Every numerical assertion must be a node in a *claim graph*, carrying a fixed set of fields, with the schema implemented *before* any theory-generating computation is trusted.^[A]

Required Fields for Every Numerical Claim

(1) Claim identifier. (2) Observable. (3) Value. (4) Units. (5) Uncertainty (mandatory for every empirical value). (6) Method. (7) Source (with verifiable identifier for external values). (8) Evidence class (A through E). (9) Tension status. (10) Reproduction command (executable).

The tenth field is the one most often missing and the one that most distinguishes an auditable claim from an assertion. To make the schema concrete, the framework's simplest ^[A]-class claim, the exact internal coupling constraint $5\kappa^2 = 3\lambda_S$ of eq. (6), is accompanied by an executable reproduction note. At eighty-digit working precision, with $\kappa = \frac{1}{2}$ and $\lambda_S = \frac{5}{12}$, the residual $|5\kappa^2 - 3\lambda_S|$ evaluates to zero:

```
python3 -c "from mpmath import mp,mpf; mp.dps=80;
k=mpf(1)/2; lS=mpf(5)/12; print(abs(5*k**2 - 3*lS))"
# -> 0.0 (residual < 1e-14: internal closure, not external agreement)
```

This is the template every numerical claim must follow: a command a reader can run, a stated precision, and an explicit reminder that the vanishing residual certifies internal consistency, not agreement with nature.^[A]

18.3 The Target-Leakage Theorem

Proposition 18.1 (Target Leakage). *Let a computational procedure P be claimed to predict a quantity q independently of its known value q^* . If q^* is accessible to P , as a hard-coded constant, a calibration target, a falsification threshold, a loss function, or any other input that P can read, then the agreement of P 's output with q^* provides no evidence that P predicts q . The agreement is, at most, a consistency check on P 's ability to reproduce a number it was given.*

The proposition is almost a tautology, which is precisely its strength: it cannot be evaded by cleverness, only by architecture. The calibrated value $\gamma = 16.339$, every quarantined museum constant, and every external benchmark value must be *inaccessible* to any code path that purports to compute γ , Δ , or related quantities from within Relational-Algebraic Pregeometry; benchmark values are stored with solver access explicitly disabled.^[A] The blind computation and the post-hoc comparison must be *two distinct programs*. The framework keeps documented instances of the failure mode — a hard-coded target ratio in a solver, a falsification condition defined as distance from the calibrated γ — both recorded as rejected target-leakage patterns.^[A] Part I extends the same discipline into the formal corpus, where it is stated as reviewable policy and enforced by continuous-integration scanning (section 3.6).

19 Empirical Baseline and Benchmark Batteries

19.1 Pure SU(3) as a Stress-Test Sector

The most disciplined way to test a candidate framework of gauge structure is to confront it first with the cleanest non-trivial gauge theory available: pure SU(3) Yang–Mills theory, rich enough to possess a mass gap and simple enough to be computed to high precision on the lattice by an external community whose results owe nothing to UIDT.^[A] The framing is a deliberate inversion of an earlier temptation: the Yang–Mills spectral gap is a *necessary* condition any serious candidate must pass, not a *sufficient* foundation from which the rest of physics follows. A stress test is something a theory survives, not something it stands upon.^[A]

Three boundaries are held firmly. Pure gauge theory is distinct from full QCD and both from G_{SM} ; success in the first arena says nothing automatic about the others.^[A] The standard pure-gauge benchmark families — the glueball spectrum [5–7, 9], the flow scales t_0 and w_0 [10], and the topological susceptibility χ_{top} [13] — are external benchmarks to be matched *after* a source audit, and never quantities the framework may quietly assume; until the audit completes for a given observable, its status is ^[B] (*pending source audit*).^[B] And the spectral gap Δ ^[B] must not be naively identified with the 0^{++} glueball mass (Limitation L12, limitation 19.1).

PI Decisions D18/D21 — Evidence-Class Determination for Δ

The value $\Delta = 1.710 \pm 0.015 \text{ GeV}$ remains at Class ^[B]: an internally consistent, lattice-compatible reference under the recorded determination, not a UIDT-internal derivation and not an exact mathematical theorem for the numerical value. Decision D21 rejects the proposed upgrade to Class A; no external counter-signature is sought because no upgrade occurs. The class of the value does not identify it with a particle state. Limitation L12 bars any 0^{++} glueball or particle-mass identification without an explicit operator and correlator matching.

19.2 Topology and the Topological Susceptibility

Among the pure-gauge benchmarks, χ_{top} occupies a special position, because topology is where a structural-realist theory of gauge fields is most exposed. The quantity $\chi_{top}^{1/4}$ is a pure-gauge topological benchmark *seed*: a target for post-

hoc comparison, carried at ^[B] (*pending source audit*) until its provenance is fully audited.^[B] For every external topological value, the framework requires: an explicit operator definition; the smoothing or gradient-flow prescription; the lattice spacings; the continuum-extrapolation procedure; units and scale-setting convention; the quoted uncertainty; and the covariance status relative to any other quantity in the same comparison.^[A] Only ratios with harmonised scale-setting are admissible, and only at ^[D].

Topology: Admissibility Conditions

A topological benchmark comparison is admissible only if it carries: operator definition; flow/smoothing prescription; lattice spacings; continuum extrapolation; units and scale-setting; uncertainty; covariance status. The SVZ leading-order route is failure documentation only. No χ_{top} agreement closes the G_{SM} -origin problem — and, per Part I, χ_{top} is a pure-SU(3) benchmark that proves nothing about SU(2), U(1), fermions, chirality, generations, or anomaly cancellation.

19.3 Calibration and the Discipline of γ

The invariant $\gamma = 16.339$ is the most load-bearing single number in the framework, and for that reason it is also the number around which the strictest discipline is enforced. Its classification is fixed and not subject to upgrade: γ is ^[A-], calibrated. That this status is fixed in the ledger is a document-state fact.^[A] The parameter is *not* Class [A], and it is not derived from renormalisation-group first principles.^[A-] To be precise about what “calibrated” means: γ is retained as the calibrated reference value recorded by the parameter ledger. Its historical kinetic-VEV motivation is documented in Appendix A; that motivation is not a first-principles derivation and is not imported as an active ontological premise. The public-facing description must read “calibrated to the kinetic VEV,” never “derived from the kinetic VEV”; the latter phrasing violates the ^[A-] discipline and is corrected wherever it appears.^[A-]

Anti-Target-Leakage Rule

A solver that has access to the value it is meant to predict cannot be said to predict it. Therefore the calibrated value $\gamma = 16.339$, together with all quarantined museum constants and all external benchmark values, must remain inaccessible to any code path that claims to compute γ , Δ , or related quantities from within Relational-Algebraic Pregeometry. Blind computation and post-hoc comparison must be separated pro-

grammatically; benchmark values are stored with solver access disabled. A derivation that imports its own target is a tautology wearing the costume of a result.

Any future derivation of γ must specify, in advance and in public, the operator whose flow it integrates; the flow equation; the regulator; the boundary conditions; an explicit proof that no target value has leaked into the computation; and an executable reproduction command. Absent any one of these, a claimed derivation of γ remains at [D] at best.^[A]

A final point of discipline concerns conflicting values. Alongside the calibrated kinetic value 16.339, a Monte-Carlo determination yields a mean of $\gamma_{\text{MC}} = 16.374 \pm 1.005$. These are not reconciled by quiet selection of the more convenient figure; they are recorded as a *tension entry* (section 23.2), displayed together with their methods and uncertainties.^[A]

The Status of γ

EVIDENCE

Classification. $\gamma = 16.339$ is ^[A] (calibrated), permanently. It is never ^[A] and never described as “derived.”

Open problem. A first-principles renormalisation-group derivation of γ does not exist; the kinetic-versus-RG discrepancy is an openly recorded limitation.

Tension. $\gamma_{\text{MC}} = 16.374 \pm 1.005$ is held alongside 16.339 as a recorded tension, not resolved by selection.

Leakage prohibition. $\gamma = 16.339$ must never enter a blind solver path — in numerical code or in the Lean corpus, where the same rule is policy (section 3.6).

19.4 The Pure SU(3) Ratio Battery

The disciplined way to compare against pure-gauge lattice results is not dimensional quantities, contaminated by scale-setting conventions, but *dimensionless ratios* in which the common scale cancels. Using the gradient-flow scale t_0 and the $\overline{\text{MS}}$ scale where a perturbative bridge is appropriate, the candidate ratios include $m_{0++} \sqrt{8t_0}$, $\chi_{\text{top}}^{1/4} \sqrt{8t_0}$, and the $\Lambda_{\overline{\text{MS}}}$ -normalised forms.^[B] (*pending external verification*) Every external ratio must arrive with a verified source and an explicit covariance model, because the ratios share inputs and are correlated; treating correlated ratios as independent would manufacture false significance.^[A] And as a direct application of proposition 18.1, these values remain inaccessible to any blind solver: post-hoc comparators only.^[A] If a ratio fails to match, the first reading is a

constraint on operator choice, truncation, or scale bridge. This diagnostic ordering is a governance rule.^[A] Only after those checks are exhausted does a persistent mismatch rise to a serious tension against the framework as a whole.^[D]

19.5 Δ , the Glueball, and the L12 Separation

The spectral gap Δ ^[B] and the mass of the lightest scalar glueball carry numerically similar values, and that proximity has repeatedly invited a conflation the framework must refuse. The distinction is one of category: the gap is a spectral feature internal to the framework’s construction; the 0^{++} glueball mass is a particle-spectrum quantity defined by the lattice through specific interpolating operators. To assert identity requires *operator matching*, and no such matching has been established.^[A]

Limitation 19.1 (Naive $\Delta \leftrightarrow 0^{++}$ Identification: L12). The Relational-Algebraic Pregeometry spectral gap Δ must not be identified with the mass of the lightest scalar (0^{++}) glueball of pure SU(3) Yang–Mills theory in the absence of explicit operator matching. Numerical proximity of the two values does not constitute identification. Any future relation between Δ and the glueball spectrum must be established by matching interpolating operators and quantum numbers, not by coincidence of magnitude.^[A]

In the formal corpus this limitation is a type-level fact: the glueball-identification conjecture is defined as `False` (section 3.2), so no Lean development can consume it. The general discipline is to keep vacuum-pole language distinct from lattice particle-spectrum language; the framework may aspire to relate them — a legitimate ^[D] target — but the relation must be earned by operator matching.

19.6 Gauge-Invariant Exact Renormalisation as Scaffold

A recurring hope in the developmental record is that a gauge-invariant exact renormalisation-group flow (GFERG) might supply the missing derivation of γ , of Δ , or of the coupling ratio K_S . The framework’s position is austere: GFERG is admitted only as a *scaffold*, to be defined precisely before any use, and is explicitly *not* a completed derivation of any of those quantities.^[A] A route counts only if it specifies, in advance: the flow equation (Wetterich or Polchinski family, used as

named scaffolds with no derivation imported); the regulator; the truncation; the boundary and initial conditions; the observables; and the failure criteria.^[A]

GFERG: Admissibility as Scaffold

GFERG is a candidate gauge-invariant exact-renormalisation route, admitted as a scaffold only.^[D] It is not a derivation of γ , Δ , or K_S . Renormalisation-group vocabulary may not be used to present a toy ODE as a flow; the previously NLO-labelled toy ODE is deleted as a claimed derivation.^[A]

19.7 The Limitation Register (L1–L15)

The framework’s known limitations are enumerated in one place. Entries L1–L7 and L12–L15 remain historically anchored. Entries L8–L11 below are newly issued limitation statements derived from the applied manuscript’s explicit formal boundaries; they are not presented as recovered quotations from the missing developmental notes. Their status is therefore provenance-clean: replacement statements, not reconstructed history.^[A]

Limitation 19.2 (L1: The geometric scale factor (ill-defined)). The framework employs an effective geometric scale factor as a calibration input bridging microscopic vacuum dynamics and macroscopic phenomenology. With reference scales explicit, the ratio is $\lambda_{\text{UIDT}}/r_{\text{conf}} = 0.660 \text{ nm}/0.197 \text{ fm} \approx 3.35 \times 10^6 \approx 10^{6.5}$; earlier versions quoted “a factor of order 10^{10} ” without specifying the compared scales, and that figure is superseded. No first-principles derivation of this factor is supplied; it is an adopted scaling assumption.^[D]

Limitation 19.3 (L2: Electron-mass discrepancy). The lepton-sector mapping retains an electron-mass discrepancy of order 23% under the universal γ -scaling architecture. The mismatch is not removed by the current formalism and must not be absorbed into the geometric scale factor of L1.^[C]

Limitation 19.4 (L3: Vacuum-energy parameter). A numerical parameter of order 2.3 invoked in the vacuum-energy discussion is presently unanchored: phenomenologically fitted, not derived.^[C]

Limitation 19.5 (L4: γ not RG-derived). $\gamma = 16.339$ is permanently ^[A-] (calibrated). A first-principles derivation does not exist; any future derivation must satisfy proposition 18.1 in full.^[A-]

Limitation 19.6 (L5: The $N = 99$ RG-step count). The choice of $N = 99$ discrete cascade steps is empirical: selected to reach the target magnitude, with no geo-

metric or mathematical principle dictating exactly 99. Downstream predictions (such as a Casimir scale near $\lambda \approx 0.66$ nm) presently lack direct experimental corroboration.^[D]

Limitation 19.7 (L6: Glueball identification withdrawn). The assertion that an exact renormalisation-group flow uniquely identifies the spectral gap with a specific physical glueball state is withdrawn from evidentiary use. The toy ODE once dressed in renormalisation-group language is deleted as a claimed result; residual mentions are historical only.^[E]

Limitation 19.8 (L7: EFT validity and the torsion scale). The dimension-5 non-minimal coupling makes the Lagrangian an effective field theory valid only below a cutoff Λ_{UIDT} , which must lie above the gap scale and is *not* identified with the torsion scale $E_T = 2.44$ MeV; the latter is a separate subatomic reference with no constructed observer, neural, biological, or macroscopic scale bridge (section 15.2). The value of Λ_{UIDT} is not fixed and a non-perturbative UV completion remains open.^[D]

Limitation 19.9 (L8: Algebraic identities do not select a partition). The block-thermodynamic relations collected in this manuscript constrain bookkeeping among block sizes, diagonal sectors, off-diagonal sectors, and entropy terms. No target-independent functional is supplied whose global minimiser uniquely selects $[3, 2, 1]$. An exact decomposition or conservation identity is therefore an internal consistency relation, not a dynamical selection theorem. Any future selector must be fixed before the candidate is evaluated and must survive sealed null controls under proposition 18.1.^[D]

Limitation 19.10 (L9: H1/H2 are conditional filters, not derived laws). The conditional $N = 6$ proposition depends entirely on H1 and H2. H1 is a design-level extrapolation from intersection-form constraints; H2 is a heuristic proxy for non-degenerate fermion masses. Their finite filtering power is auditable, but neither predicate is derived from the pregeometric kernel or from a free-energy stability theorem. Completion of the Lean checker would certify only the implication $H1 \wedge H2 \Rightarrow [3, 2, 1]$ over the declared finite domain; it would not upgrade the physical selection claim above ^[D].

Limitation 19.11 (L10: Finite $N = 6$ enumeration is not large- N dynamics). Exhaustive enumeration of the eleven sorted integer partitions of 6 establishes, at most, completeness and filter verdicts for that finite candidate set. It does not establish a thermodynamic limit, a large- N phase transition, continuum emergence,

spontaneous Wedderburn splitting, or asymptotic selection of gauge symmetries. Claims of that kind require a separately defined dynamics, scaling regime, and target-independent convergence criterion.^[D]

Limitation 19.12 (L11: The formal bridge is structure-losing and incomplete). The current bridge transfers a combinatorial block list into abstract finite-algebra and spectral-triple records through derived dimensions and axiomatic markers. It does not establish an equivalence with the classification of finite real spectral triples, and it does not endogenously supply bimodule multiplicities, chirality, Krajewski edges, a physical representation, or analytic spectral-triple axioms. Any conversion through a generic number-partition representation additionally forgets ordering or labels unless those data are restored and proved invariant. The bridge is therefore scaffolding for a formal programme, not a completeness theorem for noncommutative geometry.^[D]

Limitation 19.13 (L13: Background independence only partial). Although the framework belongs to the pregeometric tradition, axiom 3 is written as a perturbation about a fixed Minkowski metric, so $\eta_{\mu\nu}$ is inserted by hand rather than derived from a relational phase. Full background independence has not been established. The Weinberg–Witten theorem [42] bars any interpretation of the emergent metric as a composite spin-2 graviton in the naive flat-space sense — consistent with, not a refutation of, the thermodynamic reading borrowed from Jacobson [49, 50] — and the framework has not yet specified which loophole its construction exploits. Under D20 this limitation acquires a sharper address: the dynamical origin of $\eta_{\mu\nu}$ is exactly the Part-II passage (section 12.2), where it remains open.^[D]

Limitation 19.14 (L14: Scalar-glueball spectroscopy and the status of m_S). Beyond the operator-matching discipline of L12, recent exploratory lattice work using glueball, meson, and meson–meson operators has been summarised as suggesting that no scalar state below about 2 GeV can be regarded as predominantly a glueball state [8] — an explicitly hedged single study that does not by itself overturn the pure-gauge consensus near 1.5–1.7 GeV [5, 7], but sharpens the concern. The candidate resonance $m_S = 1.705 \pm 0.015$ GeV is therefore carried at ^[D] and its falsification threshold is read against the evolving lattice spectrum rather than a fixed benchmark.^[D]

Limitation 19.15 (L15: Non-Abelian Casimir geometry versus the scalar prediction). The Casimir-scale anomaly near $\lambda \approx 0.66$ nm is motivated by a scalar-field treatment. First-principles SU(3) lattice studies of the non-Abelian Casimir effect

complicate any naive scalar reading: between chromometallic plates the gluonic Casimir energy is well described by a massive effective scalar [15], but in tube and box geometries the non-Abelian force is found to be *attractive* where a massless scalar would predict repulsion [16]. The 0.66 nm figure is therefore a scalar-sector prediction awaiting both a non-Abelian-consistent derivation and direct sub-nanometre metrology.^[D]

The register is complete in enumeration. L1–L7 and L12–L15 preserve the historical anchored entries; L8–L11 are explicitly identified as newly issued replacement limitations and carry no claim of verbatim recovery from the missing notes.^[A]

20 The $d^2 = 0$ Obstruction

20.1 The Failed Scalar-Gradient Route to Curvature

The simplest conceivable way to extract a gauge field from a scalar is to identify the gauge potential with the gradient of the scalar. Its failure is fundamental, and it is stated with full precision because the framework treats it as a decisive result — and because, under D20, it is the first of the negative results on which the re-architecture rests (section 2.1).

Proposition 20.1 (Vanishing Curvature of an Exact Connection). *Let S be a smooth real scalar field and let the connection one-form be defined as its exterior derivative, $A = dS$. Then the curvature two-form vanishes identically:*

$$F = dA = d(dS) = d^2S = 0,$$

since the exterior derivative satisfies $d^2 = 0$ on any smooth manifold. Consequently the scalar-gradient construction yields only a flat, pure-gauge connection and cannot produce a non-trivial Yang–Mills field strength.^[A]

The result is elementary, and that is exactly its force. A one-form that is exact is automatically closed, and its curvature is zero by identity, not by approximation; the route fails identically, everywhere, for any smooth scalar.^[A]¹ The discrete

¹**Audit note (Clebsch loophole).** A known mathematical loophole exists via Clebsch parametrisations built from higher jets of S : with functionally independent scalars u, v constructed from

counterpart fails for the same reason: a discrete exterior calculus inherits $d^2 = 0$ in its coboundary operator.^[A] With $A = dS$, both contributions to $F = dA + A \wedge A$ vanish — the first by nilpotency, the second by $\omega \wedge \omega = 0$ for any 1-form:

```
# sympy differential-forms check (schematic):
# d(dS) = 0      (Poincare, d^2 = 0)
# dS ^ dS = 0    (omega ^ omega = 0 for a 1-form)
# => F = d(dS) + dS^dS = 0 identically, for any scalar S
```

The vanishing is an identity; no precision setting and no input data can change it.^[A]

20.2 Topological Consequences and the Interface to Part I

If F vanishes identically, every topological invariant built from the scalar-gradient connection vanishes with it: no instanton number, no non-trivial winding, nothing that could seed a topological susceptibility.^[A] This is why the topological observables of section 19.2 cannot be sourced from S alone, and why the effective sector couples S to a Yang–Mills sector that is already present.

What this does and does not establish must be stated precisely. It does *not* establish that non-abelian gauge symmetry can never arise from a scalar primitive; no such impossibility theorem exists, and the rejected “algebraic emptiness” impossibility claim stays rejected.^[A] Nor may numerical proximity be presented as a “seal” that closes the gauge-origin question; seal language is rejected wherever it appears.^[A] What the obstruction establishes is narrower and certain: the specific route $A = dS$ fails identically — and the space of non-naive routes is exactly what Part I now maps, with its localization of the obstruction to the multiplicity insertion (G1–G4), its corrected admissibility lattice, its lower bounds on any intermediate algebra, and its candidate programmes, all at their stated classes. The v3.9 fork — strict scalar versus enriched primitive — has been taken: D20 commits the research programme to the enriched arm, at ^[D], with the strict-scalar arm’s honest boundary preserved here as the permanent record of why.

S and its derivatives, the one-form $A = u dv$ yields a non-vanishing composite *abelian* curvature $F = du \wedge dv$.^[A] This does not rescue the gauge-origin programme: the construction is tensorial, providing no gauge redundancy and no gauge orbit; it offers no non-abelian structure; and it supplies no chiral matter. The obstruction remains exactly as stated: construction-specific, with no universal impossibility theorem claimed.^[A]

21 Cosmological Calibration

Every cosmological quantity in this section is Category ^[C]: calibrated to external data, not derived from the pregeometric kernel or from the historical scalar ansatz, and never presented as resolving an observational tension. The framework’s cosmological role is that of a consumer of empirical inputs, not a producer of predictions.^[C]

21.1 Empirical Inputs and the Calibration Boundary

Hubble rate H_0 . The defining feature of the current H_0 situation is a persistent $\sim 5\sigma$ discrepancy between early- and late-universe determinations. In units of $\text{km s}^{-1} \text{Mpc}^{-1}$, the *Planck* 2018 acoustic-scale analysis in base ΛCDM gives $H_0 = 67.4 \pm 0.5$ [84], while the SH0ES distance-ladder programme reports $H_0 = 73.04 \pm 1.04$ [85]; combined in quadrature the two sit roughly 4.9σ apart. The framework calibrates its macroscopic metric parameters to this empirical situation and records the disagreement in the tension ledger; it does *not* claim to resolve the Hubble tension, and any such claim is a forbidden overclaim.^[C]

Dark-energy equation of state. In the Chevallier–Polarski–Linder parametrisation $w(a) = w_0 + w_a(1 - a)$, a pure cosmological constant requires $w_0 = -1$, $w_a = 0$. The DESI Data Release 2 baryon-acoustic-oscillation measurements [86] report a preference for the quadrant $w_0 > -1$, $w_a < 0$ (evolving dark energy), with significance depending on the data combination — a mild $\sim 2.3\sigma$ BAO–CMB tension in ΛCDM , rising to $\sim 2.8\text{--}4.2\sigma$ for $w_0\text{--}w_a$ fits with supernova samples. Combination-dependent central values (e.g. $w_0 \approx -0.42 \pm 0.21$, $w_a \approx -1.75 \pm 0.58$ for DESI+CMB) are carried as Stratum-I inputs at ^[C], flagged for final confirmation against the published DESI DR2 table. The supernova-inclusive combinations trace an equation of state that crosses $w = -1$ at intermediate redshift — a phantom crossing that a single canonical real scalar cannot smoothly realise without a ghost instability, which is precisely why these are imported as external constraints rather than derived from the kernel or the historical scalar ansatz.^[C]

Matter fluctuation amplitude S_8 . The parameter $S_8 \equiv \sigma_8 \sqrt{\Omega_m/0.3}$ is currently survey-dependent: KiDS-Legacy cosmic shear reports $S_8 = 0.815^{+0.016}_{-0.021}$ (agreeing with *Planck* at $\sim 0.73\sigma$), while the DES Year-6 3×2 pt analysis reports $S_8 = 0.789 \pm 0.012$ [87, 88]. The framework treats S_8 as a survey-and-pipeline-dependent input its coarse-graining parameters are bounded by; it calibrates to the spread and does not relabel it as a resolved tension.^[C]

Table 9: Stratum-I empirical inputs to which Relational-Algebraic Pregeometry calibrates. All entries are Category [C]; none is a prediction or a tension resolution.

Parameter	Value $\pm$ uncertainty	Source	Class
H_0 (early)	$67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$	Planck 2018	[C]
H_0 (late)	$73.04 \pm 1.04 \text{ km s}^{-1} \text{ Mpc}^{-1}$	SH0ES (Riess 2022)	[C]
w_0, w_a	$w_0 \approx -0.42 \pm 0.21, w_a \approx -1.75 \pm 0.58$ (DESI+CMB; combination-dependent)	DESI DR2	[C]
S_8	$0.815^{+0.016}_{-0.021} / 0.789 \pm 0.012$	KiDS-Legacy / DES Y6	[C]

21.2 The Relational-Proliferation Scenario: Interpretive Status

The developmental corpus contains a cosmogonic narrative, retained strictly as Category ^[E] interpretation within Stratum III — a way of thinking, fed into a Category ^[C] calibration, not a derivation of cosmological data.^[E] The “beginning” is read not as a physical instant $t = 0$ but as the first logical distinction in the sense of section B.1; cosmic inflation is reinterpreted, again at ^[E], as a rapid proliferation of relational structure that, viewed under coarse-graining, appears in the effective regime as an exponentially expanding metric. The picture computes no inflationary observable, predicts no primordial spectrum, and does not touch the ^[C] cap.^[E]

22 The Verification Return Path and Final Claims

22.1 The Empirical-to-Ledger Return Path

The verification architecture traces a controlled return path from external comparators to the parameter ledger without promoting that path to ontology. One begins from the empirical and lattice stress tests of pure $SU(3)$ (^[B] pending their source and covariance audit); computes the framework-side quantities *blindly* per proposition 18.1 (^[D]); compares *post hoc* through an explicit uncertainty and covariance model; identifies structural audit facts ^[A] and open programme failures ^[D]; and interprets last, at ^[E]. The closure is governed throughout by the two structural limits this manuscript has doubled down on: the $d^2 = 0$ obstruction (proposition 20.1) and the localized G_{SM} -origin gap (section 4). The return path terminates in a disciplined accounting rather than an ontological closure. Its auditability is a software and governance fact ^[A]; its physical interpretation remains bounded by the individual claim classes.

22.2 Final Allowed Claims

Table 10: Final allowed claims, with conservative evidence classes and strata. Negative and open entries are included deliberately; classes are unchanged from the incoming ledger; D20 changes architectural role only (ONT-11).

ID	Claim	Class	Stratum
ONT-01	In the historical scalar grammar, a vanishing field value is a marked state, not the unmarked state; this is retained only in Appendix A.	^[A]	II/III
ONT-02	A morphism $f: A \rightarrow B$ presupposes distinguishable A, B ; morphism is a formal tool, not an ontological generator.	^[A]	II/III
ONT-03	With $A = dS$, the curvature $F = d^2S = 0$ identically; the naive scalar-gradient route fails.	^[A]	II
ONT-04	$\gamma = 16.339$ is calibrated, never derived from RG first principles.	^[A-]	III

ID	Claim	Class	Stratum
ONT-05	$\Delta = 1.710 \pm 0.015 \text{ GeV}$: internally consistent, lattice-compatible; determination D18 (section 19.1); not externally peer-confirmed, not derived.	[B]	I/III
ONT-06	$\kappa = \frac{1}{2}$ (exact) exact framework input (fixed-point reading a candidate condition, untested by any defined flow); $5\kappa^2 = 3\lambda_S$ closes internally to $< 10^{-14}$. The empirically calibrated reading $\kappa = 0.500 \pm 0.008$ is a separate, lower-class statement.	[A]	II
ONT-07	Pure SU(3) ratio comparisons are post-hoc stress tests with solver access disabled.	[B] (<i>pending external verification</i>)	I/III
ONT-08	Primitive $\rightarrow \mathcal{A} \rightarrow G_{\text{SM}}$ is the research programme; localization, lower bounds, and candidates are recorded in Part I.	[D]	III
ONT-09	Cosmological parameters are calibrated, capped at C; no tension is declared resolved.	[C]	I/III
ONT-10	The observer functor $F: V \rightarrow \text{Obs}$ and the compression taxonomy are bounded interpretive frameworks.	[E]	III
ONT-11	Under PI Decision D20, Part III is a verification/benchmark/ledger stratum rather than active ontology; the kernel candidate of Part I remains a research programme at [D].	[D]	III

The conservatism of table 10 is its principal feature: its firmest ground is its boundary results, and its ambitions are labelled as ambitions.^[A]

22.3 Final Forbidden Claims

Final Forbidden Claims

- (1) No derivation of the full Standard-Model gauge group G_{SM} — from the historical scalar ansatz, and equally none from the kernel candidate: Part I records selection, not derivation.
- (2) No derivation of the Born rule.
- (3) No solution of consciousness or qualia.
- (4) No validation of the historical scalar ansatz or of the kernel by psychedelic, near-death, or mystical experience.
- (5) No use of E_T as a threshold for life, death, ego, psychedelic states, or artificial general intelligence.
- (6) No identification of Δ with the 0^{++} glueball without operator matching (L12).
- (7) No use of quarantined museum artifacts as derivation or factual evidence.
- (8) No presentation of $5\kappa^2 = 3\lambda_5$ as an independent empirical discovery where λ_5 is fixed definitionally by that same constraint.
- (9) No claim that DESI, H_0 , S_8 , or w_0 - w_a data validate Relational-Algebraic Pregeometry beyond calibrated Category ^[C] status.
- (10) No claim that a lattice-compatible number establishes the Standard-Model gauge group, a confinement mechanism, or a Clay-grade Yang–Mills construction.
- (11) No presentation of the conditional $N = 6$ selection proposition (proposition 5.1) as an unconditional derivation of the Standard-Model algebra; its grade is bounded by H1 and H2.

All eleven are Class ^[A] standing constraints.

The seventh prohibition guards the integrity of all the others. The museum of refuted constructions — the numerological decomposition of γ , the failed scalar-gradient operator, the toy ODE dressed in renormalisation-group language, and now the quarantined block-repetition audit of section 4.2 — is retained for its documentary value, as evidence of method; its contents may never be cited as factual evidence for any claim.^[A]

23 Falsification Gates and the Tension Ledger

23.1 The Falsification Gates

The framework's discipline is enforced by gates through which every proposed claim, computation, or contribution must pass. They are *advisory and non-approving*: a gate can block, but a gate that is passed authorises nothing. Passing all eight is necessary hygiene, never sufficient warrant.^[A]

The Eight Falsification Gates (Advisory, Non-Approving)

F1: No target leakage. No solver may access the value it claims to predict (proposition 18.1); in the Lean corpus, no definition or hypothesis may contain the candidate partition (section 3.6).

F2: Local precision discipline. Working precision is set locally per proof-critical block; no global precision mutation; no proof-critical floating-point or rounding heuristics.

F3: Complete benchmark accounting. The requirement that every external benchmark carry a verified source, uncertainty budget, and covariance model is an internal audit rule. ^[A] Satisfaction of that external benchmark condition remains pending at ^[B].

F4: L12 separation. Δ is not identified with the 0^{++} glueball without operator matching (limitation 19.1).

F5: G_{SM} -origin honesty. The gauge-origin gap is stated as open (section 4); no derivation is claimed; selection propositions carry their conditional grades.

F6: Observer evidence boundary. No phenomenological report validates the ontology (section 17).

F7: Born-rule status. The Born rule is recorded as borrowed and open, never as a derivation (section 13.2).

F8: Governance boundary. No machine-generated verdict authorises a merge, freeze, or evidence-class upgrade — and no formal proof, once machine-checked, authorises a physics upgrade without governance.

Each gate is Class ^[A]. Passing is advisory; it approves nothing.

23.2 The Tension Ledger

Table 11: Tension ledger. Unresolved disagreements and gaps, displayed rather than smoothed.

ID	Tension / gap	Status	Resolution path
T1	$\gamma_{\text{MC}} = 16.374 \pm 1.005$ versus calibrated $\gamma = 16.339$.	[consistent within MC variance]; displayed separately, not merged by selection.	Reduce MC variance; non-leaking derivation of γ .
T2	$\Delta = 1.710 \pm 0.015 \text{ GeV}$ evidence class.	Would sit [E] (<i>pending bridge</i>) absent external work; determination D18 sets [B] (section 19.1).	External peer counter-signature.
T3	Δ versus 0^{++} glueball identity.	[A] separation enforced (L12); identity unproven.	Operator matching.
T4	SVZ leading-order topology route versus lattice topology.	Separation of the two routes is an audit fact [A]; external lattice agreement remains pending at [B]. The SVZ route is museum-only.	Audited lattice χ_{top} ratios.
T5	E_T observer/neural scale bridge.	Its absence and the stated scale mismatch are audit facts [A]; any proposed bridge remains interpretive at [E].	Derive or refute a principled scale bridge.

ID	Tension / gap	Status	Resolution path
T6	Born-rule derivation.	Absent; borrowed from Stratum II. ^[A]	Non-circular coarse-graining derivation.
T7	Cosmological tensions (H_0 , S_8).	Capped at ^[C] ; calibrated, <i>not</i> closed.	Independent prediction, not fit.
T8	[2, 2, 2] rejection (H2) rests on an unverified mass-degeneracy argument (table 3).	^[D] ; the weakest verdict in the corrected lattice; the conditional selection proposition inherits it.	Primary-source theorem or free-energy stability analysis settling the case either way.
T9	Scalar-glueball spectroscopy versus $m_S = 1.705 \pm 0.015$ GeV and the empirical grounding of Δ .	^[D] ; exploratory lattice work [8] suggests no state below 2 GeV is predominantly a glueball; the 1.5–1.7 GeV pure-gauge consensus is sharpened, not overturned, and the empirical grounding of Δ is correspondingly under-determined (L14).	Read m_S against the evolving lattice spectrum, not a fixed benchmark; monitor the sub-2 GeV picture.

Two entries warrant closing remarks. T1: the calibrated kinetic value and the Monte-Carlo mean differ by 0.035 GeV, i.e. 0.035σ given the ± 1.005 spread, and are displayed together rather than reconciled by selection.^[A] T8 is new in this manuscript and is deliberate: the formal selection layer of Part I made the weakest link of the admissibility lattice *sharper*, not weaker, and the ledger records it where

it can be seen. The ledger remains the most honest page in the manuscript: a framework that hid this page would be easier to admire and impossible to trust.^[A]

Coda: Epistemological Position and Self-Audit

The three parts of this manuscript have stated what the framework holds, at what evidence class, and under what falsification conditions. This coda states something different and equally necessary: the philosophical position from which those statements are made, the company the framework keeps, and the outcome of turning the framework's own adversarial machinery against its strongest-looking claims. Every assertion in this coda is Category ^[E], Stratum III, unless explicitly tagged otherwise, and none of it alters any evidence class established earlier.

24 Epistemological Position and Comparisons

24.1 Structural Realism, Held as a Position

The ontological proposal that physical reality is constituted by mathematical structure raises an epistemological question: does Relational-Algebraic Pregeometry claim the universe *is* a mathematical object, or merely that it is *best described by* one? The framework's working answer leans toward the former, an ontic-structural-realist position, but it is held as a philosophical *position*, not asserted as a result, and nothing in the physics strata depends on it.^[E] The proposal is narrow: not that all mathematical structures are physically real, but that one specific structure is offered as the candidate constituting structure.

What D20 Does to the Structural-Realist Position

Under PI Decision D20 the identity of the candidate constituting structure changes, and the position is thereby *weakened*, not strengthened. In v3.9.x the candidate was the self-interacting vacuum information density of eq. (6): a concrete Lagrangian, with a stated minimality argument. In this manuscript that Lagrangian describes the *effective* stratum, and the candidate constituting structure is the finite algebraic pregeometric kernel of Part I — which is held at ^[D], is not uniquely specified, and whose Wedderburn invariants, real structure, and KO-dimension are free moduli. A structural realism whose structure is a research programme is a position about a *placeholder*. The framework states this rather than concealing it: the ontological ambition of Part I exceeds its formal content by exactly the distance between ^[D] and ^[A], and that distance is the honest measure of the position's current standing.^[E]

The selection is meant to be structural rather than anthropic. For the effective sector, the Lagrangian of eq. (6) is the minimal gauge-invariant, vacuum-stable, EFT-admissible self-interaction of a real scalar non-minimally coupled to $SU(3)$ in $3 + 1$ dimensions; for the kernel, the analogous minimality claim does not yet exist, and open question 1 together with the conditional selection proposition (proposition 5.1) marks the place where it would have to be constructed. That “minimality” is in both cases a claim about the framework’s construction, not a proven uniqueness theorem, and is carried as such.^[E]

The position is explicitly bounded by its own incompleteness. The framework acknowledges that any sufficiently expressive formal system is subject to Gödel-type incompleteness, and it reads its limitation register (section 19.7) as a concrete, honest catalogue of what it cannot presently derive rather than as a temporary embarrassment. The bound is sharper than it was: the undecidability of the spectral-gap problem for general local translation-invariant Hamiltonians (Cubitt–Pérez-García–Wolf) is an external theorem showing that some questions of exactly this shape admit no algorithm and are independent of the axioms.^[A] That result is not a statement about the Yang–Mills gap and not a statement about Δ ; it is a statement about the class of questions to which such questions belong.^[A]

The framework also distinguishes itself from a simulation hypothesis: Relational-Algebraic Pregeometry posits no external simulator and no computational substrate. Whatever force the structural-realist picture has, it is ontological speculation, not a claim that the universe is being computed by something.^[E] The computational vocabulary that appears in the developmental record — interfaces, rendering, frontend and backend — is metaphor, and is barred from any argument in this manuscript.

24.2 Comparison with Alternative Frameworks

It is useful, and disciplined, to state where Relational-Algebraic Pregeometry sits relative to established programmes, always as comparison, never as a claim of superiority.

Against string theory, Relational-Algebraic Pregeometry is far more ontologically frugal but correspondingly far less developed and far less mathematically mature; frugality is not a virtue if it buys its simplicity by leaving the hard problems

unsolved, and the G_{SM} -origin gap is exactly such an unsolved problem.^[E] Against entropic gravity (Verlinde [45], Padmanabhan [44]), Relational-Algebraic Pregeometry shares the intuition that geometry is emergent and thermodynamic in spirit but does not claim to have improved on it; the relationship is one of shared intuition, not demonstrated advance.^[E] Against loop quantum gravity, Relational-Algebraic Pregeometry shares the *aspiration* to background independence but does not yet achieve it in full: the metric-emergence map (axiom 3) is written as a perturbation about a fixed Minkowski background, so background independence is at present only partial (limitation 19.13); it takes the opposite route to geometry (a continuum coarse-graining limit rather than quantised geometric structures); and like LQG it has not recovered the full Standard-Model particle content — indeed Relational-Algebraic Pregeometry has not recovered it at all, which is the content of the open gap.^[E]

Two comparisons are new to this manuscript, and both are unflattering. Against the noncommutative-geometry programme of Connes, Chamseddine, and collaborators, Part I is a strict consumer: every structural constraint it uses — Krajewski combinatorics, multiplicity matrices, the KO-dimension-6 real structure, the classification of irreducible geometries — is imported, and Relational-Algebraic Pregeometry has added no theorem to that literature. Against the matrix-model programme (IKKT, BFSS, and their condensation studies), the framework’s dynamical-selection search returned an honest null. The relationship in both cases is that of a framework standing on external results, and the correct posture is acknowledgment rather than positioning.^[E]

The framework’s one genuine methodological commitment here is an epistemic razor: it admits no unobservable extra dimensions, no fundamental supersymmetry, and no infinite landscape, and it ties its claims to a small set of parameters with explicit falsification thresholds (section 23). This makes the framework vulnerable, and that vulnerability, not any claimed explanatory triumph, is what the framework offers as its scientific virtue.^[E] The earlier developmental claim that UIDT “eliminates” dark matter or dark energy is withdrawn: cosmology is capped at ^[C] (section 21), and no such elimination is asserted.

24.3 Relation to the Information-Theoretic and Structural-Realist Tradition

The general move that Relational-Algebraic Pregeometry makes — taking an information-theoretic primitive as ontologically basic and recovering spacetime, and aspirationally gauge structure, as emergent — is not itself novel, and intellectual honesty requires locating it explicitly within an established tradition rather than presenting it as unprecedented.

The lineage begins with Wheeler’s “it from bit” programme, in which information is treated as the substrate from which physical things derive [47]. It continues through the thermodynamic and entropic-gravity derivations of geometry (Jacobson [43]; Padmanabhan [44]; Verlinde [45]), the matrix-model emergent-geometry programme [75, 76], and group field theory, in which spacetime emerges from a condensate. Within holography, Van Raamsdonk argues that the emergence of classically connected spacetimes is intimately related to the entanglement of degrees of freedom in a non-perturbative description of quantum gravity, and that disentangling the degrees of freedom associated with two regions causes those regions to pull apart and pinch off, quantifiable by standard entanglement measures [95]. That statement is a Stratum-II^[B] result *within* AdS/CFT; Relational-Algebraic Pregeometry has no holographic dual and no boundary conformal field theory, so the generalisation “spatial connectivity *is* entanglement” is a Stratum-III reading at^[E] and is not inherited by this framework.^[E] A particularly close ontological cousin is the Carroll–Singh proposal [48], in which the only fundamental objects are a vector in Hilbert space and a Hamiltonian; UIDT under D20 has moved *toward* that austerity, replacing a concrete real-scalar field with a finite algebraic candidate, while sharing the aim of a maximally frugal primitive.

Prior-Art Register (Thermodynamic-Gravity Bridge)

EVIDENCE

RMC-01 In AdS/CFT, disentangling boundary degrees of freedom disconnects the bulk geometry [95]. ^[B], Stratum II.

RMC-01b The generalisation “spatial connectivity *is* entanglement” beyond holography. ^[E], Stratum III. Not inherited by Relational-Algebraic Pregeometry.

RMC-02 Time as a local informational calibration measure. ^[E], Stratum III.

RMC-03 Continuous gravity as an effective field theory; the thermodynamic-gravity results of Jacobson, Padmanabhan, and Verlinde are *precedent* that emergent geometry

is a respectable position, not derivations imported into Relational-Algebraic Pregeometry. ^[A] re: corpus (as a statement about what Relational-Algebraic Pregeometry does and does not derive); ^[E] for the mapping.

Against this background, what is specific to Relational-Algebraic Pregeometry — and what is offered as its actual contribution, as distinct from the shared tradition above — is narrow and concrete: the particular Lagrangian of eq. (6) with its non-minimal $S F^2$ coupling at a stated cutoff; the identification of the spectral gap as a contraction fixed point; the calibrated dimensionless invariant $\gamma = 16.339$ ^[A]; the localization of the G_{SM} -origin obstruction to the four conditions G1–G4 with quantitative lower bounds on any admissible intermediate algebra (section 4.4); and the explicit evidence-grading discipline under which every claim in this manuscript is carried. None of these is a claim to have solved the emergence problem that the tradition leaves open; they are the specific, falsifiable form in which Relational-Algebraic Pregeometry states its version of it. ^[E]

25 Scientific Context and Reciprocal Demarcation

The developmental chat is treated here as a question inventory, not as a scientific source. Its useful content consists of candidate comparisons, objections, and missing bridges. Every external statement below is grounded in a primary or peer-reviewed source rather than in the chat response that first raised it. ^[A]

25.1 The Non-Anthropocentric Lens as Method

The “non-anthropocentric lens” is retained as a methodological instruction: do not infer primitive ontology from perceptual salience, biological utility, mathematical familiarity, aesthetic preference, or human-scale language. ^[E] It is not a theorem that space, time, objects, causality, or continuity are illusions. UIDT has not derived the conditions under which those structures emerge or fail. ^[A]

The defensible content is type discipline. If a concept is absent from the primitive signature, it cannot be used at that level without a declared bridge. ^[A] This rule blocks background spacetime, observers, temperatures, particles, and continuum

fields from being smuggled into the Phase-6 kernel. It does not establish that the present finite algebraic candidate is the final ontology.^[E]

The stronger developmental thesis that all scientific progress consists in the destruction of category errors is rejected. A narrower historical claim is sustainable: progress often requires separating variables, objects, and explanations that belong to different domains or effective levels.^[B] UIDT applies that diagnostic to itself by preserving its scalar history while removing that history from the active primitive layer.^[E]

25.2 Structural Realism and Underdetermination

Ontic structural realism interprets modern physics as favouring relations and structures over independently individuated fundamental objects [100, 101].^[E] UIDT is compatible with that orientation, but it neither follows from nor confirms structural realism. OSR selects no `BlockPartition`, spectral triple, Krajewski diagram, thermodynamic functional, or parameter value.^[A]

The reciprocal relation is methodological. Structural realism supplies a vocabulary for structure-dependent objecthood; UIDT supplies a concrete but unverified formal candidate on which that vocabulary can be stress-tested. ^[E] Underdetermination remains binding: predictive equivalence or compression efficiency does not uniquely select ontology.^[E]

25.3 Pregeometry and Quantum-Gravity Neighbours

Causal-set theory, loop quantum gravity, spin foams, group field theory, tensor networks, holography, matrix models, and noncommutative geometry contain serious programmes in which continuum spacetime is non-fundamental, discrete, algebraic, quantum, or emergent [75, 76, 95, 102–106]. ^[B] Their existence makes non-fundamental-continuum research scientifically legitimate. It provides no evidence for UIDT's specific kernel or bridge.^[A]

The differences are structural. Causal sets begin from causal order and local finiteness; loop approaches quantize geometric observables; group field theory uses field-theoretic quanta of geometry; holographic and tensor-network models relate

entanglement to geometry in specified regimes; matrix models derive candidate geometries from matrix dynamics; UIDT instead proposes a finite relational-algebraic kernel and leaves the continuum passage open.^[B] No equivalence functor, shared universality class, or common renormalisation trajectory has been constructed between UIDT and any of these programmes. ^[D]

Jacobson, Padmanabhan, and Verlinde are precedents for thermodynamic or entropic reformulations of gravitational dynamics [43–45].^[B] UIDT does not recover their horizon assumptions, thermodynamic variables, or Einstein dynamics from its kernel.^[A]

25.4 Noncommutative Geometry, Division Algebras, and the Gauge-Origin Gap

Noncommutative geometry supplies the mature language from which UIDT imports finite spectral triples, real structures, KO-dimension, first-order conditions, Krajewski combinatorics, and spectral-action methods [60, 80, 89].^[B] Standard-Model structures can be recovered within that programme under substantial geometric and representation-theoretic assumptions. UIDT has not derived those assumptions from a prior featureless substrate.^[A]

Division-algebra programmes use complex, quaternionic, and octonionic structures to organise Standard-Model representations and symmetry breaking [92, 94, 120, 121].^[B] These results are candidate constraints on an internal algebra. They do not show that UIDT generates the division algebras, selects exactly three generations, or satisfies G1–G4 without inserted target structure.^[A]

The scalar-gradient obstruction is narrow and exact: for an ordinary scalar function, $A = dS$ implies $F = dA = d^2S = 0$.^[A] It excludes one route from a scalar gradient to non-trivial curvature. It does not refute scalar fields, scalar-tensor gravity, Higgs sectors, inflationary models, or emergent gauge fields produced by richer systems.^[A]

25.5 Information, Computation, and the Status of the Meta-Equation

Shannon information theory defines entropy, channel capacity, coding limits, and communication rates for specified ensembles and channels [107].^[A] It does not entail that information is a substance or that the universe is fundamentally a communication channel. ^[A] Wheeler’s information-first language remains a historically important research heuristic, not a derivation of UIDT.^[E]

The spectral-gap undecidability theorem of Cubitt, Pérez-García, and Wolf applies to a precisely constructed class of infinite, translationally invariant quantum lattice systems [96].^[A] It does not prove that all physical systems, all Yang–Mills gaps, or the UIDT kernel are computationally irreducible.^[A]

The developmental expressions $K(\mathcal{A}_E) \gg C(O)$ and an $\arg \min D_{\text{KL}}$ “master equation” are therefore not admitted as equations of motion or derived bounds. Their variables, probability spaces, units, dynamics, existence conditions, and uniqueness conditions were never defined.^[A] They may be retained only as a prose schema: a finite subsystem generally uses reduced representations of a larger environment. Any quantitative inequality remains an open construction. ^[E]

Ramsey theory guarantees ordered substructures in sufficiently large graphs under specified colouring conditions.^[A] It does not establish that observers are Ramsey cliques, that cognition is forced by a Ramsey number, or that the UIDT kernel generates biological observers.^[A]

25.6 Relational Time and the Scope of Analogue Experiments

The thermal-time hypothesis and Page–Wootters-type constructions show that internal or state-dependent temporal order can be formulated in precise operator-algebraic and quantum settings [97, 118].^[B] A 2026 cold-atom experiment constructed an entropic internal time for a partitioned Bose–Einstein condensate and used it to order subsystem dynamics [119].^[B] This is a controlled analogue test of a relational-time construction. It is not evidence that UIDT’s kernel generates time, and it does not resolve the manuscript’s non-circular state-selection problem.^[A]

25.7 Quantum Probability and the Born-Rule Boundary

Gleason-type theorems derive the density-operator probability form from specified Hilbert-space, projection, additivity, and compositional assumptions [115–117].^[A] These results clarify what a mathematical Born-rule theorem requires. They do not derive the Hilbert space, projection lattice, composition rule, or probability measure from the UIDT pregeometric kernel.^[A] The Born rule therefore remains a Stratum-II import unless a separate bridge is constructed.^[D]

25.8 Reciprocal Demarcation

Descriptions of established external programmes are carried at ^[B]. UIDT-specific mappings are carried at ^[E]. Statements that no bridge is present in the current corpus are document-level facts at ^[A].

Programme	Established contribution	UIDT relation and non-inference
Ontic structural realism	Relation-first metaphysical interpretation	Compatible vocabulary; no selection or validation of the UIDT kernel.
Noncommutative geometry	Finite spectral triples and spectral action	Primary formal source language; UIDT does not derive its assumptions.
Causal sets / loop / GFT	Discrete or quantum-geometric programmes	Neighbouring research directions; no shared continuum theorem.
Tensor networks / holography	Entanglement–geometry relations in defined regimes	Useful coarse-graining comparator; no universal entanglement–space identity.
Matrix models	Matrix dynamics and candidate emergent geometries	Closest dynamical neighbour; no target-independent UIDT selection theorem.
Thermodynamic gravity	State-equation and entropy-based gravitational reformulations	Precedent only; no Einstein-equation derivation from UIDT block data.
Division algebras	Structured particle-representation programmes	Candidate constraints; no endogenous selection or generation theorem.
FEP / predictive processing	Statistical boundaries and generative control	Functional observer analogy; no fundamental observer or qualia theorem.
IIT / GNWT	Empirically testable consciousness theories	Neutral comparators; no identification with UIDT observer language.
Orch OR	Quantum-biological objective-reduction proposal	Neither adopted nor refuted; no microtubule or quantum-gravity bridge.
Psychedelic research	Reports and neural correlates of altered experience	Phenomenology only; no fundamental-physics evidence.
Gleason programmes	Quantum probability from stated structural assumptions	Clarify missing premises; do not close the UIDT Born-rule gap.

Table 12: Reciprocal scientific demarcation. Similarity, precedent, or shared vocabulary does not constitute confirmation.

26 Scope, Contingency, and Interpretive Minimality

26.1 From “Unreasonable Effectiveness” to a Modest Minimality

Wigner’s remark on the unreasonable effectiveness of mathematics invites a strong reading: that if the universe *is* mathematical structure, the effectiveness is no longer surprising. The framework records this as a tempting interpretive move and explicitly declines its strong form. To call the effectiveness a “tautology” or the structure “inevitable” would be exactly the kind of rhetorical overreach the manuscript’s discipline forbids; the honest statement is far more modest. Within the framework’s own axioms and within $3 + 1$ dimensions, the $SU(3)$ Lagrangian of eq. (6) is presented as the *minimal* self-interacting structure compatible with gauge invariance, vacuum stability, and EFT admissibility. That is a statement about minimality under stated constraints, not a proof of cosmic necessity, and the framework does not claim the universe *had* to be this way.^[E]

The question “why this universe and not another?” is therefore answered, if at all, by structural minimality rather than by modal necessity or anthropic selection. Under D20 even that answer contracts: the minimality argument applies to the effective sector, and no minimality argument exists for the kernel. The spectral gap $\Delta = 1.710 \pm 0.015 \text{ GeV}$ remains, in this reading, a structurally determined feature of the construction’s fixed point, but its evidence class is unchanged by any of this philosophy: it is ^[B], and the philosophical gloss adds no evidential weight.^[E]

A Framework Cannot Be Made Irrefutable

A recurring temptation, in a research programme that has survived several audit cycles, is to seek a formulation so guarded that it can no longer be wrong. That ambition is incoherent, and axiom 4 forbids it. A framework that cannot fail is not a physics framework; it is a vocabulary. What can be sought, and what this manuscript seeks, is a formulation whose claims are so precisely bounded that a hostile reader finds no gap between what is asserted and what is shown. That is not irrefutability. It is the opposite: it is the condition under which refutation becomes possible and informative.^[A]

26.2 Reception Risk and Non-Evidential Context

Finally, the framework reflects, again purely as ^[E] context, on how a radically simplifying ontology is liable to be received. A framework whose evidentiary classes are not kept explicit can be misread in both directions: dismissed wholesale on account of its speculative stratum, or embraced wholesale on the strength of its calibrated agreements.^[E] The framework's response is not to claim vindication but to insist on the opposite safeguard: the evidence-classification system and the verification discipline exist as bulwarks against *both* premature rejection *and* uncritical acceptance. The right attitude toward a radical simplification is neither enthusiasm nor dismissal but exactly the falsification-first scrutiny the manuscript has tried to embody; by that standard, the framework's simplifications must earn their place by surviving tests, not by being elegant.^[E]

Philosophical Context: Standing Status

EVIDENCE

This entire coda is ^[E], Stratum III, except where a tag states otherwise. Structural realism is a position held, not a result; under D20 its candidate structure is a ^[D] research programme, which weakens the position rather than strengthening it. Minimality is a construction claim about the effective sector, not a uniqueness proof, and no minimality argument exists for the kernel. "Inevitability" and "tautology" language is withdrawn. Comparisons with string theory, entropic gravity, LQG, noncommutative geometry, and matrix models are stated without claim of superiority. The "eliminates dark matter/energy" claim is withdrawn. No philosophical statement here alters any evidence class.^[E]

27 The Adversary Test: A Self-Audit

The framework maintains an internal adversary test: a formalised attempt to break its own strongest-looking claim. The discipline of this edition requires that the test be run honestly, which means reporting where the adversary *wins*, not staging a defence that quietly declares victory. This manuscript runs the test twice: once against the v3.9 scalar claim it inherits as history, and once against the D20 Phase-6 rebase.

27.1 First Round: The Origin of γ (Limitation L4)

Adversarial argument. If γ is a phenomenological fit parameter rather than a derived constant, then any vacuum-energy or cosmological calculation scaled by a power of γ risks tautology: it would scale by precisely the amount needed to match observation, draining the result of independent explanatory power. The adversary presses further: the developmental record once decomposed γ numerologically ($49/3 + 17/3000$, with the integer 16339 prime), which is positive evidence that the value was *constructed* to a target rather than found.

Honest adjudication. The framework does not claim the adversary is refuted. On the contrary: the numerological decomposition is quarantined in the museum precisely because the adversary is right about it — that construction is not evidence and is barred from use. The rejection is *methodological* before it is arithmetical: the term $17/3000$ was obtained by working backwards from the calibration target, and the primality of 16339 is the evidence that no mechanism could have produced the split. A solver whose stopping criterion reads γ is corrupt regardless of the elegance of its output.^[A] As for the broader charge, the defence is partial and is recorded as partial. γ is fixed by matching to a historical internal kinetic reference documented in Appendix A, independently of cosmology, so a vacuum-energy calculation using γ is not circular *with respect to the cosmological data*; but this does not make γ derived. The honest verdict is that the adversarial argument *stands* on its central point: γ is ^[A-], calibrated, not derived, and the framework's protection against tautology is not a first-principles derivation but the structural separation of strata together with the anti-target-leakage rule (proposition 18.1).^[A-]

27.2 Second Round: The D20 Phase-6 Rebase

Adversarial argument. The Phase-6 rebase, says the adversary, is an evasion dressed as rigour. Faced with a primitive that cannot generate gauge structure, the framework has not solved the problem; it has renamed the primitive. Three charges follow. *First*, the kernel is unestablished: Part I offers a ^[D] candidate whose block partition, real structure, and KO-dimension are free moduli, so the ontological office has been transferred from one unproven occupant to another. *Second*, the technical move that appears to rescue the programme — replacing

the exterior derivative by an inner derivation $\delta(\cdot) = [\mathbf{S}, \cdot]$, so that the nilpotency $d^2 = 0$ no longer applies — buys nothing: the algebra generated by a single self-adjoint operator is commutative by the spectral theorem, so δ vanishes identically on it, and a non-vanishing commutator requires an ambient noncommutative algebra that has been *inserted*. *Third*, the one crisp positive result of Part I, the uniqueness of the $[3, 2, 1]$ partition at $N = 6$, is conditional on a design-level filter and a heuristic filter, and its Lean proof obligation is open at the audited head.

Honest adjudication. All three charges are conceded.^[A] On the first: the transfer is real, and section 24.1 states it in those terms; what the re-architecture buys is not a foundation but a sharper obstruction — G1–G4 with quantitative lower bounds, where v3.9 had only a declaration that the gap was open. On the second: the adversary’s algebra is correct, and the framework adopts the point as its own result rather than resisting it; the passage to inner derivations evades the $d^2 = 0$ identity but relocates the obstruction to the insertion of the ambient algebra, which is condition G1. Any claim that this passage “lays the foundation for gauge groups” is withdrawn. On the third: the conditionality is stated in proposition 5.1 and its grade is ^[D]; discharging the Lean obligation would upgrade the formal layer and nothing else. The framework’s answer to all three is the same: these are the honest boundaries, and the manuscript exists to make them explicit rather than to argue past them.

What the self-audit establishes. The adversary test is not passed by winning the argument; it is passed by surviving the argument with the evidence classes intact. γ remains ^[A-]; the museum construction stays refuted; the calibration is non-circular with respect to cosmology but is not a derivation; the kernel remains ^[D]; the inner-derivation escape is a relocation, not a closure; and the selection proposition remains conditional. This is the appropriate outcome of an honest self-audit: the framework’s weakest load-bearing points are identified, the adversary’s valid points are conceded, and nothing is relabelled to make the defence look stronger than it is.^[A]

Adversary Test: Outcome

Round 1 (v3.9, γ). The strongest attack targets the origin of γ (L4). The numerological construction is conceded to the adversary and stays quarantined; the rejection is

methodological (backward construction), with primality as the evidence. The calibration of γ is non-circular with respect to cosmology but is *not* a derivation; γ remains [A-].

Round 2 (D20). All three charges conceded: the kernel is an unestablished [D] candidate; the inner-derivation route relocates the $d^2 = 0$ obstruction rather than closing it, and any “foundation for gauge groups” reading is withdrawn; the $N = 6$ selection proposition is conditional on a design-level and a heuristic filter, with its formal obligation open.

Verdict. The test is “passed” only in the sense that the evidence classes survive the attack unchanged, not in the sense that either adversary is refuted.[A-]

28 UIDT v4.0 Research-Gap Atlas

The atlas distinguishes missing definitions, missing theorems, missing physical bridges, missing data, and unresolved governance actions. A gap is not evidence against the programme by itself; concealing or misclassifying it would be.[A]

ID	Open object	Closure condition	Class
GAP-01	Primitive selection	Target-independent construction or a non-uniqueness theorem over an explicit admissible class.	[D]
GAP-02	G_{SM} origin	Gauge algebra, representations, chirality, anomalies, and generations derived without target insertion.	[D]
GAP-03	G1–G4 joint satisfiability	Constructed mechanism satisfying all four conditions with proofs and falsification cases.	[D]
GAP-04	Generation number	Endogenous selection of exactly three generations.	[D]
GAP-05	Kernel dynamics	Target-blind action, state space, transition law, and measure.	[D]
GAP-06	Continuum limit	Controlled convergence of observables and a universality or renormalisation result.	[D]

ID	Open object	Closure condition	Class
GAP-07	Dimension emergence	Selection theorem that does not encode the requested dimension.	[D]
GAP-08	Lorentzian signature	Target-independent mechanism yielding causal and signature structure.	[D]
GAP-09	Time and state selection	Non-temporal selection principle for the state whose flow has physical temporal meaning.	[D]
GAP-10	UV/IR decoupling	Proof that microscopic structure yields the required low-energy locality and suppresses contamination.	[D]
GAP-11	Observer map	Explicit domains, morphisms, functor laws, physical implementation, observables, and interventions.	[D]
GAP-12	Born rule	Kernel-to-quantum bridge satisfying the assumptions of a Gleason-type theorem, or permanent import status.	[D]
GAP-13	Consciousness boundary	Falsifiable observer-level model without a qualia, collapse, or fundamental-experience overclaim.	[E]
GAP-14	Biological scale bridge	Physical channel from subatomic quantities to neural observables with noise and decoupling analysis.	[D]
GAP-15	Yang–Mills operator matching	Operator and correlator map from the internal gap object to a consensus Yang–Mills observable.	[D]
GAP-16	γ origin	Blind, preregistered mechanism independent of the calibrated value.	[D]
GAP-17	Effective-sector derivation	Coarse-graining theorem explaining why an EFT and its parameters follow from the kernel.	[D]

ID	Open object	Closure condition	Class
GAP-18	Cosmological dynamics	Equations and likelihood predictions for CMB, BAO, supernovae, lensing, and structure growth.	[D]
GAP-19	Matter–antimatter asymmetry	Sakharov-complete baryogenesis or leptogenesis mechanism.	[D]
GAP-20	Black-hole sector	Defined singularity, entropy, evaporation, and information-recovery model.	[D]
GAP-21	Vacuum extraction	Protocol compatible with passivity, quantum inequalities, conservation, and backreaction.	[D]
GAP-22	Lean reconciliation at f1ae5486	Five-part snapshot and line-by-line adjudication completed for the named commit; every later head requires a fresh reconciliation.	[A]
GAP-23	NORM-01	Single-class decomposition of compound claims; closed by this content-finalization patch, subject to Claude diff audit.	[A]
GAP-24	PROV-002	Immutable TeX image digest, source lock, negative fixtures, and deterministic report.	[A]
GAP-25	Governance enforcement	Non-empty PR-range gates, protected branch, human PI sign-off, and required external counter-signature.	[A]
GAP-26	Source authority	Claim graph separating chat, hypothesis, source, raw data, code, derived artefact, and manuscript statement.	[A]
GAP-27	External replication	Independent formal and numerical reproduction on a separately controlled system.	[D]

ID	Open object	Closure condition	Class
GAP-28	FSS/P2 scientific re-sult	Active use is quarantined under Decision D24; the scientific question remains open pending raw data.	[D]

Table 13: UIDT v4.0 research-gap atlas.

Gap Closure Rule
No gap is closed by analogy, numerical proximity, AI agreement, successful compilation alone, or agreement with a theory that starts from different primitives. Closure requires the mathematical, empirical, provenance, or governance object stated in the relevant row.

29 **Reproduction Note**

Every quantitative and formal statement in this manuscript is reproducible from the following commands and artefacts. Nothing in this section constitutes an authorization.

Numerical claims. The exact coupling identity retained from the historical scalar/EFT record is reproduced from Appendix A; it is an internal ledger relation, not a premise of the active ontology. Working precision is set locally, never globally:

```
python3 -c "from mpmath import mp,mpf; mp.dps=80; assert mp.dps==80
k=mpf(1)/2; lS=mpf(5)/12; print(mp.nstr(abs(5*k**2 - 3*lS), 80))"
# -> 0.0    internal closure < 1e-14; NOT external agreement
```

No other numerical value in this manuscript is computed here; ledger values are reproduced at their recorded classes, and no new eighty-digit value is introduced, estimated, or interpolated.

Symbolic claims. proposition 20.1 is an identity of differential forms and requires no numerics; the schematic check is given at section 20.1. The rigidity results underlying G1–G4 are representation-theoretic identities: the maximal compact

subgroup of $GL(1, \mathbb{R})$ is $\{\pm 1\}$; loop phases factor through $H_1(\Gamma) = \pi_1(\Gamma)^{\text{ab}}$; Skolem–Noether gives $\text{Aut}(M_N(\mathbb{C})) = PU(N)$; and the rank budget for level-one simply-laced affine algebras yields $\text{rank} = 1$.

Formal corpus. Part I now records two non-interchangeable repository states. The historical table at 61bc24da remains a static whole-word scan for which no build was executed. The reconciled table at f1ae5486 records the supplied commit tree, root import graph, environment-qualified 18-module build, axiom inventory, and whole-repository proof-debt scan. Reproduction requires fetching the named commit and repeating those exact operations. Neither snapshot is projected onto master or a later head, and the successful build does not constitute a physics result.

Empirical values. All Stratum-I values (table 9 and the benchmark discussion of section 19) are quoted from the cited external sources with their published uncertainties. They are post-hoc comparators, stored with solver access disabled (proposition 18.1); none enters any computation whose output it is meant to test.

30 Reference and DOI Check

The bibliography of this manuscript is the verified bibliography of UIDT Ontology v3.9.9, carried over without addition. Two corrections were made in the present audit pass and are marked in place:

1. **Attribution correction.** The survey arXiv:1904.12392, cited in the v3.9 bibliography as by van Suijlekom alone, is by *Chamseddine and van Suijlekom*. The first author has been restored. The claim this source supports in remark 4.1 — that the Standard-Model algebra with each summand appearing once carries a KO-dimension-6 real structure — rests, in this manuscript, on Chamseddine–Connes–Marcolli [19], with the survey cited as context. *No theorem number is asserted for a source whose theorem statement was not read in the present pass.*
2. **Bibliographic completion.** The page range and place of publication for Ćačić [90] (Wiesbaden, pp. 9–68) were verified and added. The source’s own abstract confirms the framing used in Part I: the structure theory of finite real

spectral triples due to Krajewski and to Paschke–Sitarz is there generalised to arbitrary KO-dimension and to the failure of orientability and Poincaré duality, and that theory reduces the study of finite spectral triples largely to *multiplicity matrices*. This is the primary-source basis of the bimodule-multiplicity reading of G1.

No source is cited in this manuscript for more than it claims. Preprints are carried at the class of their strongest independent confirmation, not at the class of their abstract. Any bibliography entry required by future additions to this manuscript must pass a DOI or arXiv resolution check before insertion; entries that cannot be resolved are marked UNVERIFIED and may not carry evidential weight.

31 Consolidated Claims Register

The three per-part registers (table 6, table 8, table 10) are authoritative in their respective domains. The table below is a navigational summary of the evidence distribution and is not a substitute for them.

Class	Part I	Part II	Part III ledger
[A] (proof/identity/audit fact)	K-02, G-01–G-04, L-01, L-02	O-01, O-03, Q-01, E-01, P-01	ONT-01–ONT-03, ONT-06
[A-] (calibrated)	—	—	ONT-04 (γ)
[B] (benchmarked)	lattice verdicts, table 3	—	ONT-05 (Δ , D18), ONT-07
[C] (cosmology cap)	—	—	ONT-09
[D] (programme)	K-01, G-05, S-01, S-02, C-01, C-02, L8–L11	T-01, T-02	ONT-08, ONT-11
[E] (interpretive/museum)	quarantined audit doc	O-02, O-04	ONT-10, L6

Table 14: Evidence distribution across the three parts. The firmest entries are boundary and negative results; every positive structural ambition sits at [D].

Read as a whole, the distribution is the point. Not one positive claim about the origin of gauge structure rises above [D] in any part of this manuscript. The classes above [D] are occupied by identities, obstructions, calibrations, audit facts, and governance commitments. A framework whose strongest statements are the statements of its own limits has not thereby proved anything about nature; it has, at least, made itself checkable.

32 Ledger Record: PI Decision D19

D19 — Ledger Record: Ontological Re-Architecture

Decision. The continuous real scalar field $S_{\text{legacy}}(x)$ is demoted from ontological primitive to a historical macroscopic effective-field model. A finite relational-algebraic pregeometric kernel is the active candidate under investigation at evidence class ^[D].

Binding conditions. (i) Part I is a formal research programme at ^[D], not an established foundation. (ii) No target block partition is permitted in core definitions, axioms, hypotheses, losses, or acceptance conditions. (iii) Historical ledger values retain their recorded classes: $\kappa = \frac{1}{2}$ (exact) ^[A], $\lambda_S = 5/12 \approx 0.4167$ ^[A], $v = 47.7 \text{ MeV}$ ^[A], $\gamma = 16.339$ ^[A-], $\Delta = 1.710 \pm 0.015 \text{ GeV}$ ^[B] under D18/D21, and cosmological entries at no higher than ^[C].

Status. CLOSED on 23 July 2026 under the PI standing instruction for scientifically determinate decisions. No constant is reclassified by D19.

33 Ledger Record: PI Decision D20

D20 — Ledger Record: Architectural Reassignment

Decision. The quantitative content of the former scalar/EFT Part III loses its role as active ontology and is reorganized as a verification, benchmark, and parameter-ledger stratum. The historical scalar model is retained in Appendix A and is not imported into the active architecture. The finite algebraic pregeometric kernel remains a candidate under investigation at evidence class D.

Binding conditions. (i) Part I remains a formal research programme at evidence class D, not a theorem corpus; no machine-checked statement upgrades a physics claim without separate governance. (ii) No Standard-Model-like block partition is hard-coded into a core formal structure; partitions enter as inserted candidate instances under GLBC. (iii) All incoming evidence classes remain unchanged exactly as recorded in the preceding D19 ledger entry; D20 changes architectural role only. (iv) PI-DELTA-P6, PI-ET-P6, PI-MUSEUM, and P2 were separate decisions at D20 and are subsequently closed by D21–D24.

Application record. The MSS-000 rebase was audited against blob 601cc57d and is present in the supplied watermarked baseline blob 7962e5e7. This statement records manuscript provenance only; final integration and merge remain PI-only.

34 Closed PI Dispositions D21–D25

D21 — Δ . Upgrade rejected; D18 retained at 1.710 ± 0.015 GeV and ^[B]. No particle-mass or glueball identification.

D22 — E_T . Scale firewall confirmed; class retained at ^[E]; no neural, biological, cognitive, macroscopic, or life/death application without a separately audited scale bridge.

D23 — Museum.

Route B approved as the final v4.0 historical-retention route. The appendix is non-imported and reversible.

D24 — P2. FSS/P2 placed in archival quarantine because raw data and run provenance are absent. No active claim survives the quarantine.

D25 — Name.

Active programme and manuscript names approved by Decision D25; UIDT remains the historical lineage name.

Open Obligations Carried Forward

The following are recorded so that no reader of this manuscript mistakes its completeness for closure.

1. Lean findings F1 and F3 are reconciled for the named head f1ae5486: the flagship declaration has no sorryAx, and the eight remaining sorry occurrences are all registered. F2 remains open because no dedicated target-shape scan was supplied. Four modules remain outside the successful root build, and every later head requires a fresh reconciliation. These facts alter only the formal software record; physical claims remain bounded by their hypotheses.
2. Hypothesis H1 (intersection-form filter) is design-level: its derivation from spectral-triple axioms is the stated upgrade path. Hypothesis H2 (mass non-degeneracy) is a heuristic proxy for a claim about the Yukawa sector. The $[2, 2, 2]$ rejection rests on H2 alone (tension T8).

3. The Relocation Necessity Conjecture (open question 2) and the staircase conjecture (open question 1) are open and independently publishable.
4. The thermodynamic passage of Part II has no construction. The three no-go points — absence of a dimensional-emergence theorem, UV/IR mixing, and thermal-time circularity — stand.
5. The Born rule, a physical E_T scale bridge, the H_0 and S_8 tensions, and the origin of G_{SM} remain unresolved. Decision D22 closes only the misuse boundary, not the missing scale bridge; Decision D24 closes only the active use of P2, not its scientific question.
6. No “66 $\rightarrow$ 7” Krajewski reduction is part of this manuscript without the complete starting inventory and elimination ledger required in section 3.5. Division-algebra stabilizers remain imported comparators, not a derivation from the RAP kernel.

Historical Appendix: Non-Imported Scalar/EFT Museum Record

Museum Route B — Final Reversible Retention under D23

The material in this appendix is preserved to prevent silent loss, support regression audit, and document the superseded v3.9/D19 scalar and EFT grammar. It is not imported into Parts I–III. Occurrences of $S_{\text{legacy}}(x)$, $\partial_\mu S$, $\eta_{\mu\nu}$, the effective metric ansatz, the scalar potential, or scalar correlators below are historical model content, not active primitives, operators, acceptance conditions, or derivations. Decision D23 fixes Route B for v4.0: the record is retained, non-imported, and may not be silently deleted or reactivated. Any later relocation requires a new separately audited PI record.

A Historical Effective-Scalar Axiomatics

The v3.9 effective-sector record that associated the spectral gap Δ , the calibrated invariant γ , the coupling κ , and the self-coupling λ_S owes the reader the Lagrangian in which those quantities actually live. This section preserves the four historical axioms, the effective Lagrangian, and the dimensional bookkeeping for audit. None of these equations is imported as a premise of the active Phase-6 architecture. One axiom — and only one — changes under D19; the change is stated first and in full.

A.1 The Four Axioms in Demoted Form

Axiom 1 (Historical Effective-Field Postulate (D19 record)). There exists a real scalar effective field $S_{\text{legacy}}(x)$, the vacuum information density, of canonical mass dimension $[S_{\text{legacy}}(x)] = 1$. Under PI Decision D19, $S_{\text{legacy}}(x)$ is *not* an ontological primitive: it is postulated as the macroscopic, order-parameter-level description in which the kernel programme’s continuum limit (section 12) is expressed, and within the grammar of this Part it is the sole dynamical scalar degree of freedom of the effective sector. The v3.9 reading of this axiom — $S_{\text{legacy}}(x)$ as sole fundamental degree of freedom, with all structure emergent from its dynamics (Variant B) —

is recorded in section B.2 as the superseded grammar. Its preservation in this appendix is a document-state fact.^[A] The former ontological status remains a historical interpretive commitment at ^[E].

Axiom 2 (Gauge-Invariant Self-Interaction). The dynamics of $S_{\text{legacy}}(x)$ are governed by a Lagrangian density $\mathcal{L}_{\text{UIDT}}$ invariant under $\text{SU}(3)$ gauge transformations and respecting vacuum stability ($V''(v) > 0$). It is an effective field theory valid below a cutoff Λ_{UIDT} . The non-minimal coupling $(\bar{\kappa}/\Lambda_{\text{UIDT}}) S F_{\mu\nu}^a F^{a\mu\nu}$ has mass dimension 5 and is therefore not renormalisable by strict power counting; its validity is confined to the EFT domain (Limitation L7). A full non-perturbative UV completion is an open research question.^[A]

The minimal realisation satisfying axiom 2 is the effective Lagrangian

$$\mathcal{L}_{\text{UIDT}} = -\frac{1}{4} F_{\mu\nu}^a F^{a\mu\nu} + \frac{1}{2} (\partial_\mu S)(\partial^\mu S) - \frac{1}{2} \mu^2 S^2 - \frac{\lambda_S}{4!} S^4 + \frac{\bar{\kappa}}{\Lambda_{\text{UIDT}}} S F_{\mu\nu}^a F^{a\mu\nu}, \quad (6)$$

where $\bar{\kappa}$ is the dimensionless EFT-normalised non-minimal gauge-scalar coupling and λ_S the scalar self-coupling; the dimension-5 operator carries the dimensionful prefactor $\bar{\kappa}/\Lambda_{\text{UIDT}}$, in agreement with table 15. Because S is a real gauge singlet, its kinetic term uses the ordinary derivative ∂_μ . In the framework's ledger and renormalisation-group statements this dimensionless coefficient is written simply κ (so $\kappa \equiv \bar{\kappa}$, with ledger value $\kappa = 0.500$). The exact internal coupling constraint of the framework is $5\kappa^2 = 3\lambda_S$ (a candidate fixed-point condition, untested by any defined functional renormalisation-group flow), which for $\kappa = \frac{1}{2}$ gives the exact value $\lambda_S = 5/12 \approx 0.4167$ and the vacuum expectation value $v = 47.7 \text{ MeV}$; like λ_S , v is fixed by this exact constraint rather than independently measured, which is why it carries no uncertainty. This constraint closes to a residual below 10^{-14} at eighty-digit working precision (proposition 18.1 discipline applies: the closure is an internal-consistency certificate, not an external agreement).

Table 15: Dimensional and EFT status of the terms entering axiom 2. The dimension-5 operator marks the sector as an effective field theory.

Term	Op. dim.	Coupling dim.	Renormalisability
$F_{\mu\nu}^a F^{a\mu\nu}$	4	dimensionless	Strictly renormalisable Yang–Mills kinetic term ^[A]
$(\partial_\mu S)(\partial^\mu S)$	4	dimensionless	Canonical scalar kinetic term ^[A]
$\mu^2 S^2$	2	+2	Relevant scalar mass operator ^[A]
$\lambda_S S^4$	4	dimensionless	Power-counting renormalisable self-interaction ^[A]
$(\bar{\kappa}/\Lambda_{\text{UIDT}}) S F_{\mu\nu}^a F^{a\mu\nu}$	5	−1	Non-renormalisable; EFT-only below Λ_{UIDT} ^[A]

Axiom 3 (Geometric Emergence). The effective spacetime metric $g_{\mu\nu}^{\text{eff}}$ is not fundamental but a derived quantity determined by the vacuum configuration of $S_{\text{legacy}}(x)$. A candidate parametrisation is

$$g_{\mu\nu}^{\text{eff}} = \eta_{\mu\nu} + \alpha \partial_\mu S_{\text{legacy}}(x) \partial_\nu S_{\text{legacy}}(x) + \beta S_{\text{legacy}}(x)^2 \eta_{\mu\nu}, \quad (7)$$

with $\eta_{\mu\nu}$ the Minkowski metric and α, β dimensionful constants ($[\alpha] = -4$, $[\beta] = -2$) fixed by the requirement that the emergent geometry reproduce general relativity in the low-energy limit. The specific form of eq. (7) and the values of α, β are model-dependent and carried at predictive status; under D19 this axiom describes emergence *within the effective stratum*, with the deeper origin of $\eta_{\mu\nu}$ deferred to the Part-II passage and Limitation L13.^[D]

Axiom 4 (Falsifiability). The theory defines a finite set of experimentally testable predictions (section 23), each with an explicit falsification threshold. If any critical prediction is excluded at the specified confidence level, the corresponding UIDT mapping, calibration, or prediction must be revised or abandoned.^[A]

Remark A.1 (Falsifiability and critical rationalism). Including falsifiability as a formal axiom is a methodological commitment to the critical-rationalist tradition: a theory that cannot in principle be wrong has no claim to being right. axiom 4 operationalises this by assigning every central prediction a quantitative falsification threshold. The axiom is not a physical law but a meta-theoretical commitment to epistemic honesty, continuous with the principle that internal consistency is not external confirmation (section 18.1).

Historical Effective-Scalar Record: Preserved Status

Axioms. Preservation of the Effective-Field Postulate is a document-state fact ^[A]; its historical ontological content is ^[E] (D19 form). Gauge-Invariant Self-Interaction ^[A] (EFT, dimension-5 operator); Geometric Emergence ^[D] (model-dependent metric); Falsifiability ^[A] (meta-theoretical).

Lagrangian. eq. (6) with $\kappa = 0.500$, $\lambda_S = 5/12 \approx 0.4167$, $v = 47.7 \text{ MeV}$; internal coupling constraint $5\kappa^2 = 3\lambda_S$ closes internally to $< 10^{-14}$ at eighty digits.

Status. The Lagrangian is an effective field theory below Λ_{UIDT} ; renormalisability is explicit per table 15; the gauge sector is coupled, not generated (section B.4).

B Unmarked State, First Distinction, and Superseded Variant B (Historical Record)

B.1 Why $S(x) = 0$ Is Not the Unmarked State

A foundational theory must be exact about its own beginning, and the most common error at the beginning is to confuse *absence of value* with *absence of structure*. UIDT takes the unmarked state, “nothing” in the precise sense of Spencer-Brown’s calculus of distinctions, as the condition in which no field-value assignment exists at all. It is essential to see that this is *not* the same as the field configuration $S(x) = 0$.^[A]

The point is structural, not rhetorical. To write $S(x) = 0$ is already to presuppose a great deal: that there is a field S ; that there is a domain indexed by x over which it is defined; that evaluation of the field at a point is a meaningful operation; and, most importantly, that the value 0 is *distinguishable* from other possible values. A vanishing field value is a marked state: it is the assertion “the field here is zero,” and an assertion of that form lives entirely within an already-constituted system of field, domain, and distinguishability. The unmarked state precedes all of this. It is not the number zero; it is the absence of the apparatus within which numbers could be assigned.^[A]

Because the unmarked state is not a physical configuration, it cannot be assigned a physical history, a temperature, a fluctuation spectrum, or a cosmological epoch. One cannot say that “in the beginning $S(x) = 0$ and then it became non-zero,” because that sentence smuggles a temporal ordering and a pre-existing field into a

condition that, by construction, contains neither. The framework explicitly refuses cosmological, theological, and metaphysical elaborations of the unmarked state. It is a logical boundary, not a primordial substance, and the prohibition on treating it as a substance is as binding as any numerical constraint in the theory.^[A] The result is unaffected by D19: it concerned the grammar of beginnings, not the identity of the primitive, and it survives the primitive's replacement intact.

B.2 Variant B, Recorded and Superseded

The relationship between the first distinction and the scalar field $S(x)$ admits two readings, and the developmental record was right to treat the choice between them as a potential source of circularity. Under Variant A, the distinction is strictly prior: first-distinction, then $S(x)$, then relations, then geometry. Under Variant B, the field $S(x)$ is itself the primitive, and the first distinction is identified with its first non-trivial realisation, $S(x) \neq 0$. The v3.9 manuscript adopted Variant B as a matter of *internal ontological grammar* — explicitly not as a physical proof that the field is the unique primitive of nature. That adoption is a document-state fact.^[A] The choice itself was an interpretive commitment at ^[E]. Any inference of unique physical primitiveness from that grammar remains at ^[E].

Under D19 that grammar is superseded. The office of candidate primitive transfers to the pregeometric kernel of Part I, held at ^[D]; the first distinction is no longer identified with $S(x) \neq 0$, and the question of what, in a finite algebraic primitive, plays the role of first non-trivial structure is recorded as open rather than answered by relabelling. Two elements of the Variant-B analysis are retained at full force because they were never specific to the scalar. The first is the anti-circularity discipline itself: naming the primitive choice, and labelling it as a grammar rather than a proof, is what keeps the definitional circle from closing unnoticed, and D19 is written in exactly that register. The second is the terminology guard.

Terminology Guard: $S(x)$ and "Information"

The UIDT information density $S(x)$ is not Shannon, semantic, Bateson, algorithmic, thermodynamic, or epistemic information. It is a real scalar effective field (D19). "Information" in UIDT is the emergent counting-measure of difference, attaching at the first distinction. Under no reading is $S(x)$ identified with consciousness; that identification is a category error and is prohibited — and the same prohibition applies, unchanged, to the kernel candidate of Part I.^[A]

B.3 The Corrected Ontological Sequence: Morphism as Formal Tool, Not Generator

The developmental dialogue at one stage placed “morphism” within the vertical sequence of ontological stages, as though a morphism were a step in the genesis of being. The honesty audit corrected this, and the correction survives D19 unchanged because it is a statement about mathematics, not about the primitive. The corrected architecture rests on a separation between two axes: a vertical *ontological* axis, the genesis of reality, and a horizontal *formal* axis, the mathematical apparatus used to describe that genesis.^[A]

The decisive observation is elementary and therefore robust. A morphism $f: A \rightarrow B$ presupposes a distinguishable domain A and a distinguishable codomain B ; the very act of writing down a morphism requires that two distinct objects already exist to be related. A morphism therefore cannot be the generator of distinction, because it presupposes distinction.^[A] It belongs to the formal representation layer, not to the primitive genesis layer. Information docks at the first distinction, as the emergent counting-measure of difference; morphisms dock at relations, as the computational interface. Their placement outside the formal representation layer is a structural bookkeeping statement.^[A] The claim that neither is a constituent of genesis remains interpretive at ^[E]. Under D19 the vertical line is re-read with the kernel candidate at its base and the effective field far downstream; the placement of the formal apparatus on the horizontal axis is unchanged, and the major categorical and relational programmes — topos theory, causal sets, constructor theory, QBism — are catalogued here as external lenses.^[A] Their exclusion from the definition of the primitive remains an interpretive boundary at ^[E].

B.4 Relations, Graph Discipline, and the Effective-Field Boundary

The graph stage of the vertical line is a *theorem target*, not a completed construction: the honest form of the proposal is not “the world is a graph” but “reconstructing an effective spacetime from a correlation network is a legitimate theorem target.”^[D] No quantitative graph claim — no count of edges, no degree distribution, no spectral property — is admissible until nodes, edges, weights, and dynamics are

explicitly defined; graph notation represents relational structure and is not that structure; and the schematic $I \rightarrow R(I)$ is toy notation, a recorded direction of construction, not a defined map.^[E] The candidate primary observable $G(x, y) = \langle S(x) S(y) \rangle$ is recorded as a candidate whose canonical status must be confirmed against the ledger before any load-bearing use.^[D] One prohibition carries Class A force and is inherited by Part II's boundary: there is no direct inference from phenomenology to graph structure.^[A]

Geometry, in this architecture, is a coarse-grained representation of underlying structure — the pregeometric tradition's position, with the thermodynamic derivation of the Einstein equations from horizon entropy (Jacobson) [43] and the later entropic-gravity programme [44–46] invoked as Stratum-II precedent that emergent geometry is a respectable position, not as a derivation imported into UIDT.^[E] Downstream of geometry lies the effective field regime, whose most important sentence must be stated plainly, because it is the honest boundary this Part inherits from v3.9 and now shares with Part I: the framework couples the scalar $S(x)$ to a Yang–Mills sector that is *already present*. The gauge fields are part of the effective description, not something the scalar has been shown to generate.^[A]

Historical Effective-Field Boundary

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Regime. The gauge sector is an effective-field-theory regime, in which $S(x)$ is coupled to an already-present Yang–Mills sector. It is not a generative regime.^[A]

Negative result. UIDT does *not* derive the Standard-Model gauge group G_{SM} from a structureless real scalar $S(x) \in \mathbb{R}$ — and, per Part I, no known route derives it from the kernel candidate either; on every known route it is selected, not forced.^[A]

Consequence. The transition from primitive to G_{SM} is a research programme, not a result; its localization (G1–G4), lower bounds, and candidate algebras are the subject of Part I.^[D]

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- [4] **Bibliography audit note.** The external references below were resolved to verifiable primary sources (DOI or arXiv) in a dedicated literature audit. In keeping with the framework's evidence discipline, citation of these sources changes no UIDT evidence class: external literature supports the quantitative pure-gauge benchmarks, is contested on the interpretive physics, explicitly identifies open chiral-fermion obstructions, is interpretive for the consciousness frameworks, and refutes any MeV-to-neural coupling. A small number of secondary venue/page details remain flagged for final primary confirmation before submission.
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